

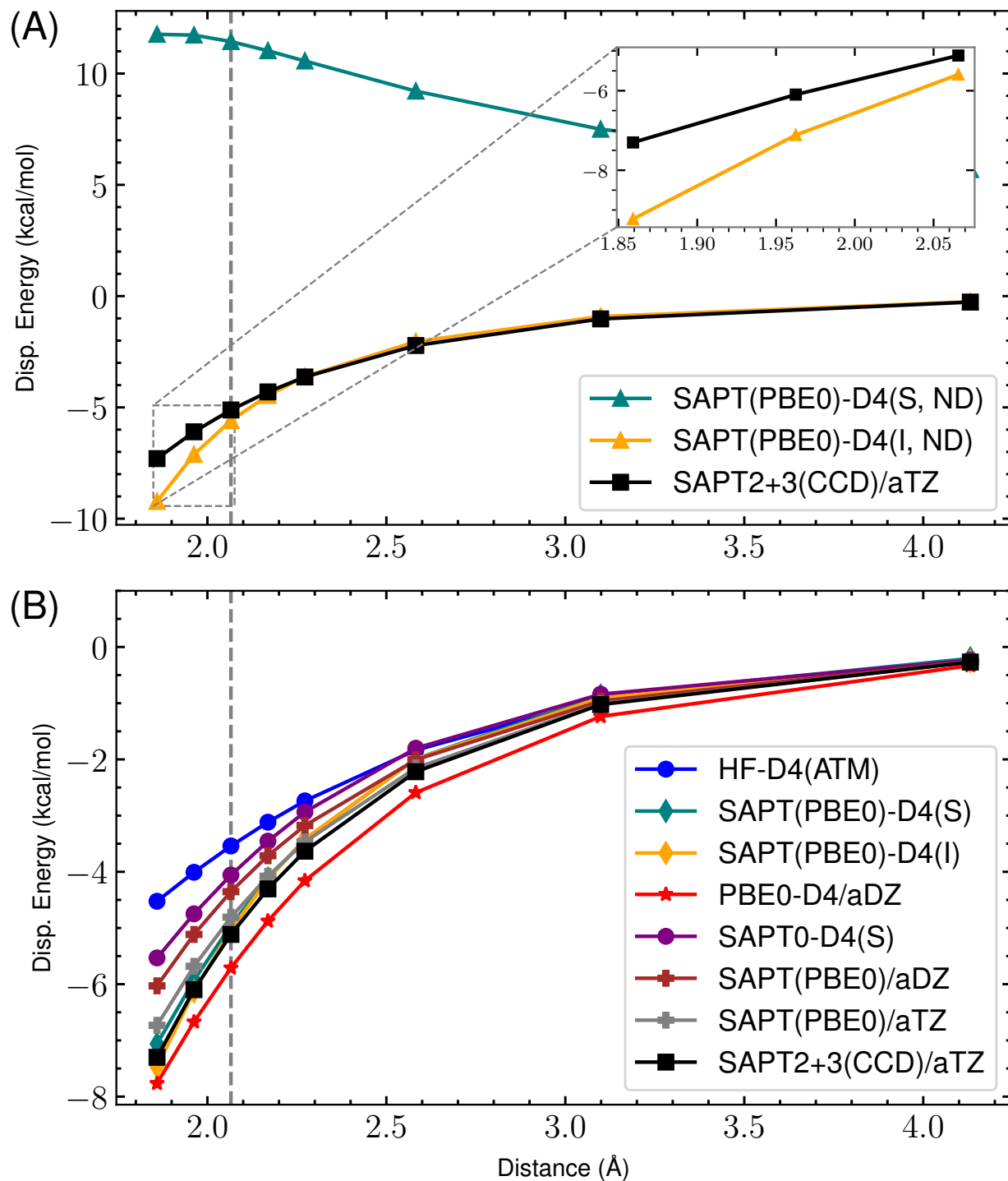


Figure 1: **s66x8 14_Peptide-MeNH2**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

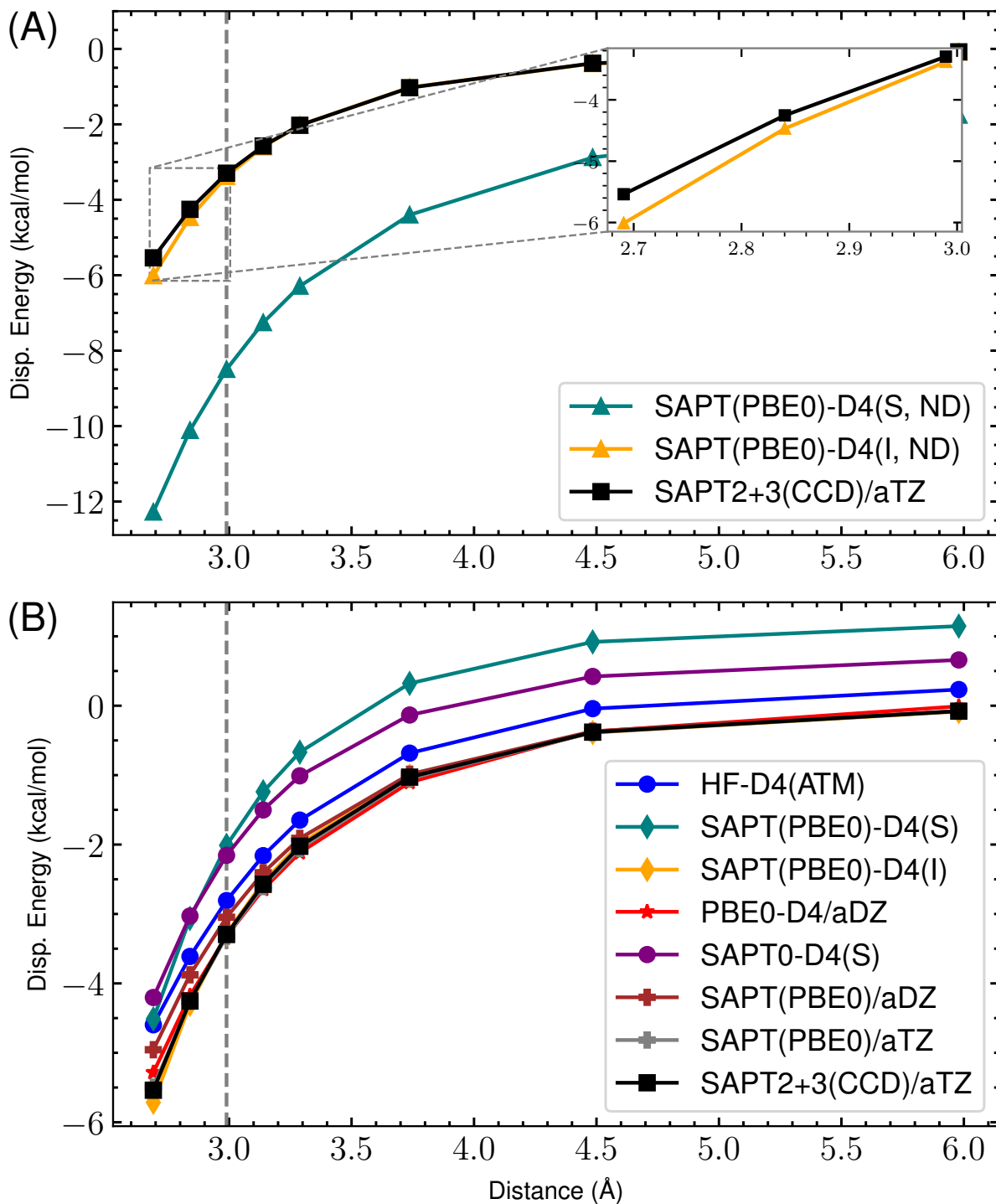


Figure 2: **s66x8 45_Ethyne-Pentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

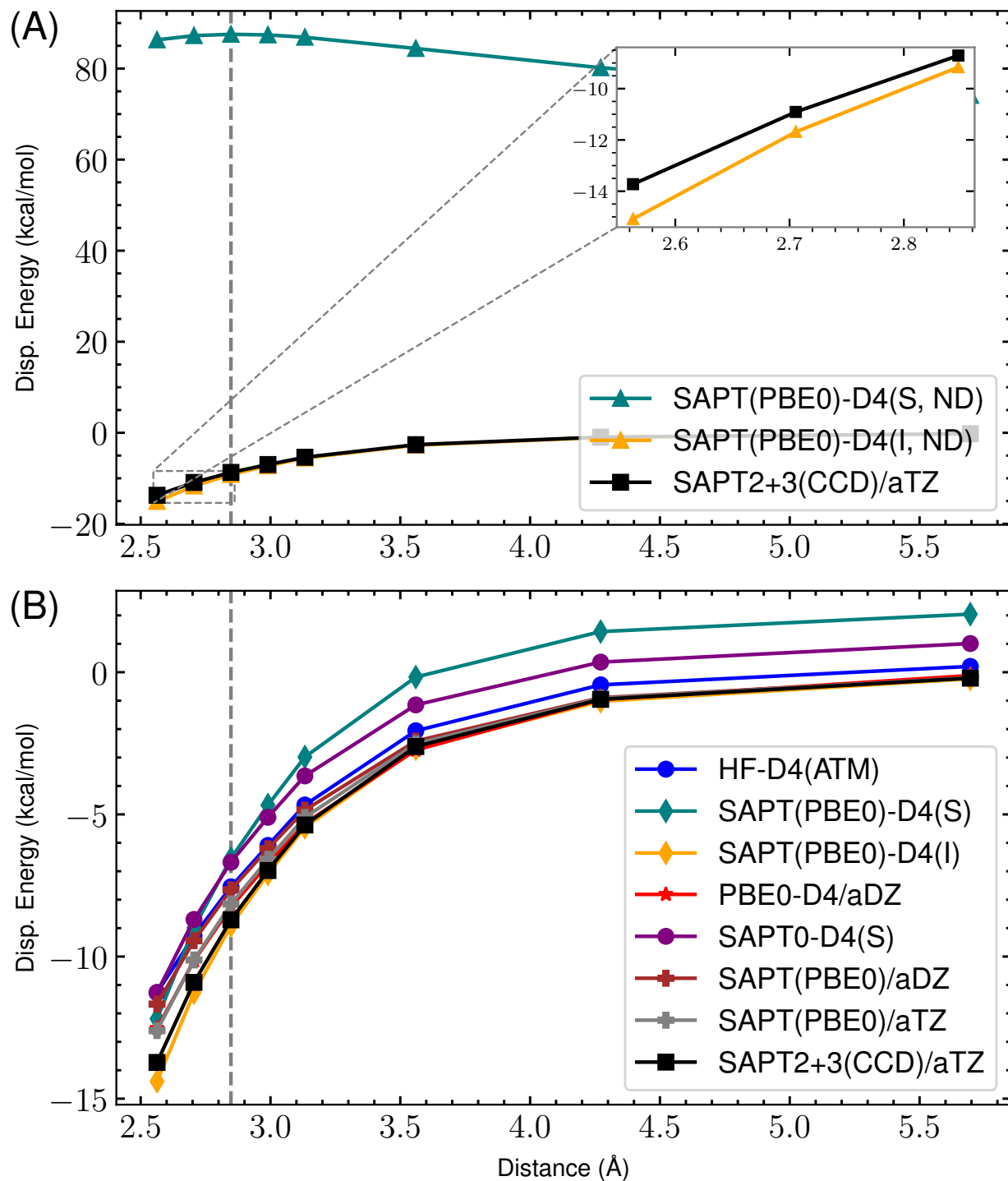


Figure 3: **s66x8 41_Uracil-Pentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

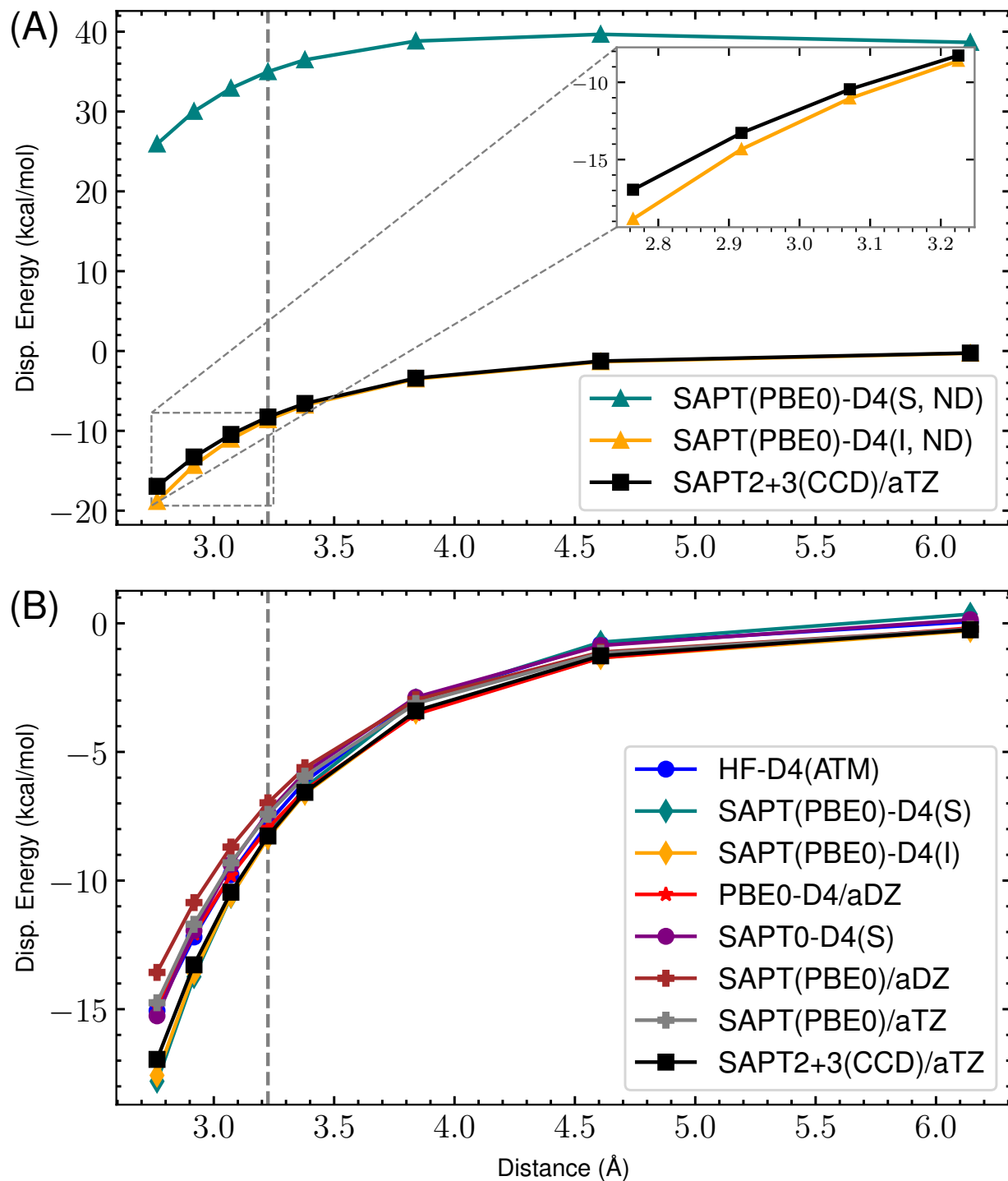


Figure 4: **s66x8 28_Benzene-Uracil**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

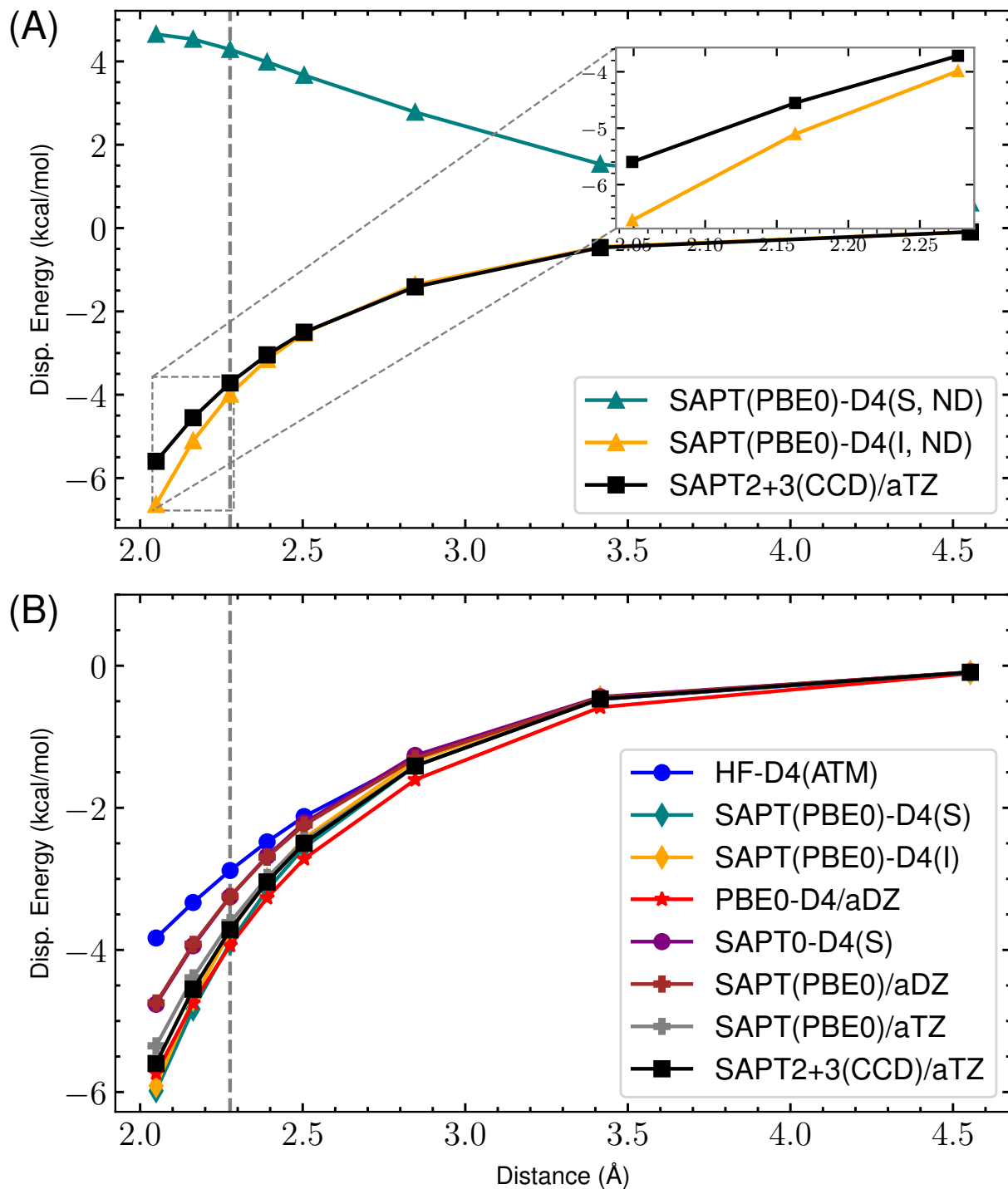


Figure 5: **s66x8 10_MeNH₂-MeNH₂**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

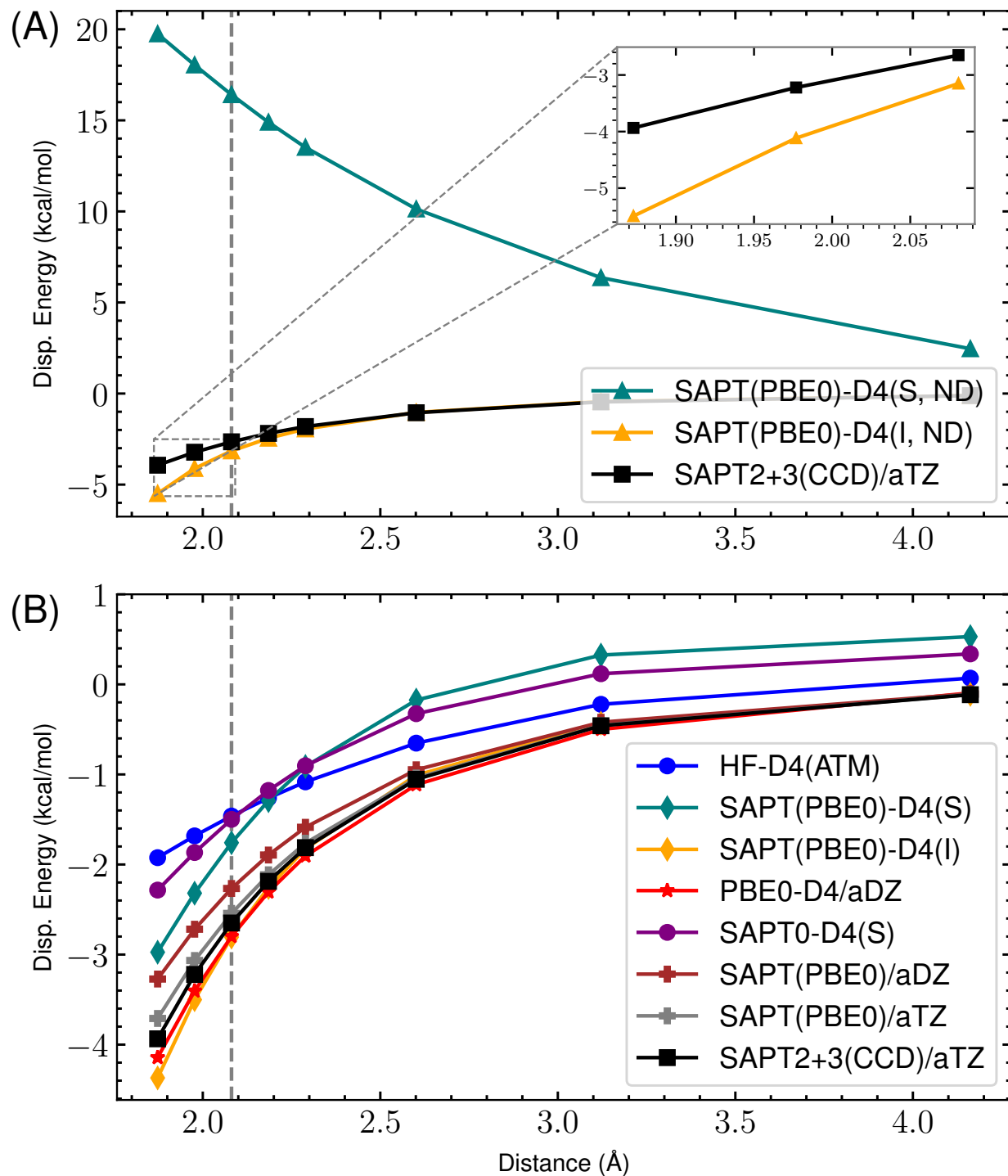


Figure 6: **s66x8 16_Peptide-Water**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

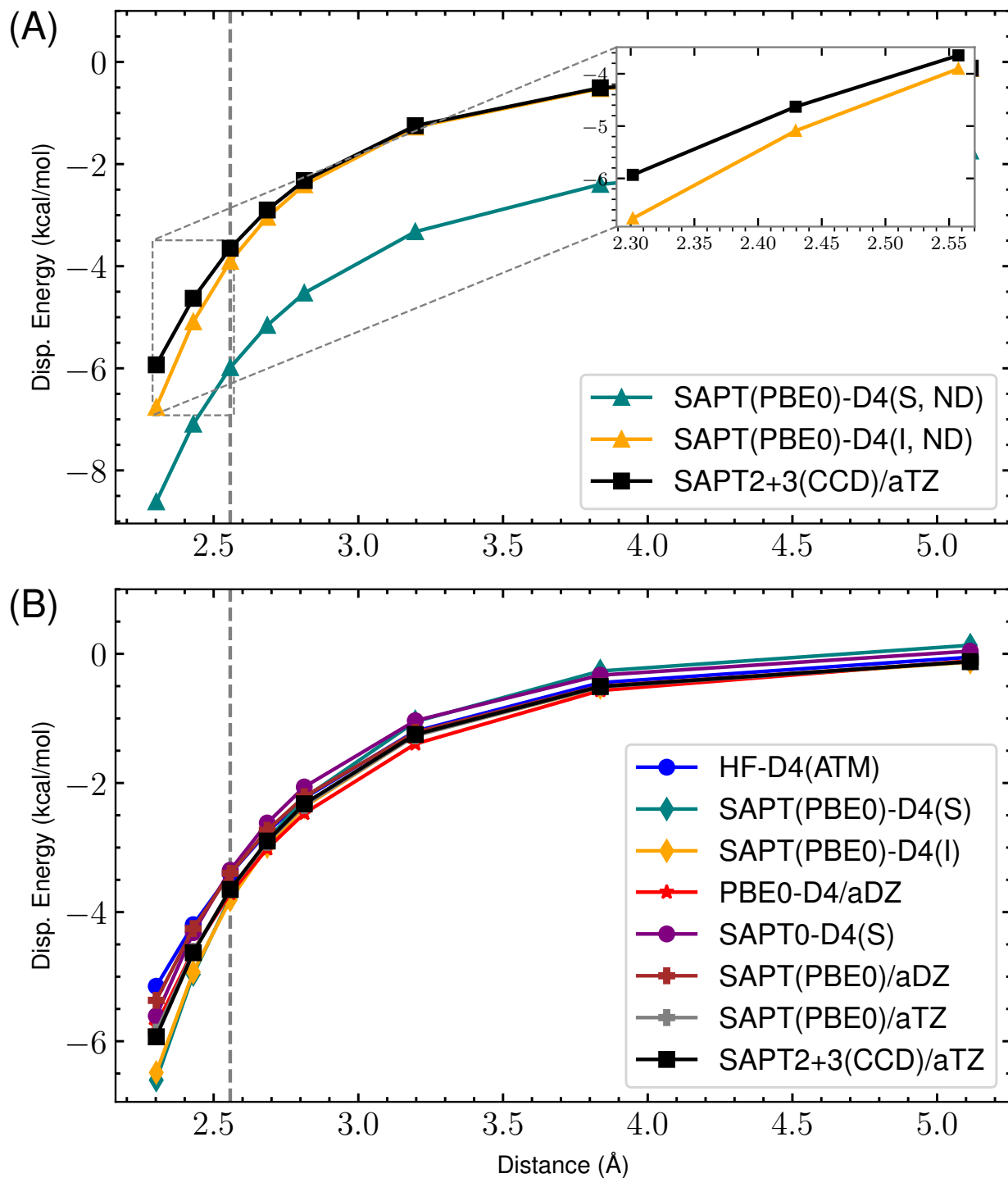


Figure 7: **s66x8 44_Ethene-Pentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

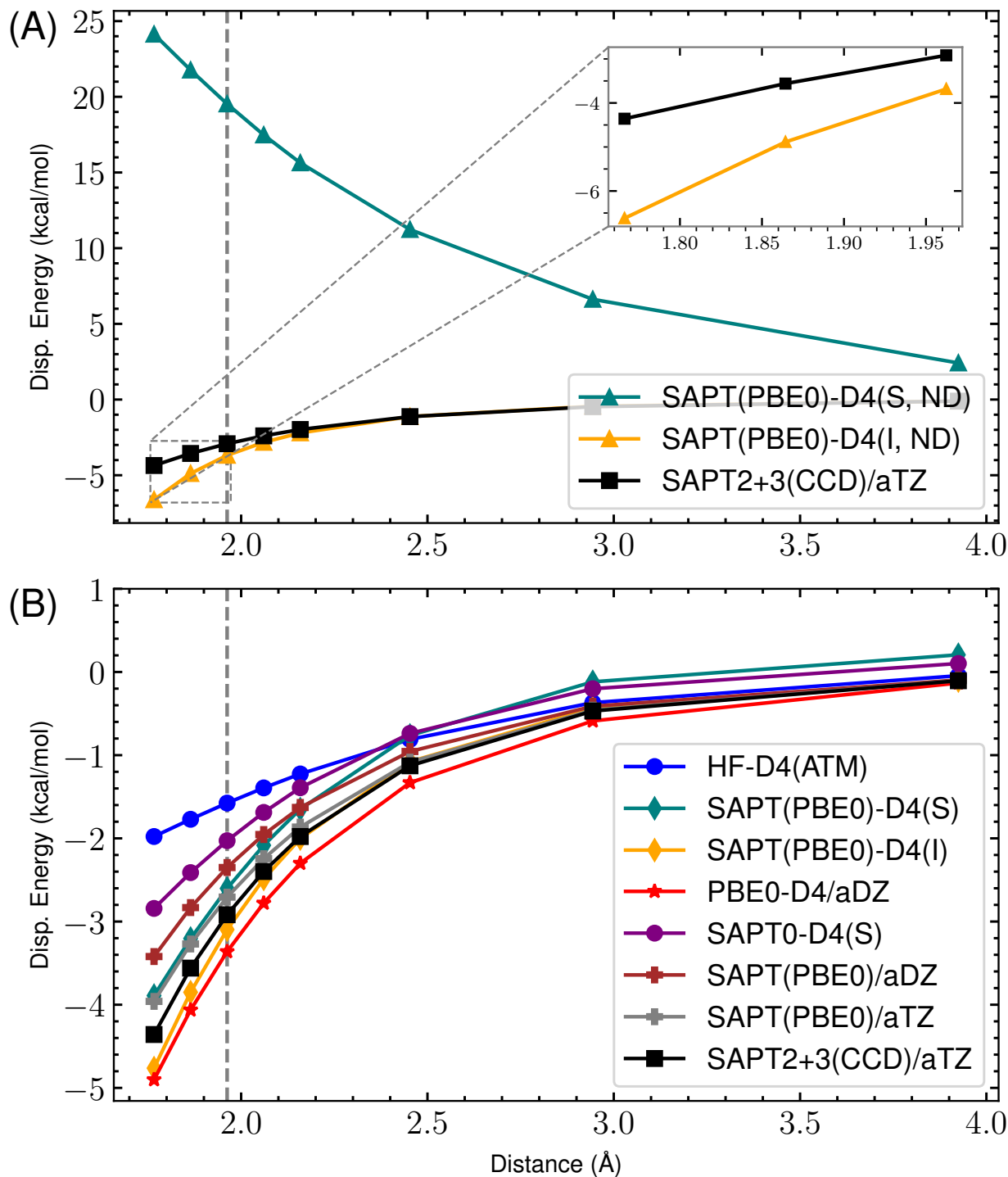


Figure 8: **s66x8 02_Water-MeOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

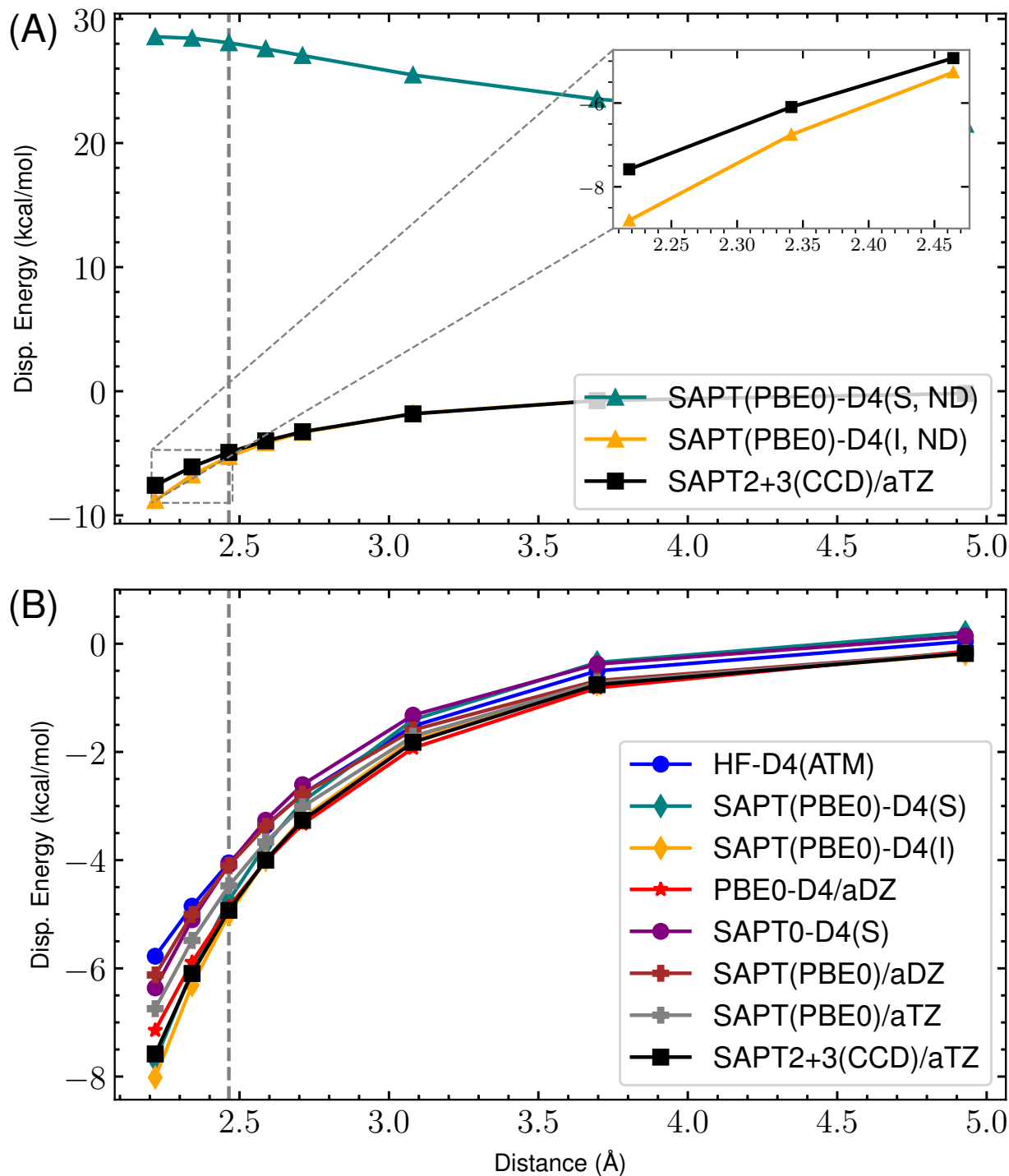


Figure 9: **s66x8 52_Benzene-AcOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

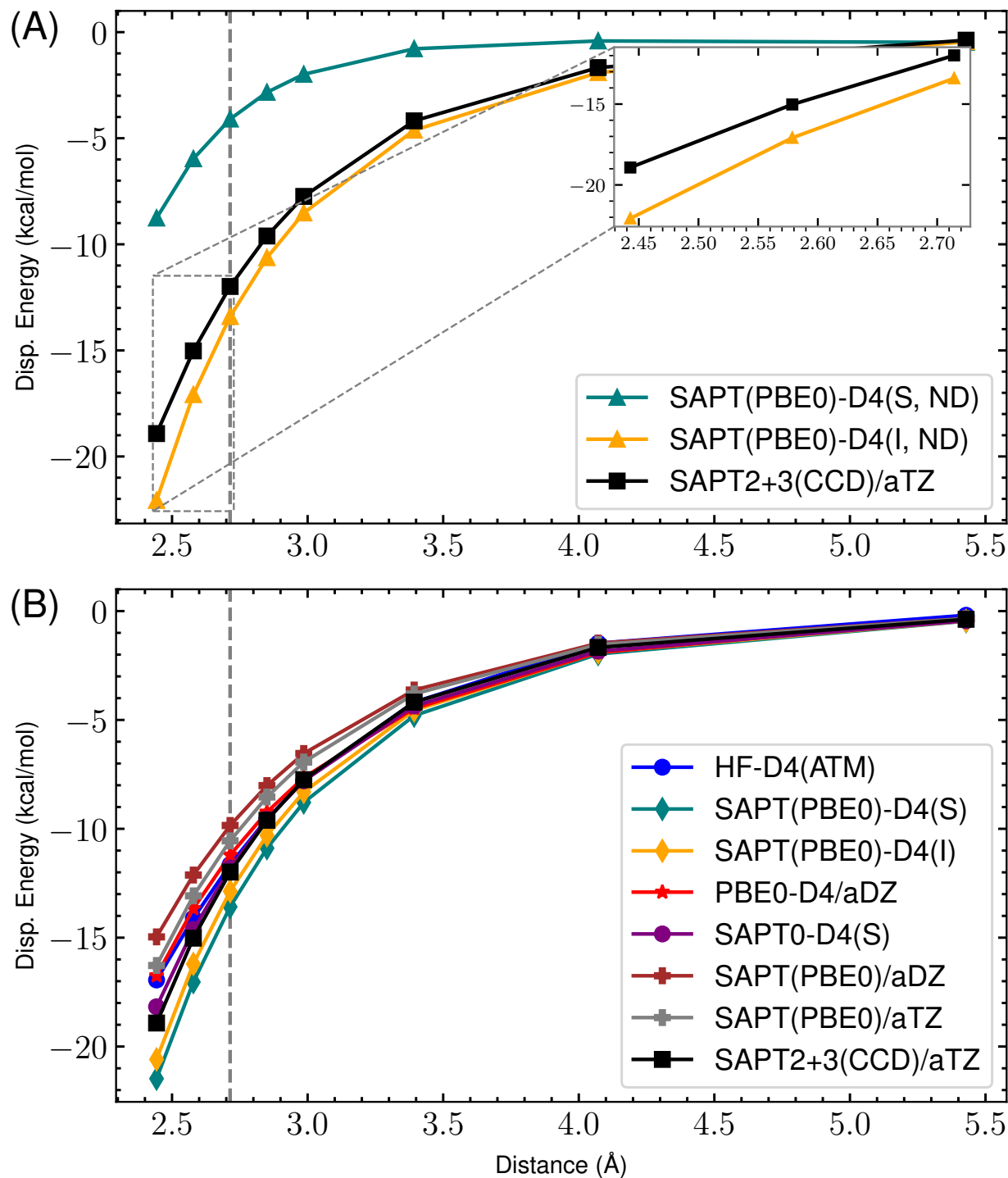


Figure 10: **s66x8 26_Uracil-Uracil**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

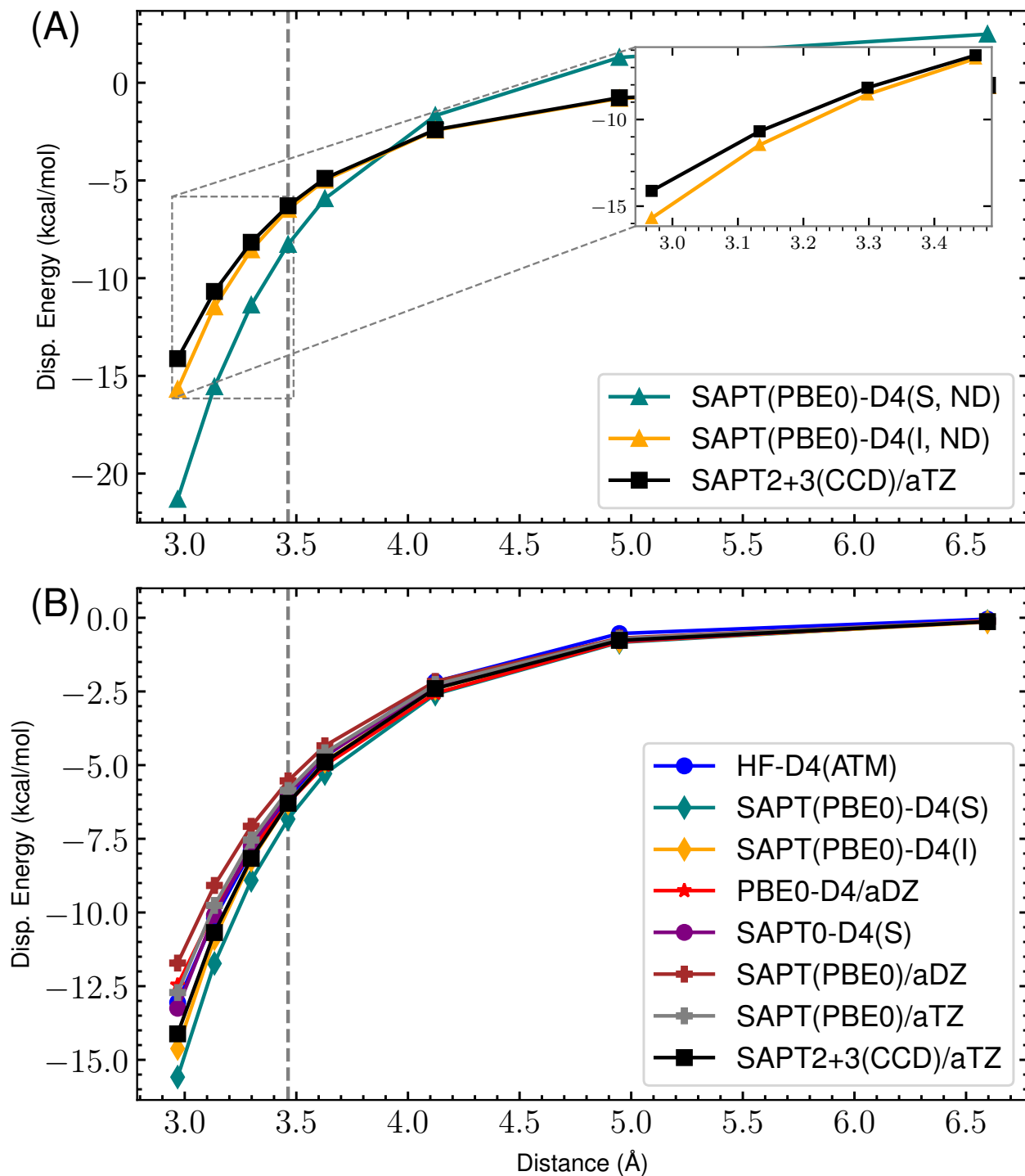


Figure 11: **s66x8 27_Benzene-Pyridine**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

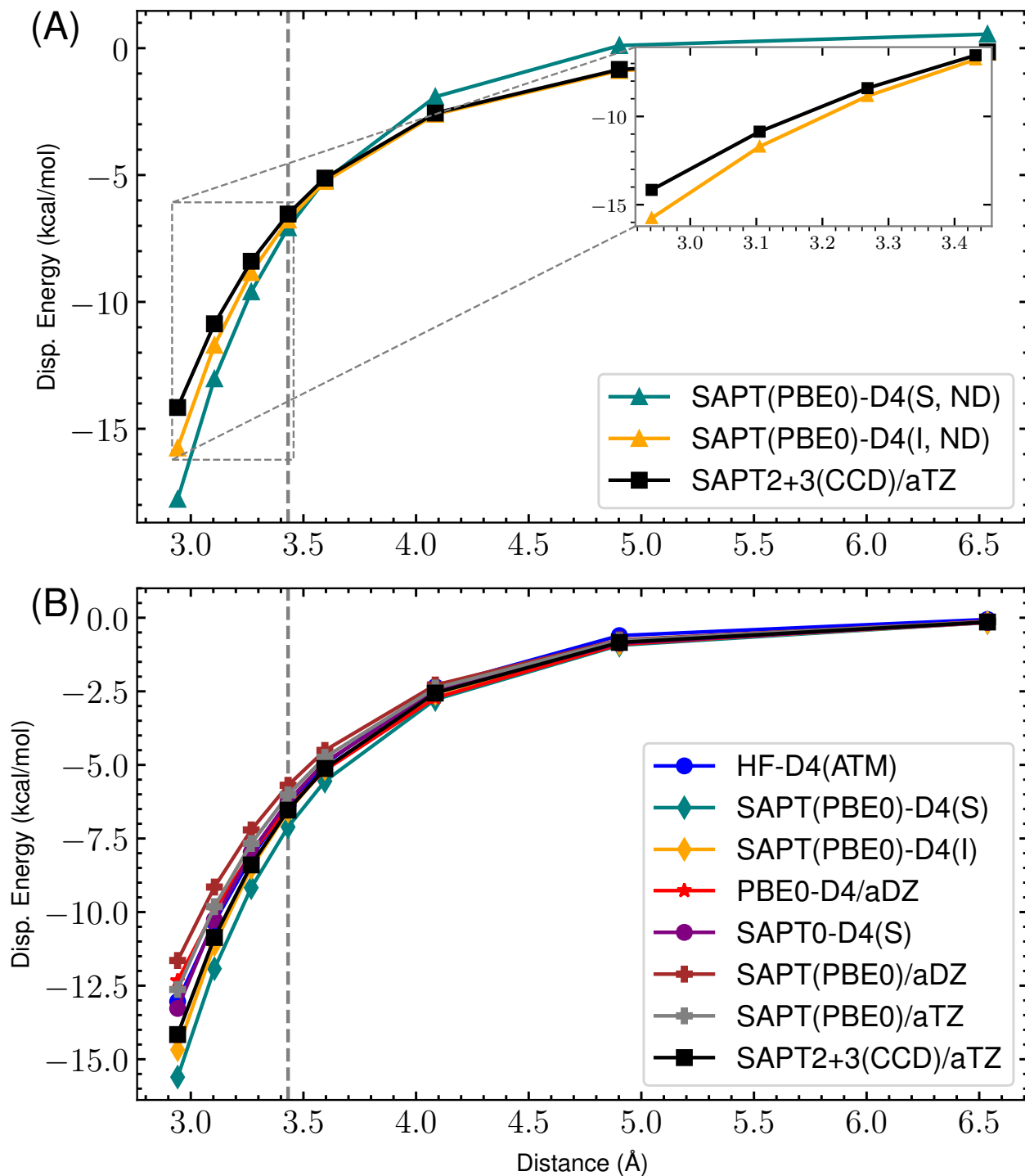


Figure 12: **s66x8 25.Pyridine-Pyridine**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

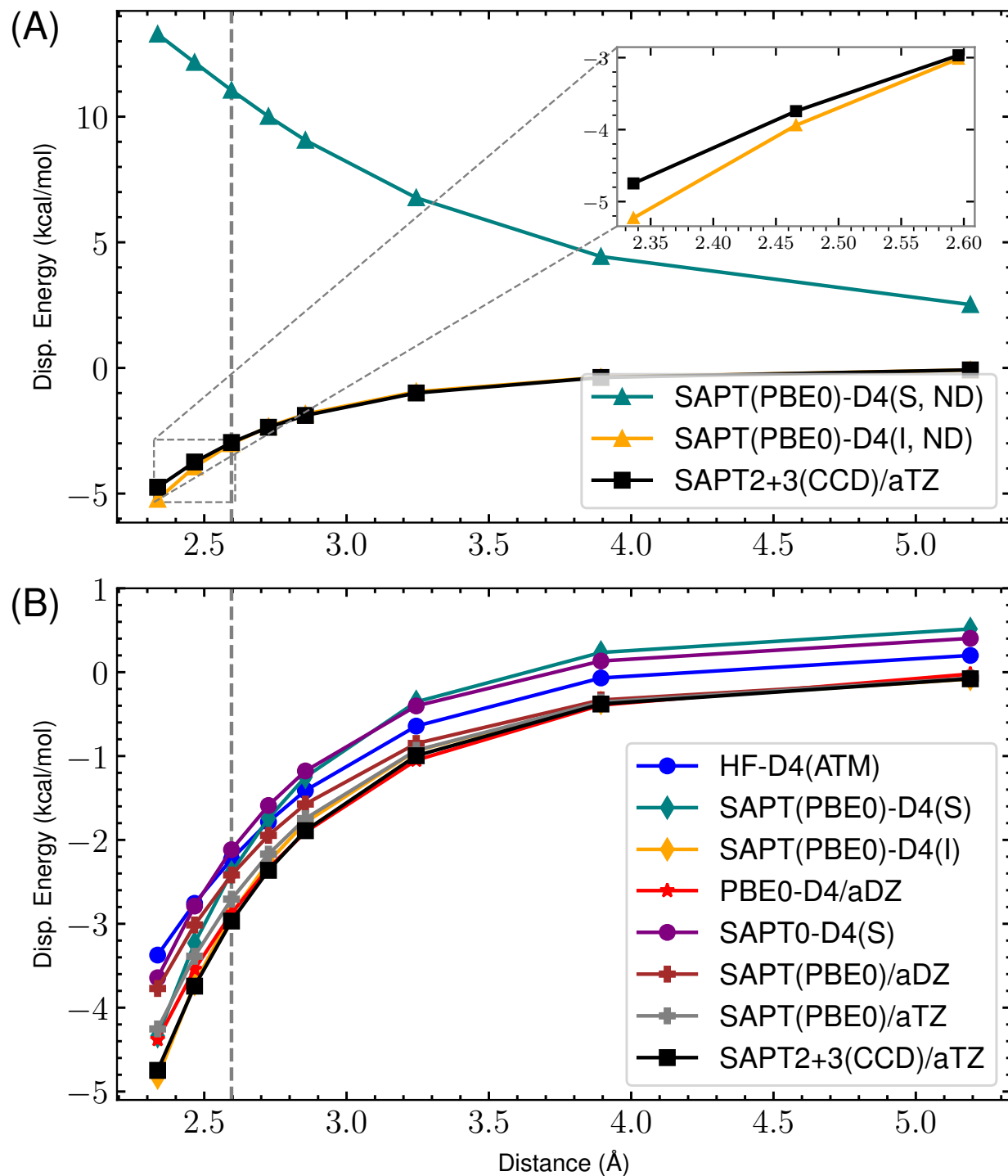


Figure 13: **s66x8 54_Benzene-Water**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

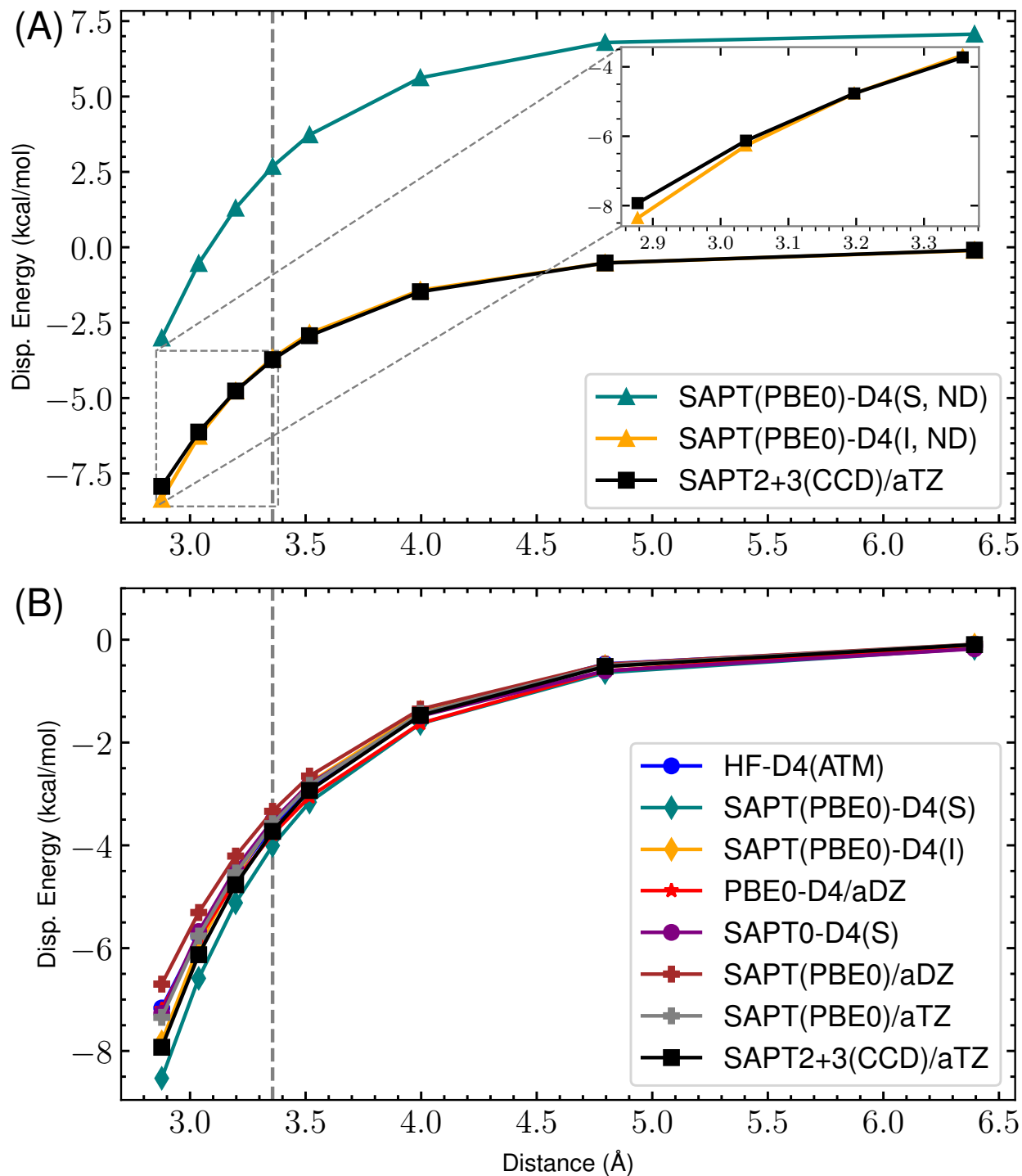


Figure 14: **s66x8 33_Pyridine-Ethene**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

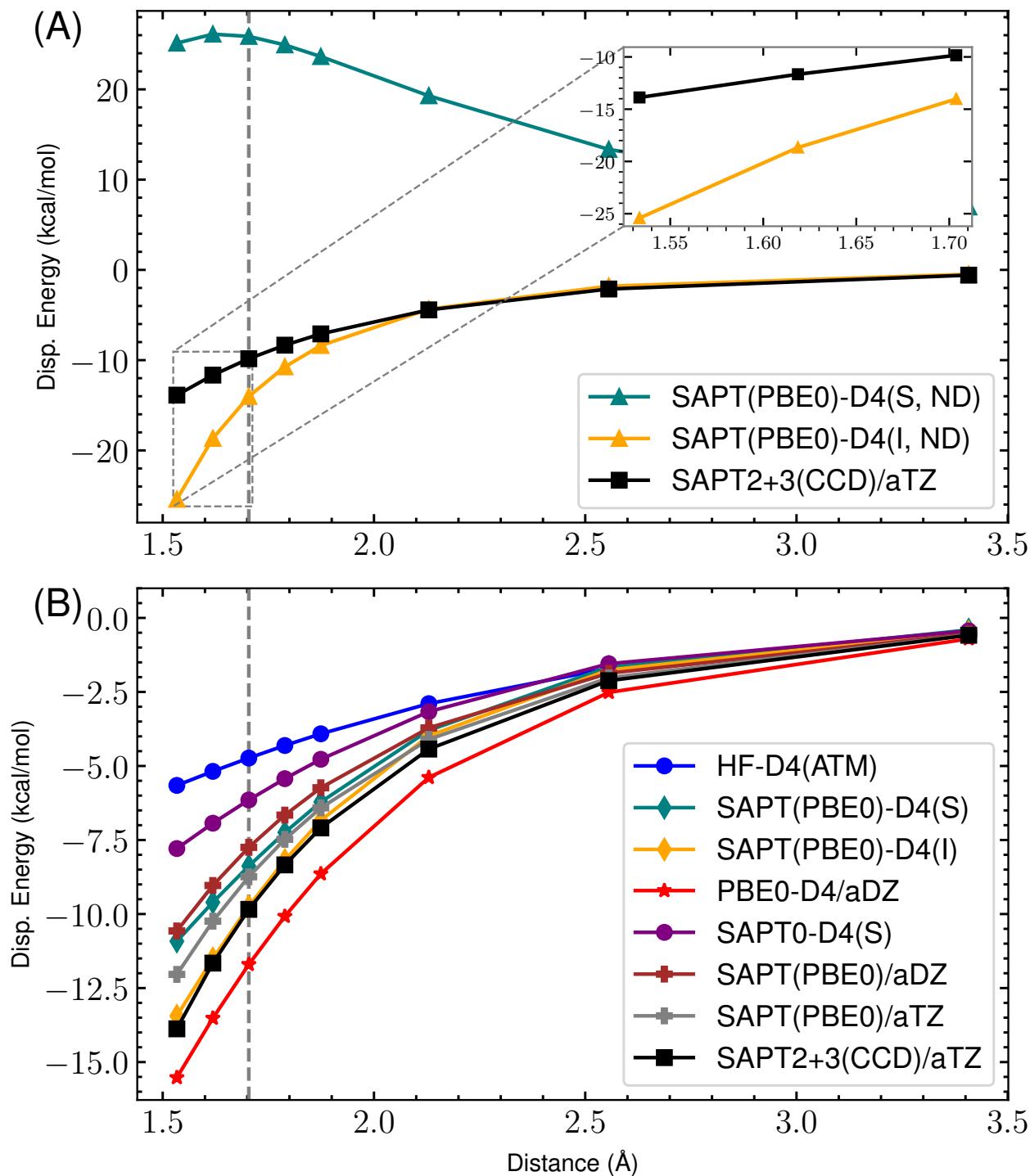


Figure 15: **s66x8 22_AcOH-Uracil**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

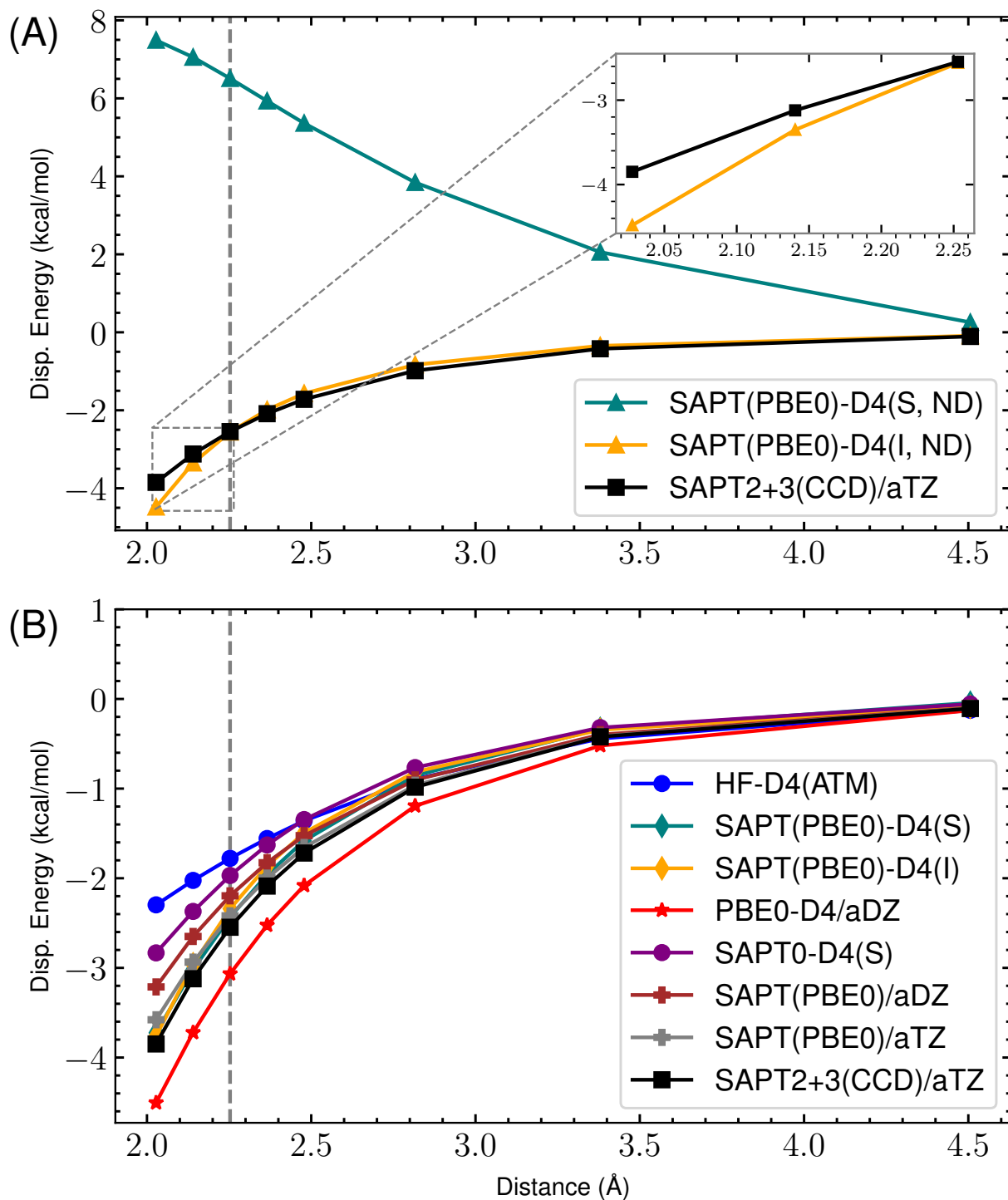


Figure 16: **s66x8 65_Pyridine-Ethyne**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

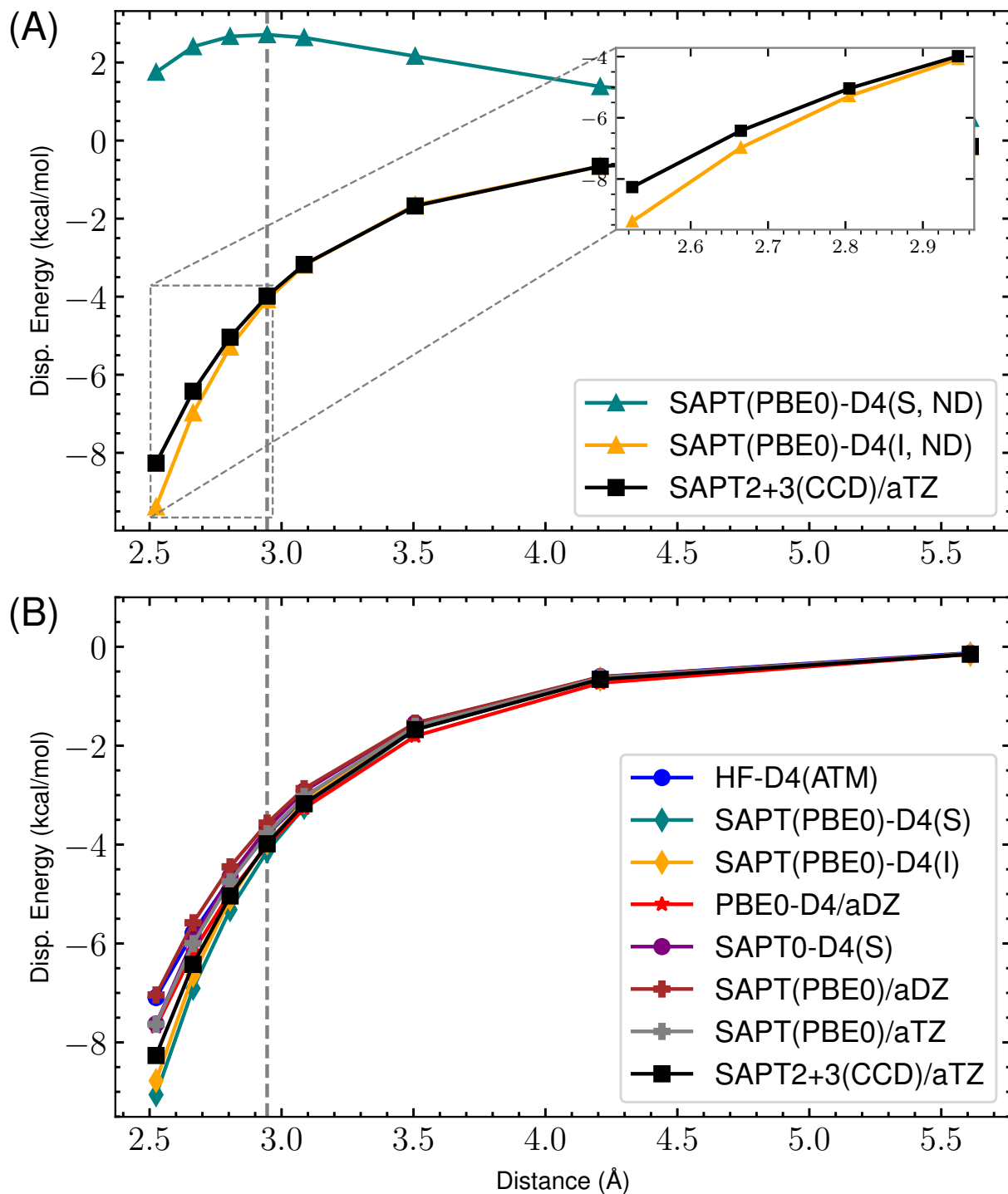


Figure 17: **s66x8 47_Benzene-Benzene**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

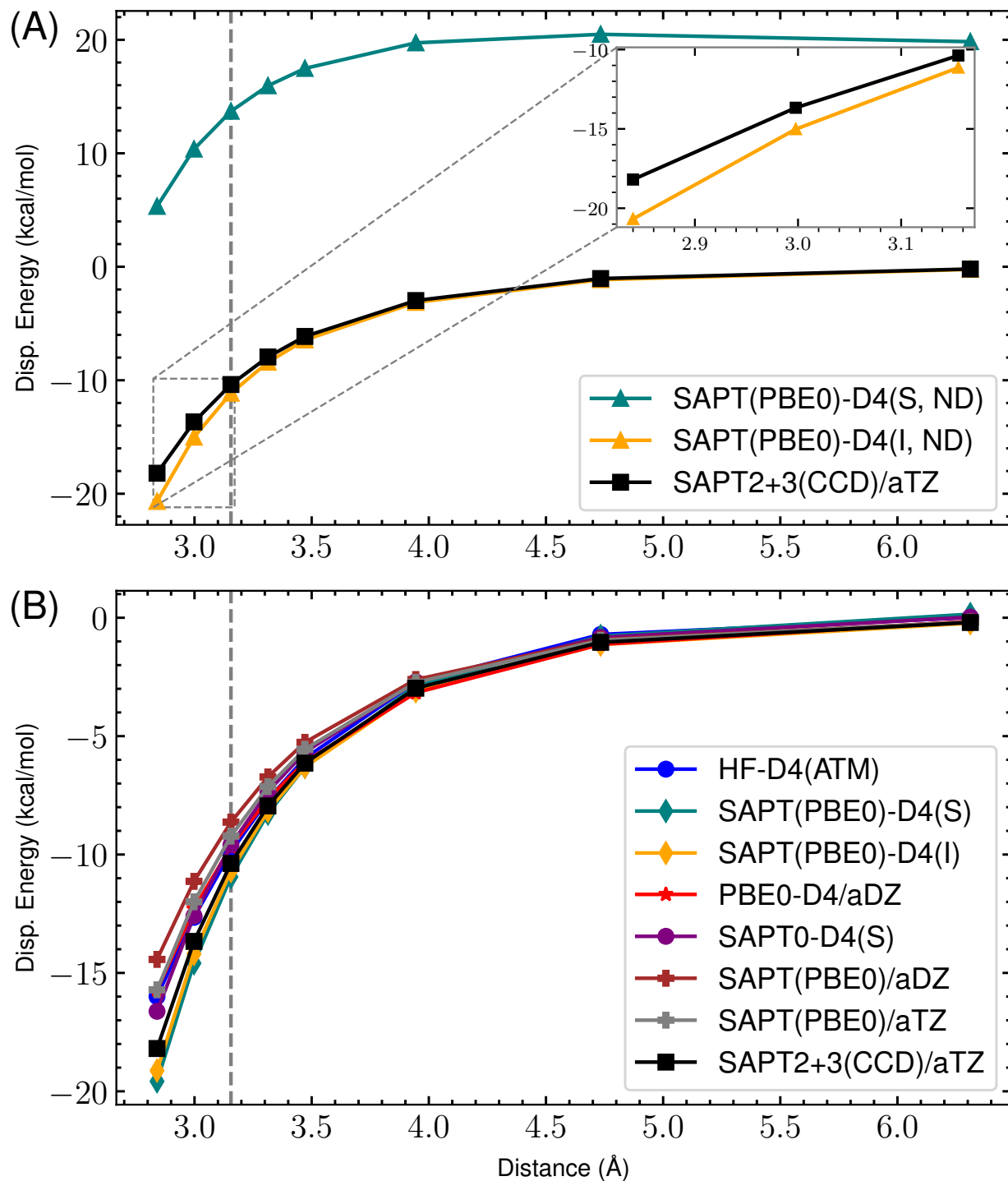


Figure 18: **s66x8 29_Pyridine-Uracil**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

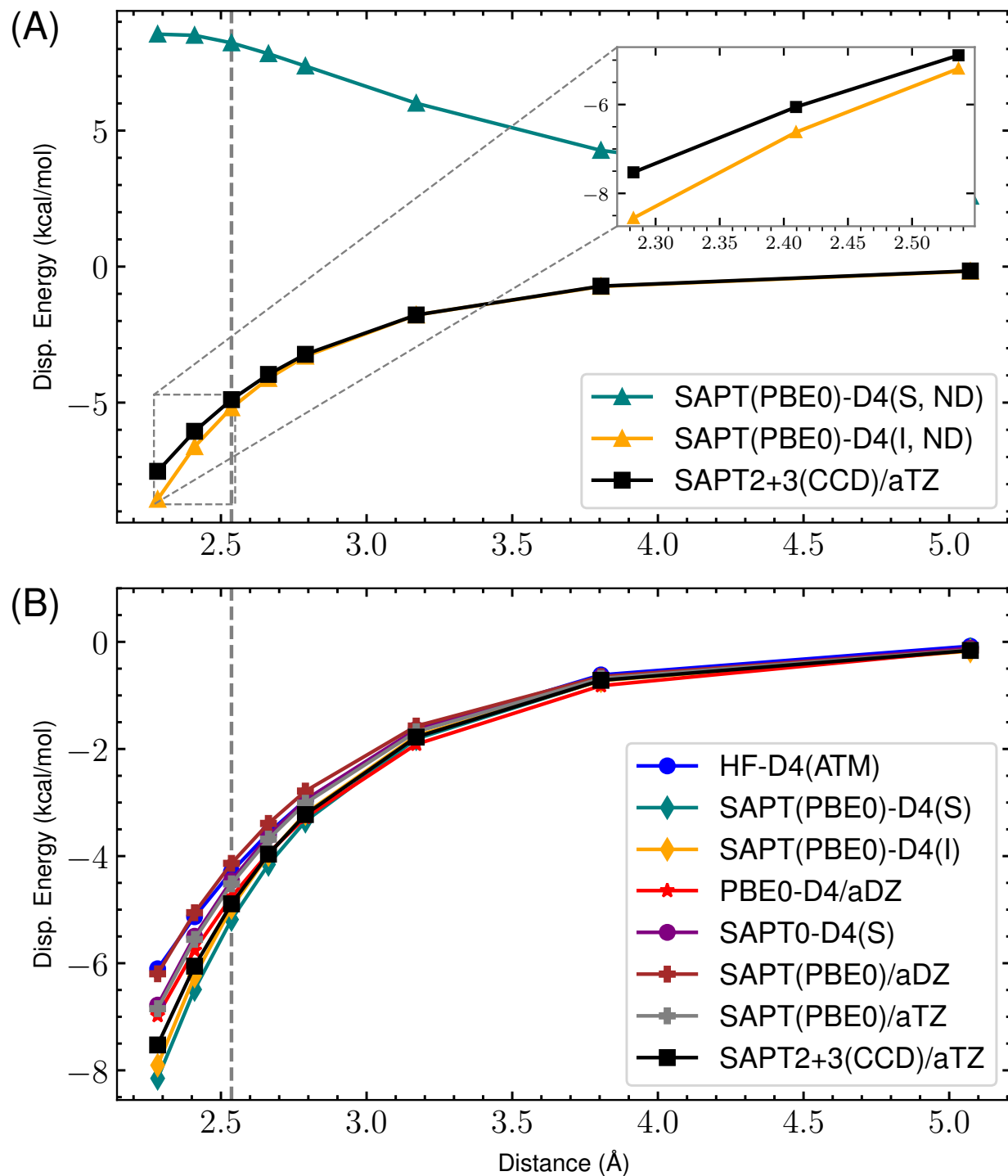


Figure 19: **s66x8 55_Benzene-MeOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

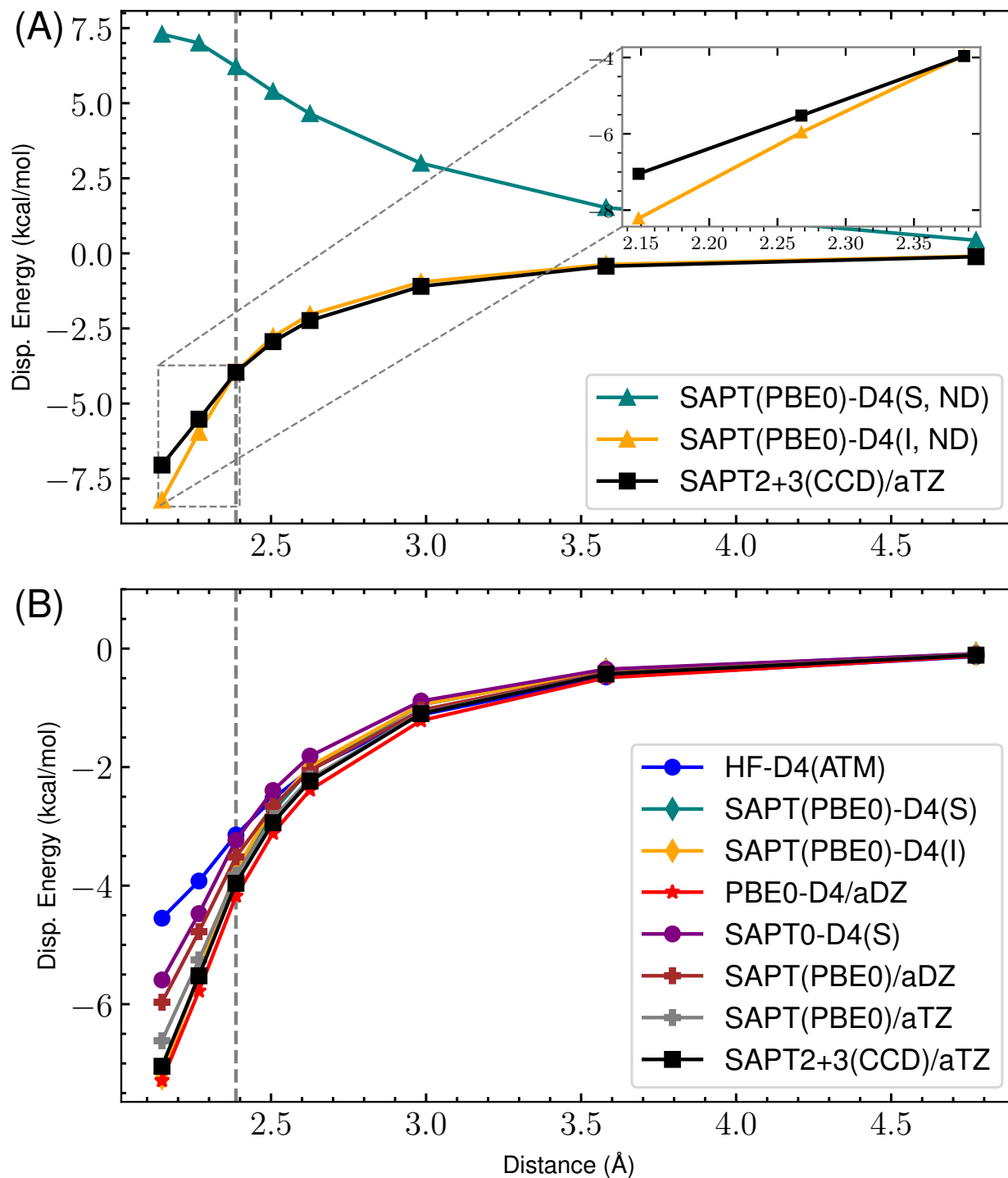


Figure 20: **s66x8 58_Pyridine-Pyridine**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

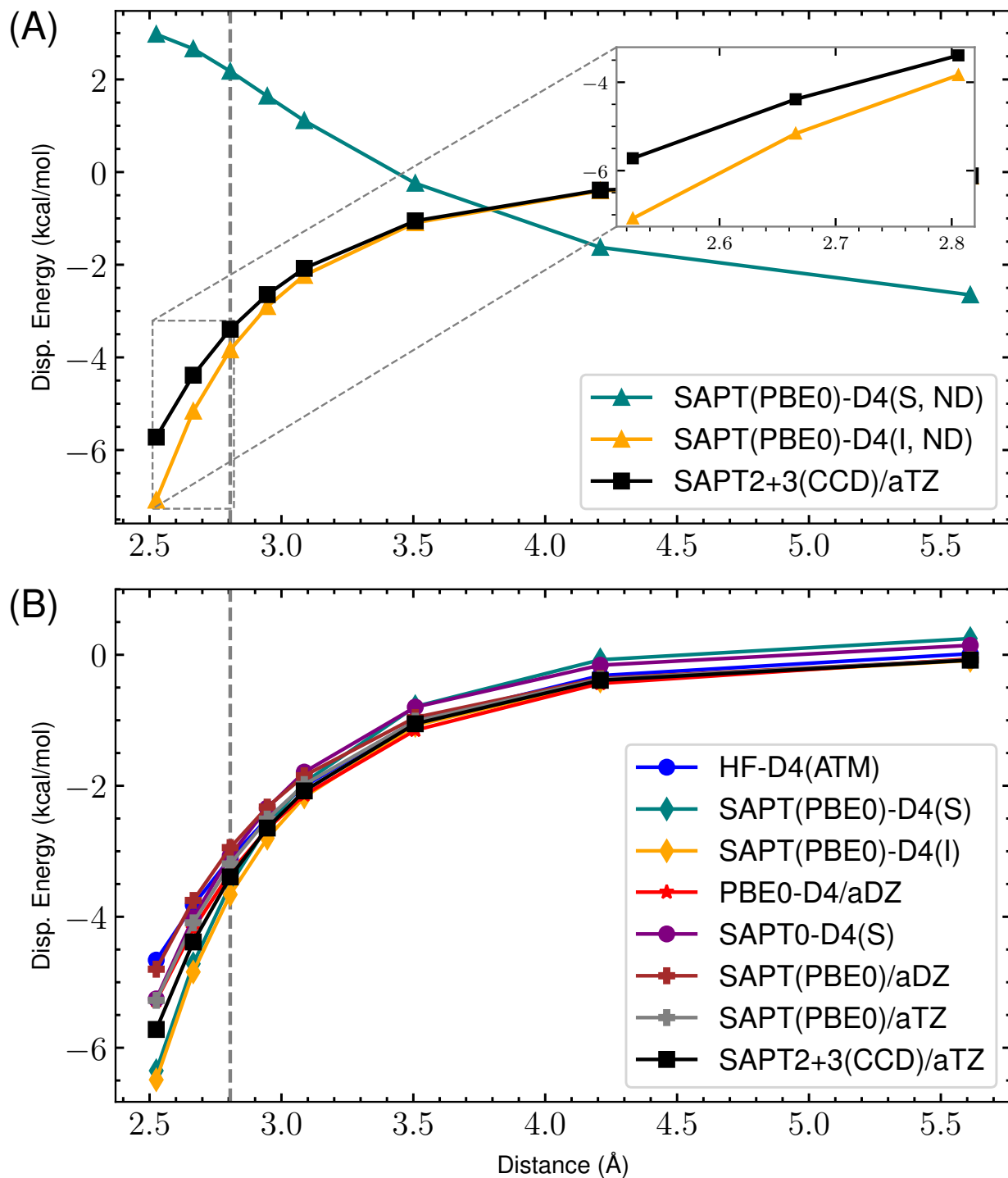


Figure 21: **s66x8 50_Benzene-Ethyne**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

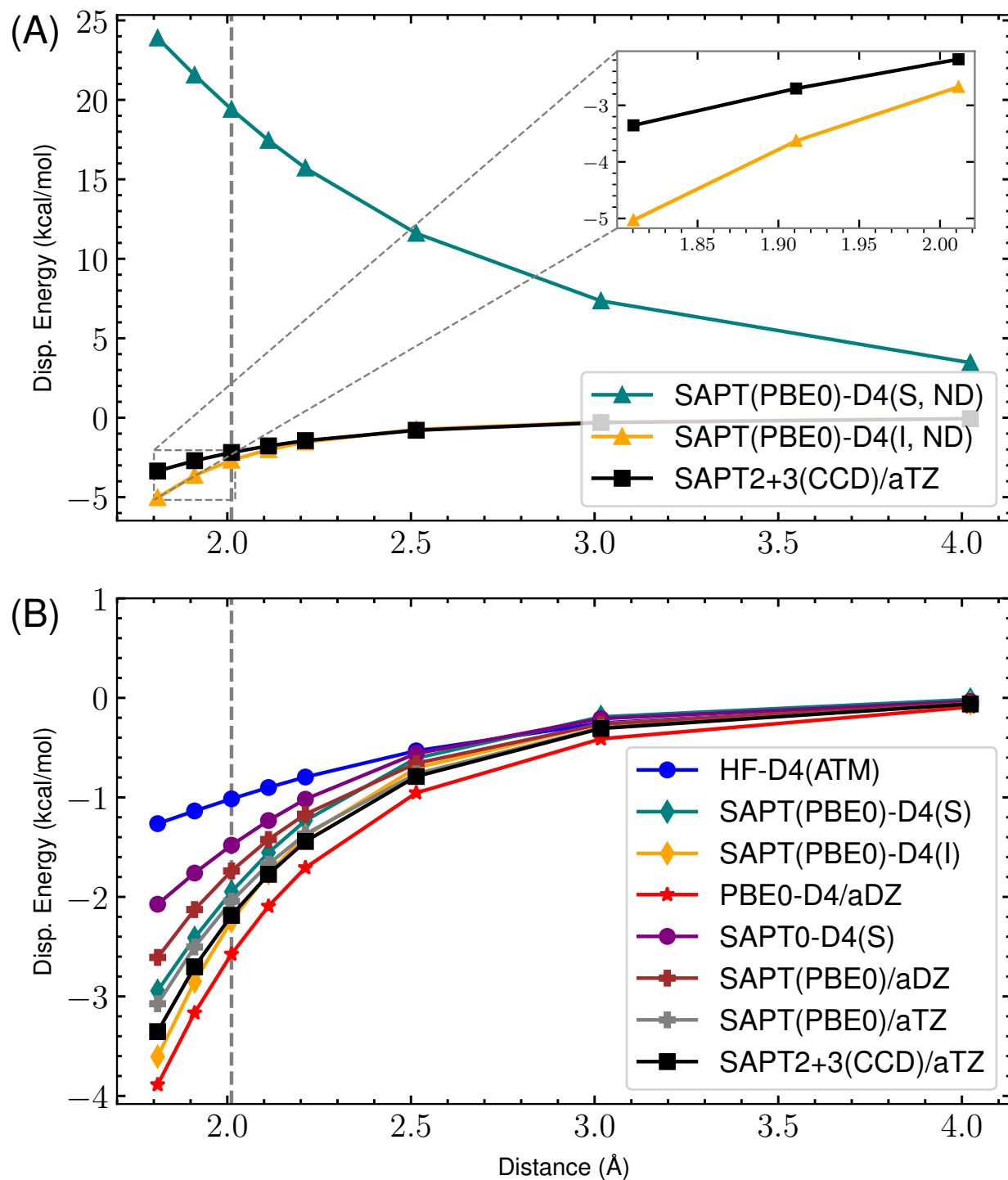


Figure 22: **s66x8 01_Water-Water**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

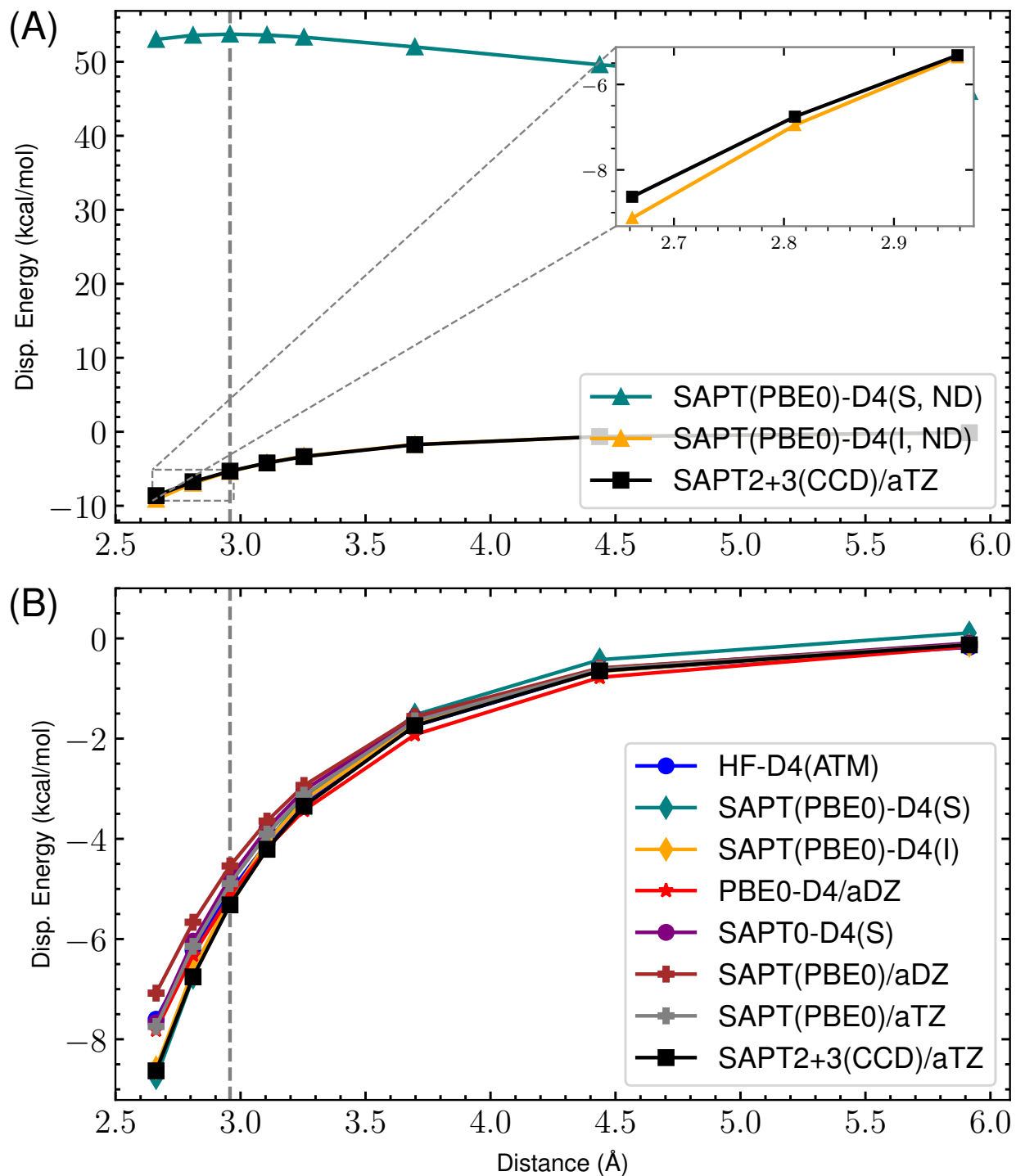


Figure 23: **s66x8 31_Uracil-Ethene**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

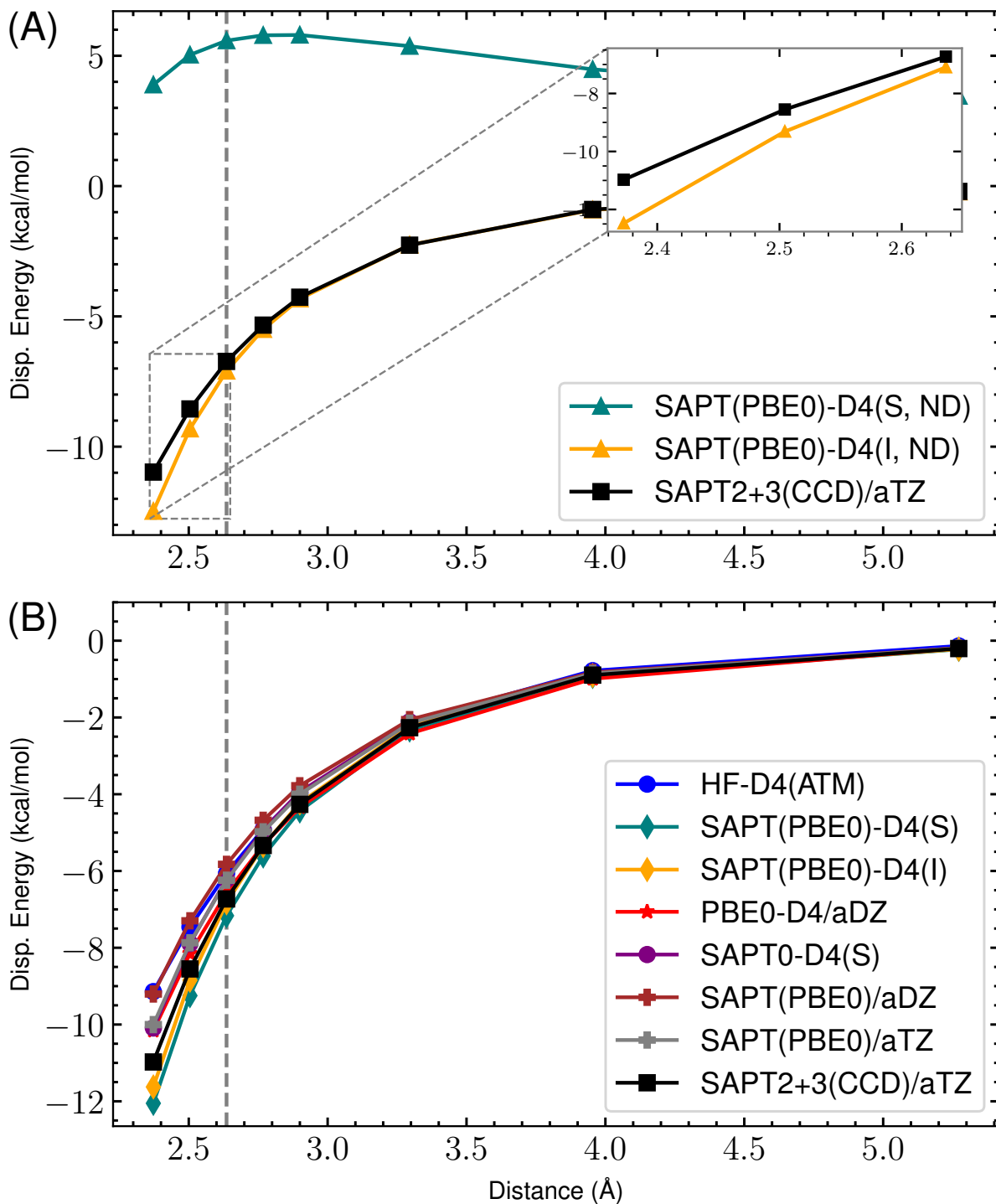


Figure 24: **s66x8 57_Benzene-Peptide**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

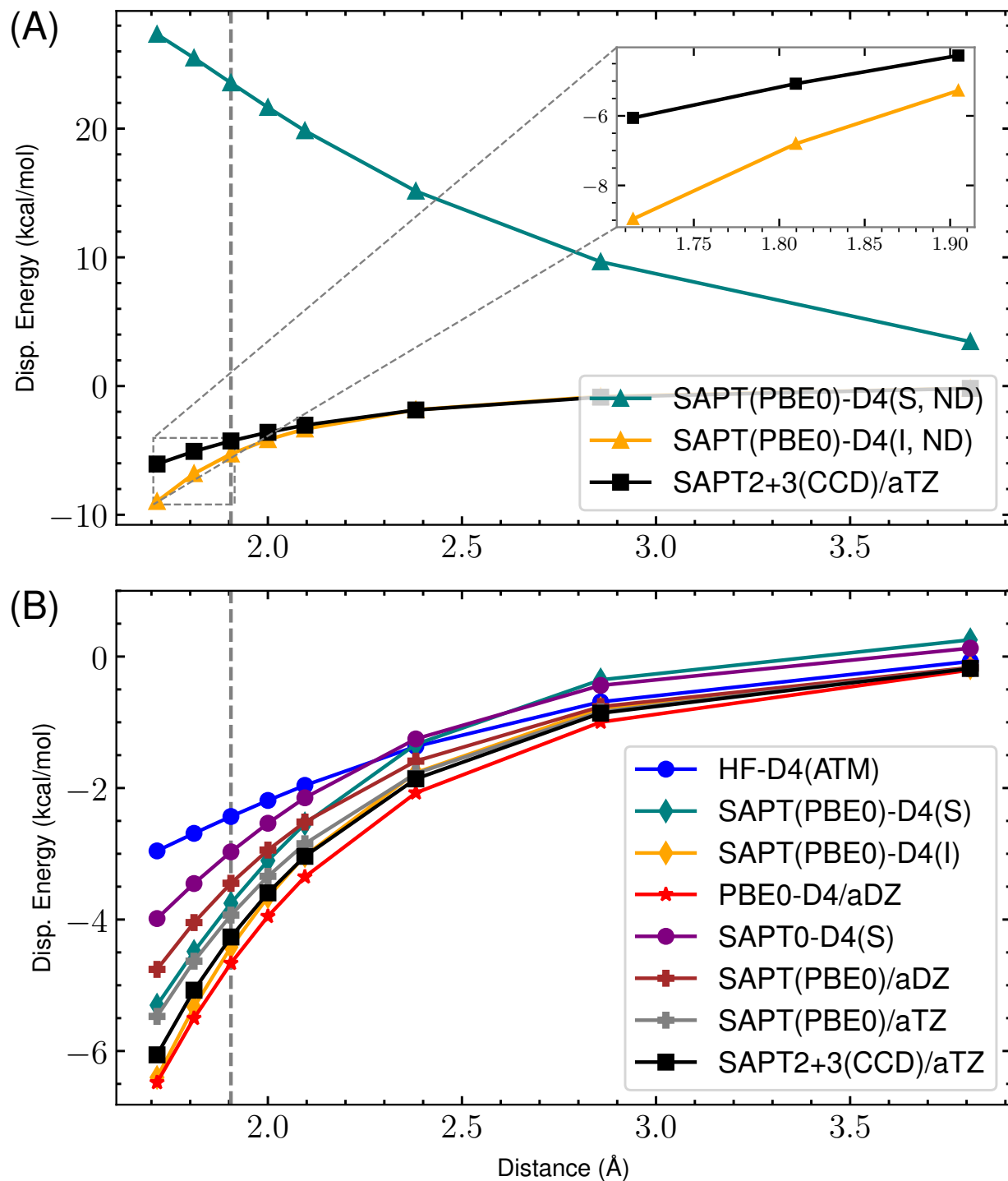


Figure 25: **s66x8 04_Water-Peptide**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

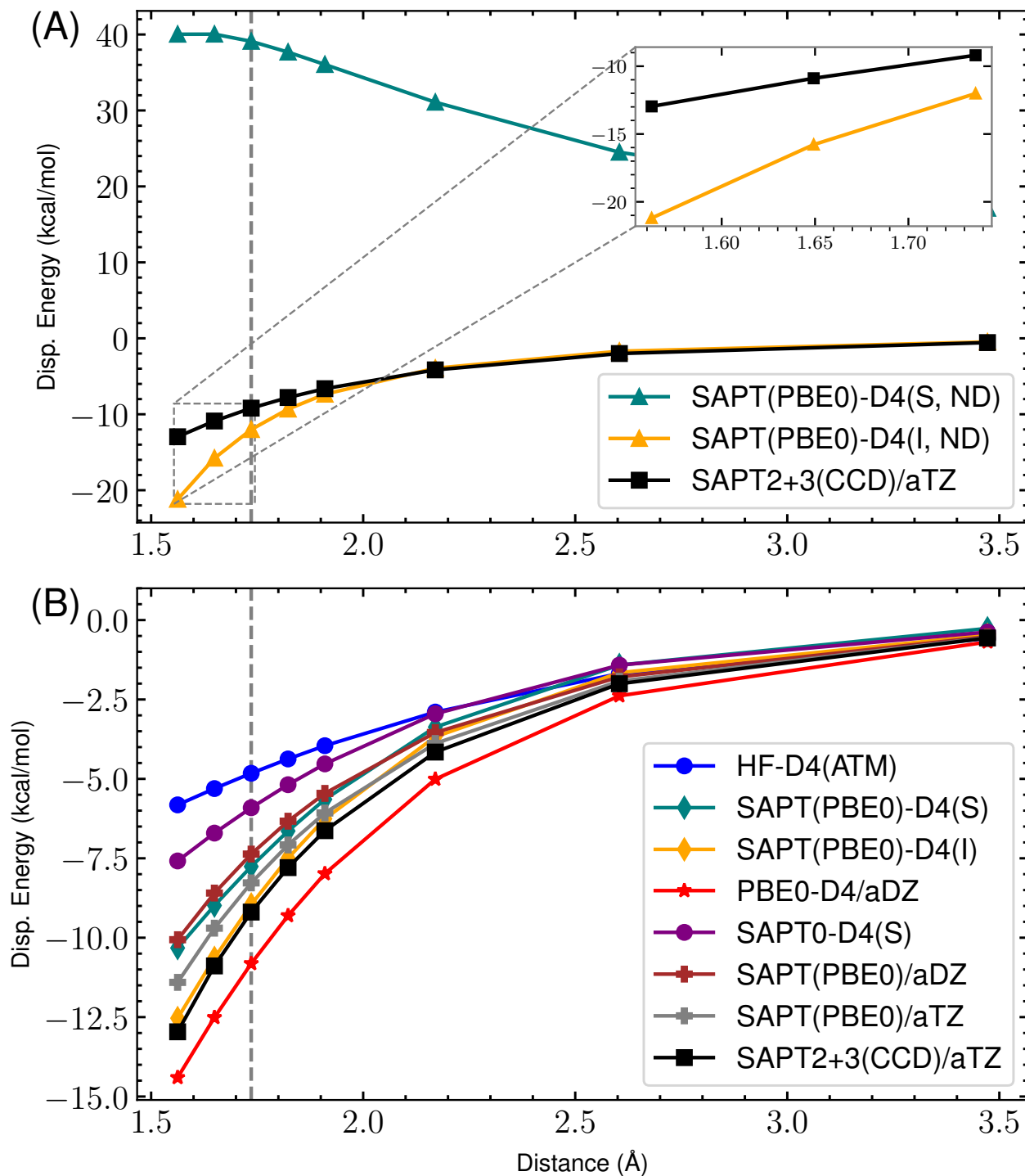


Figure 26: **s66x8 23_AcNH2-Uracil**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

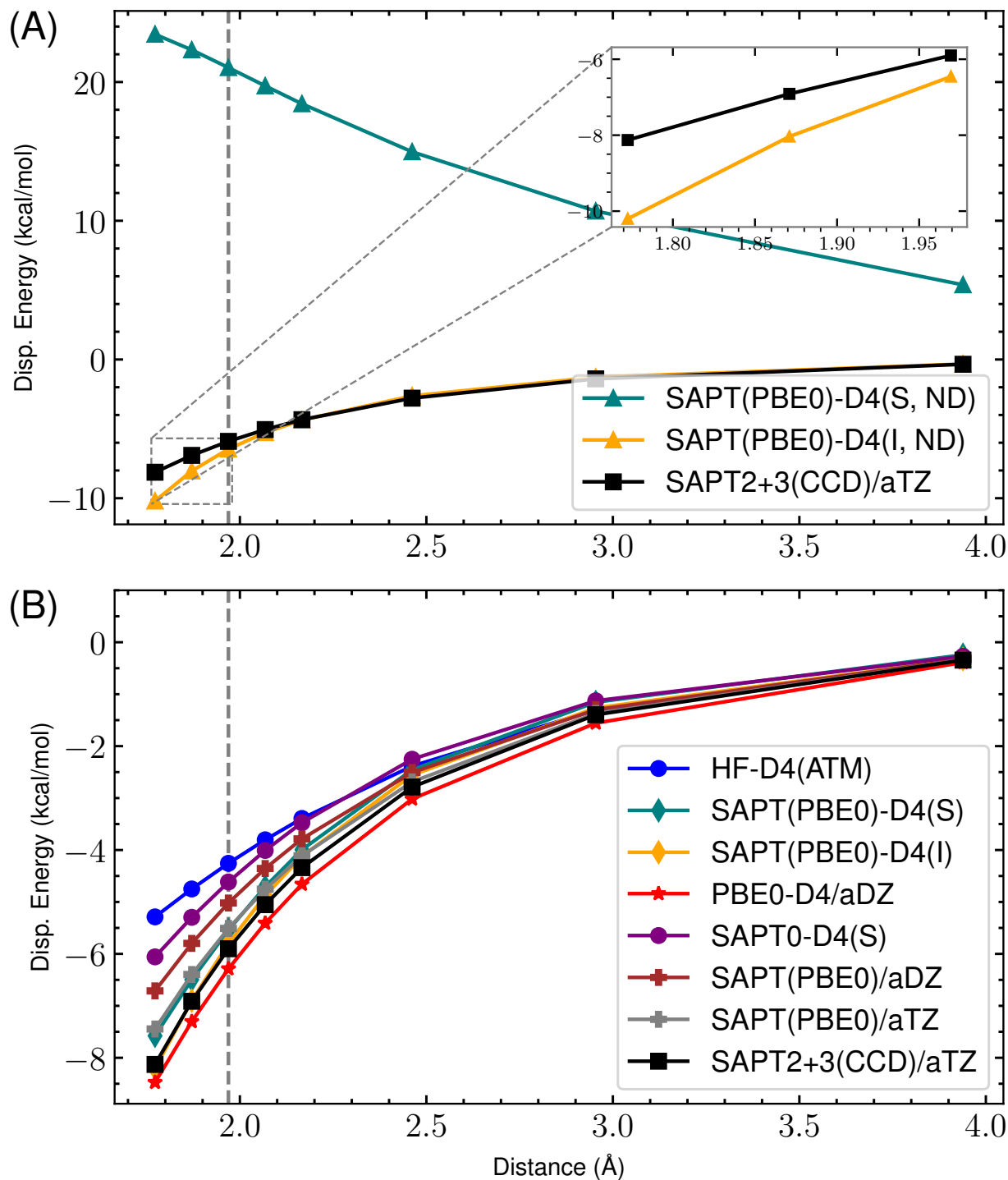


Figure 27: **s66x8 15_Peptide-Peptide**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

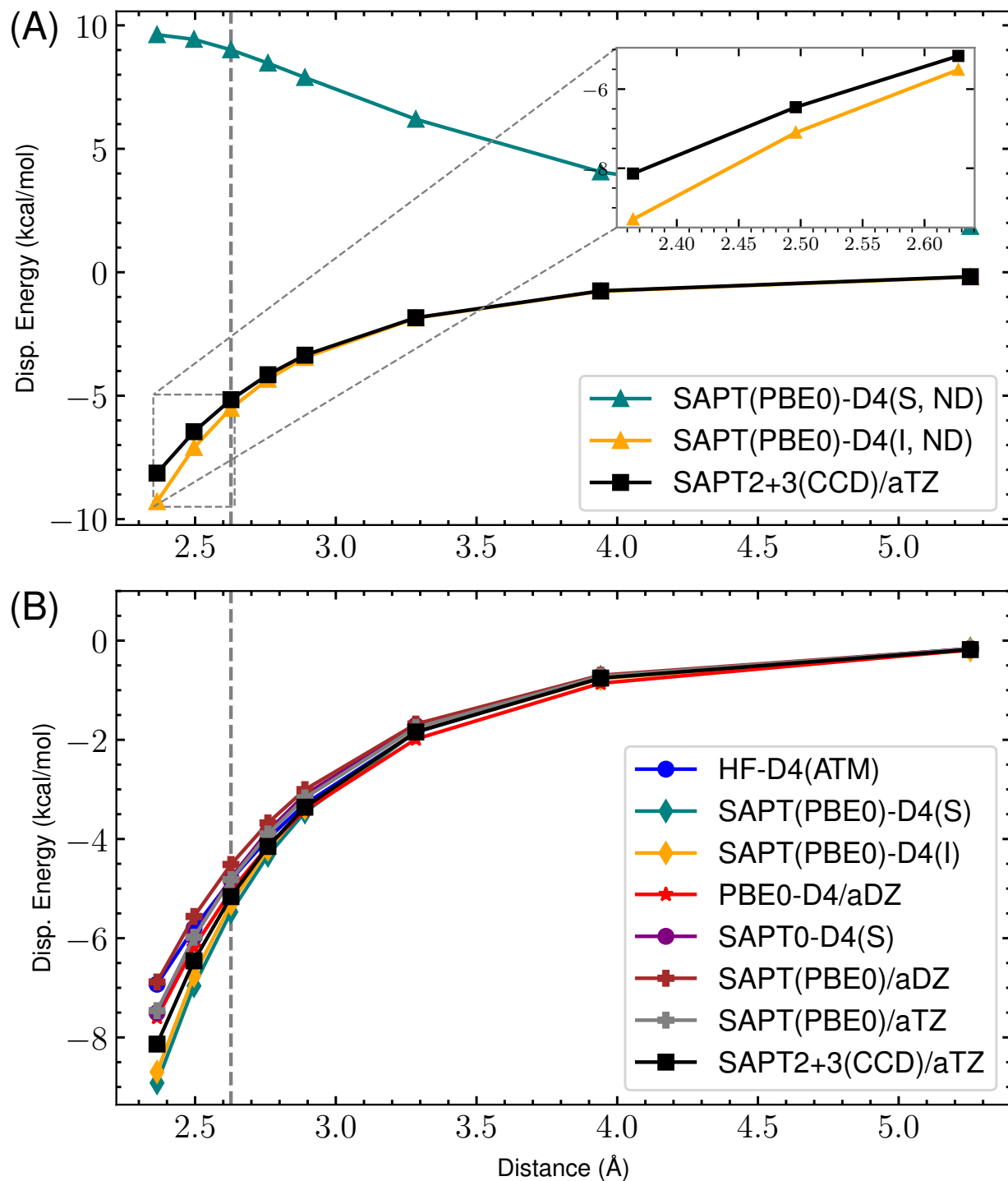


Figure 28: **s66x8 48_Pyridine-Pyridine**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

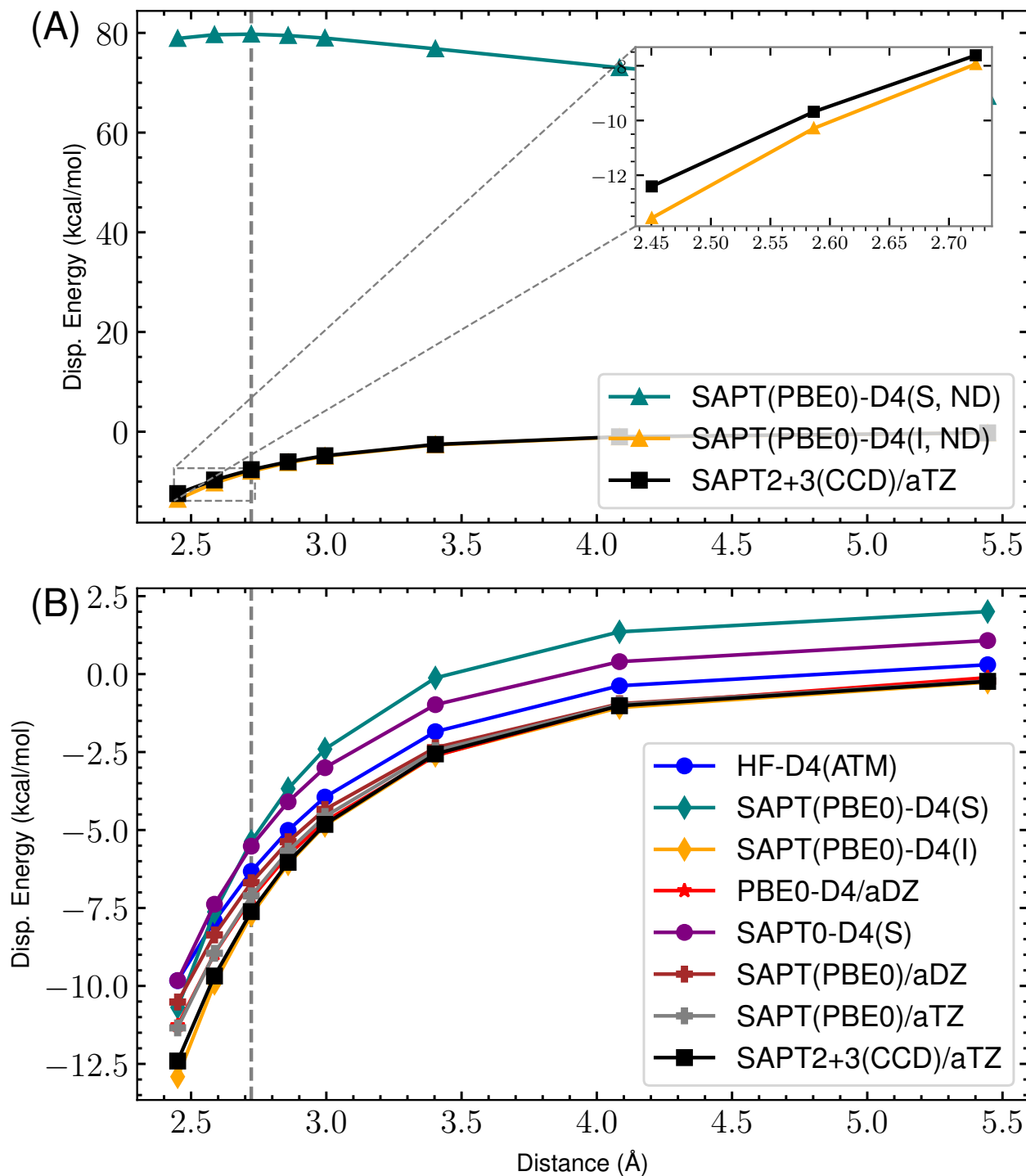


Figure 29: **s66x8 42_Uracil-Cyclopentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

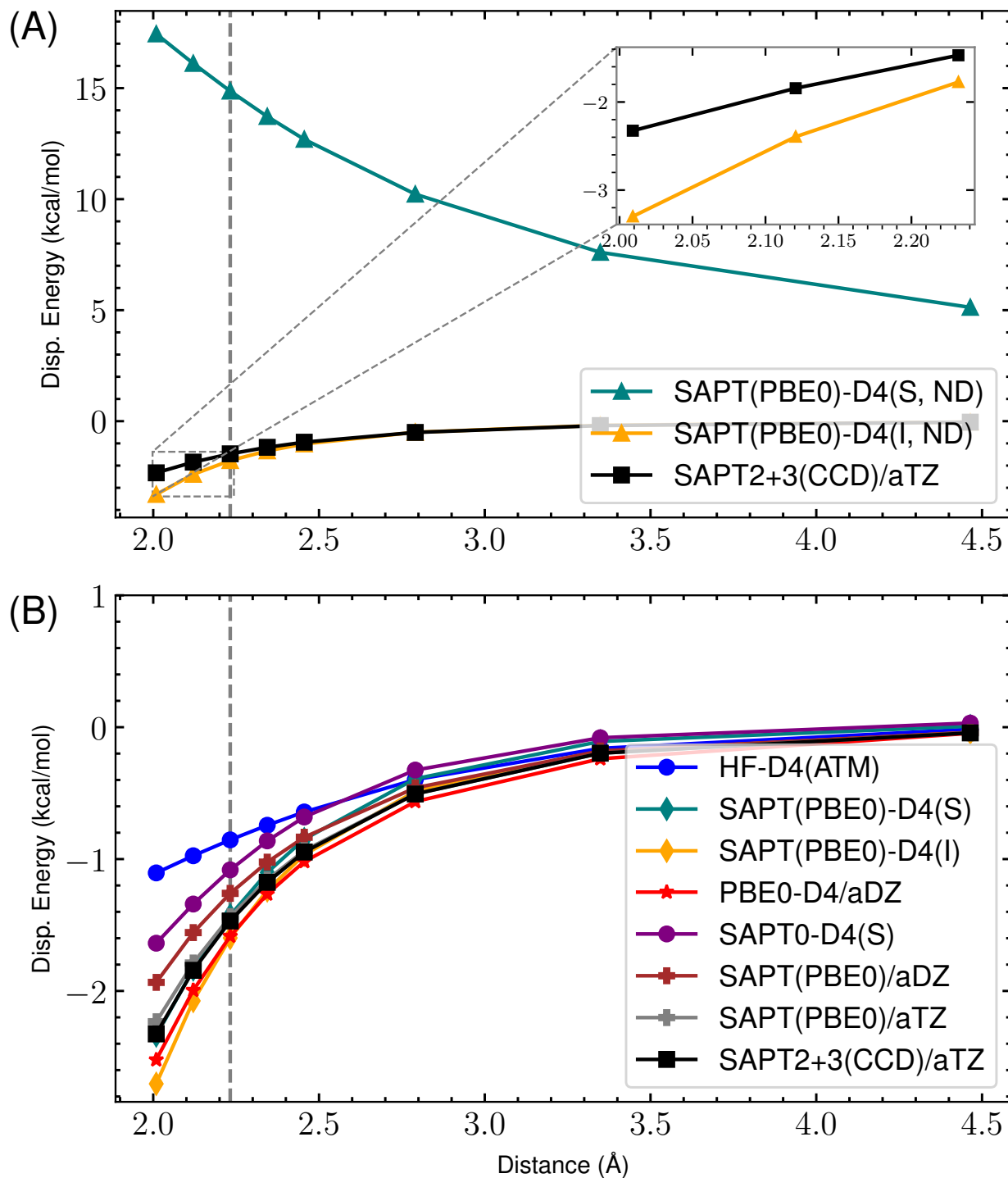


Figure 30: **s66x8 59_Ethyne-Water**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

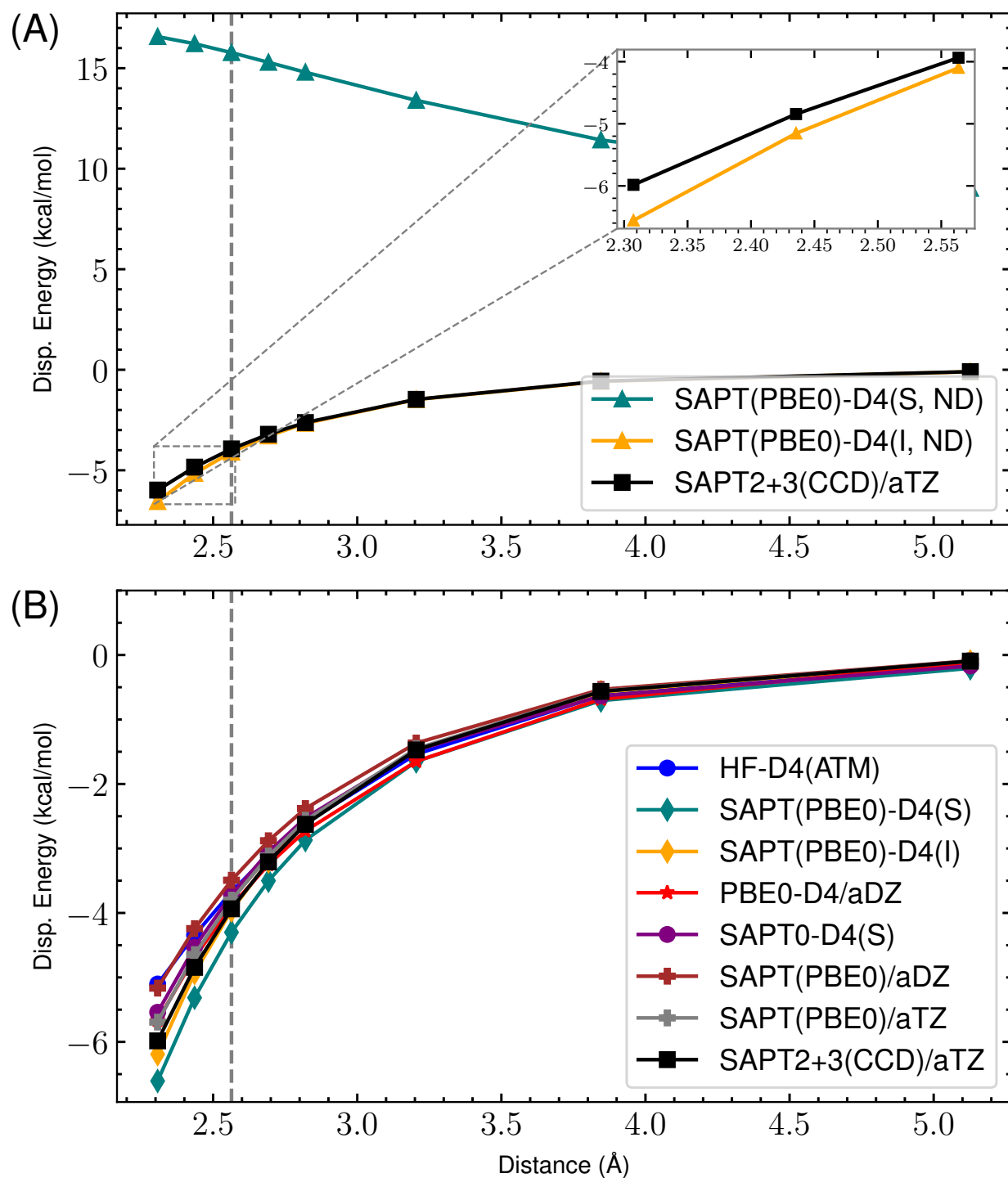


Figure 31: **s66x8 64_Peptide-Ethene**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

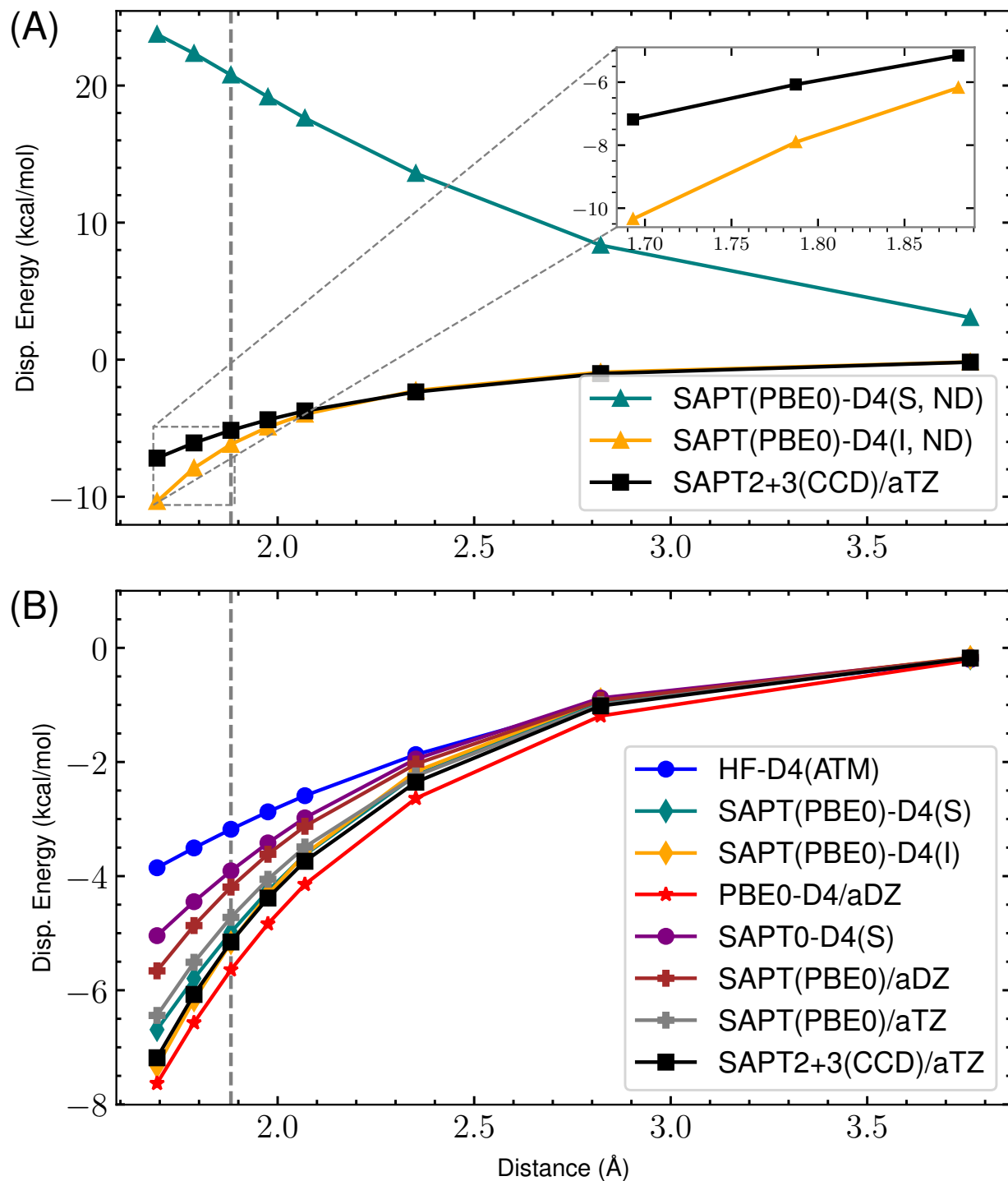


Figure 32: **s66x8 07_MeOH-Peptide**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

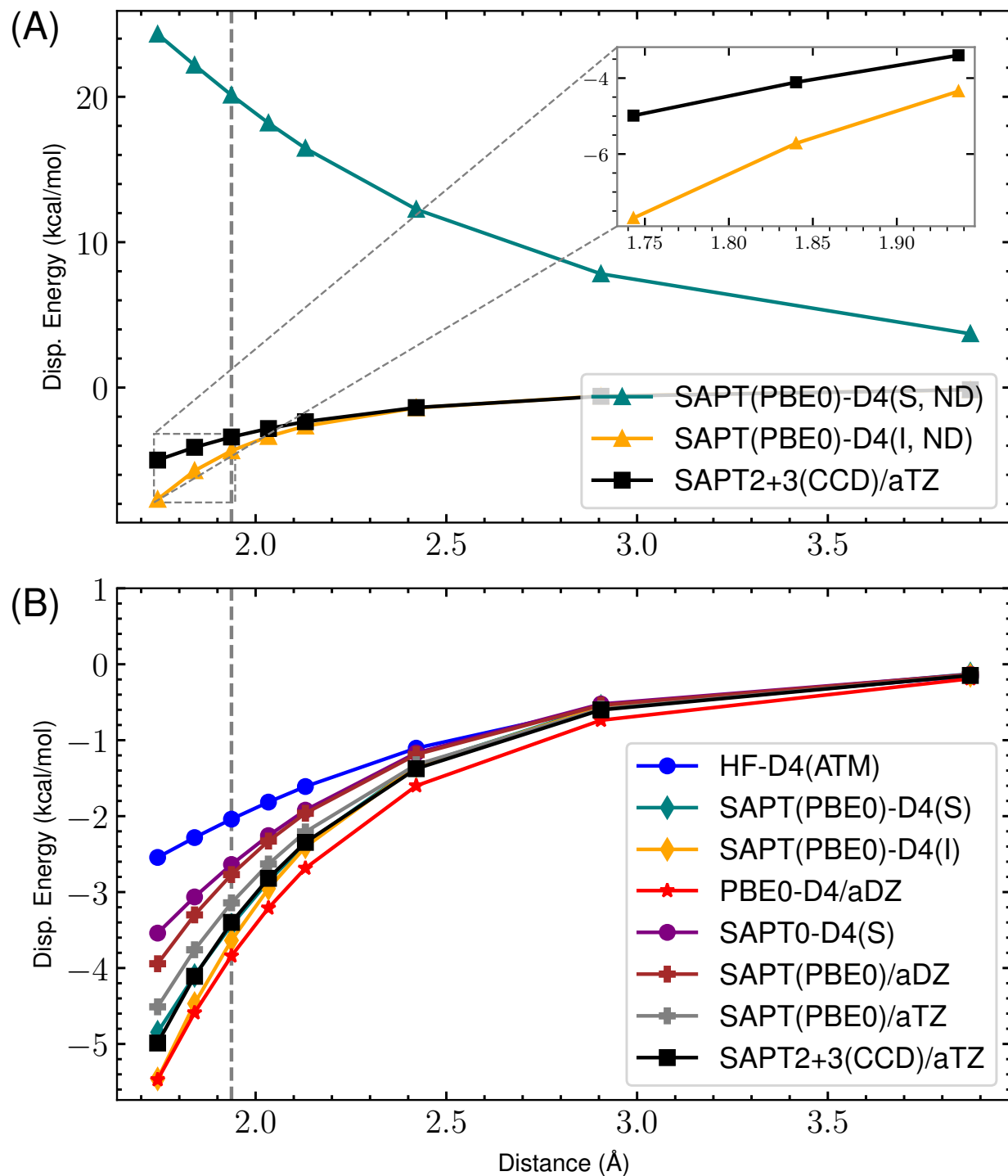


Figure 33: **s66x8 05_MeOH-MeOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

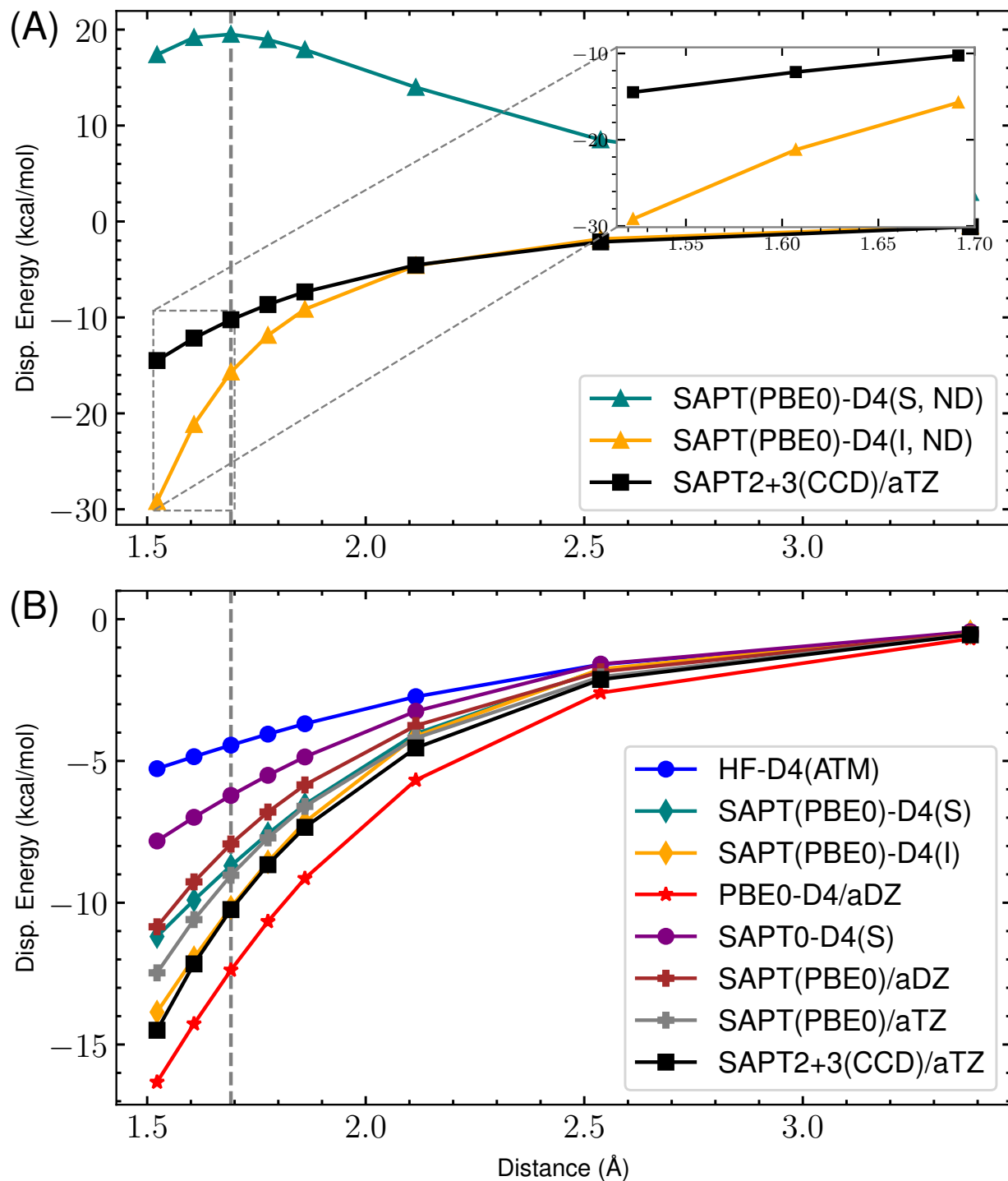


Figure 34: $s66x8$ 20_AcOH-AcOH: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

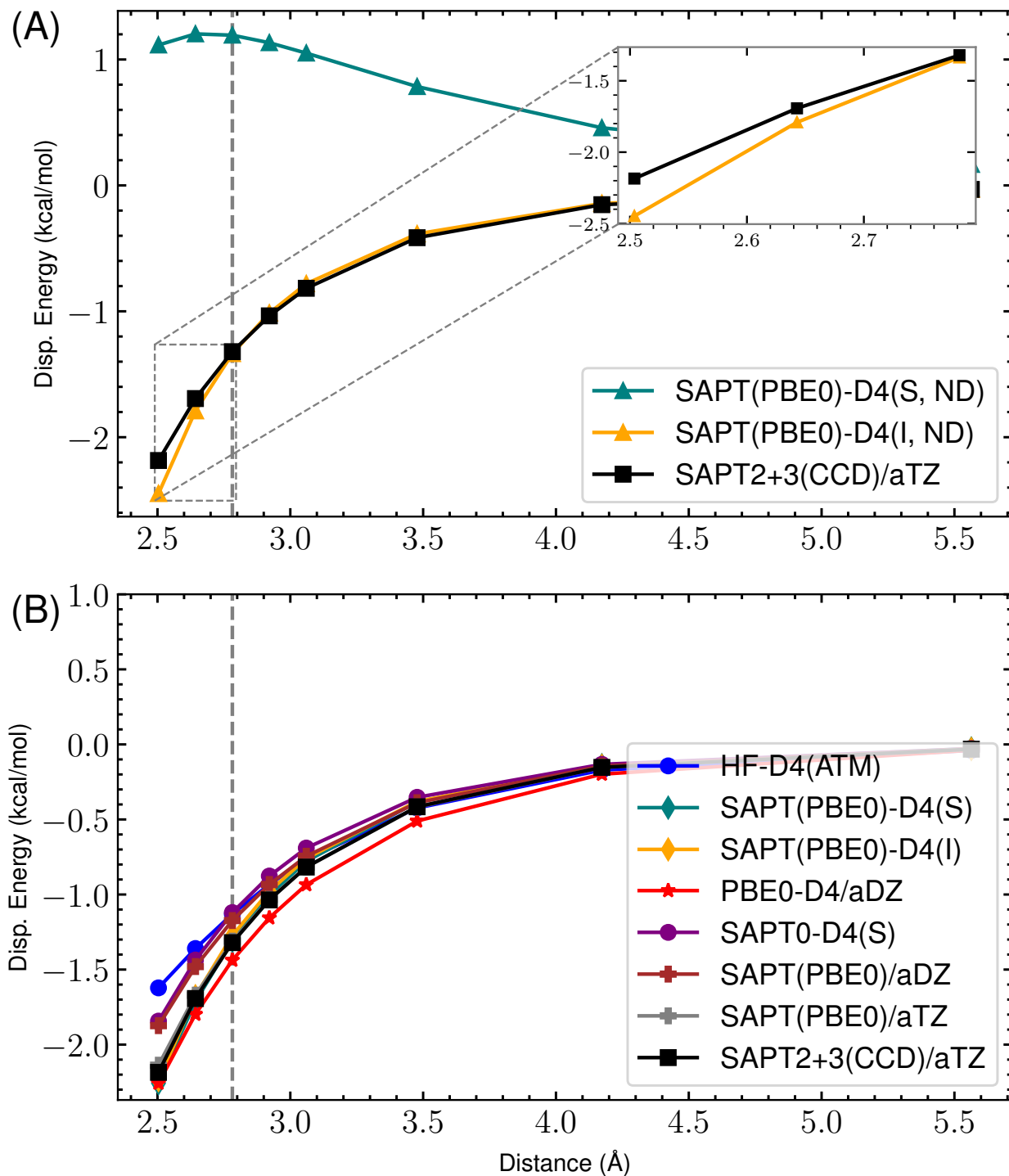


Figure 35: **s66x8 51_Ethyne-Ethyne**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

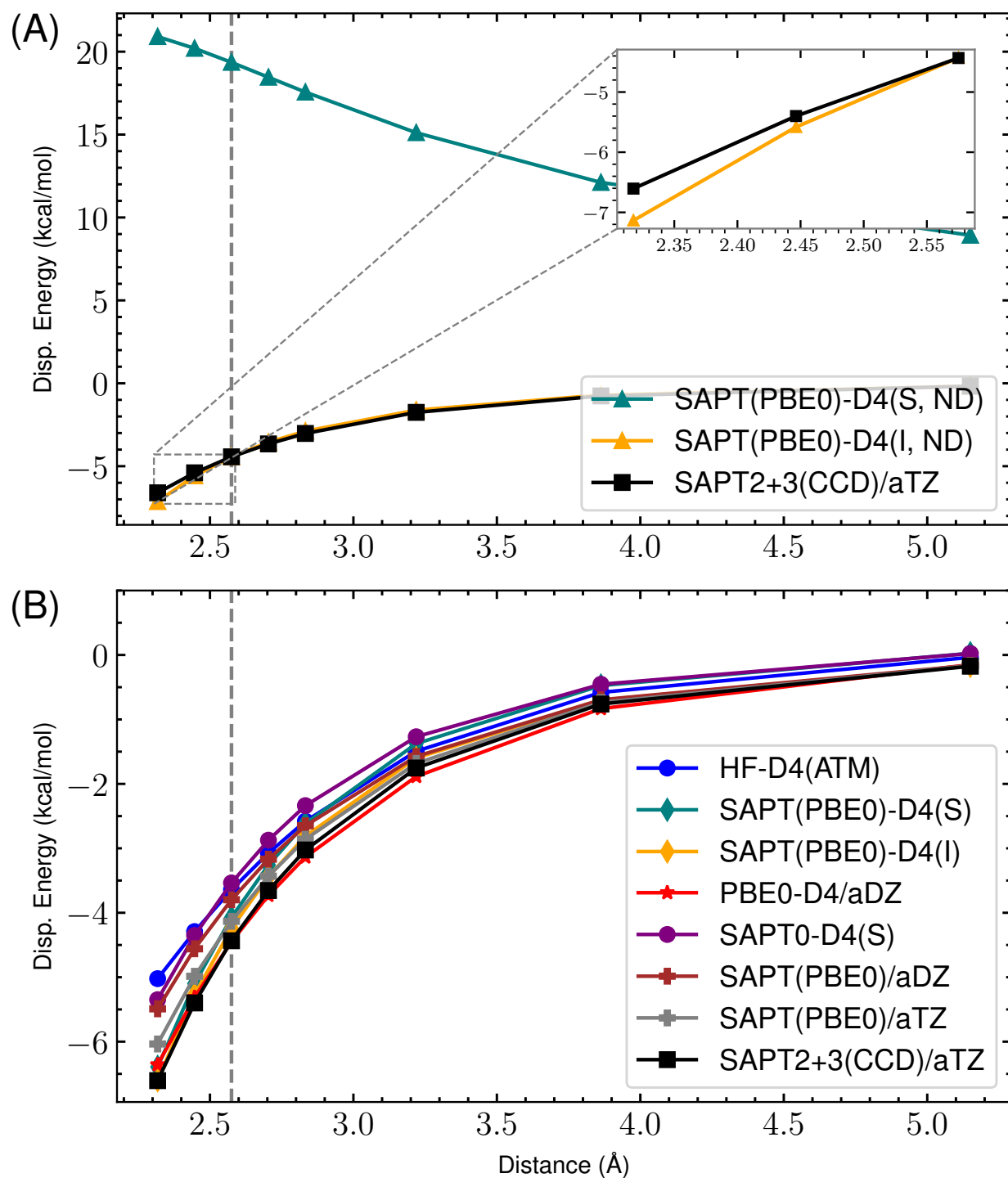


Figure 36: **s66x8 53_Benzene-AcNH2**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

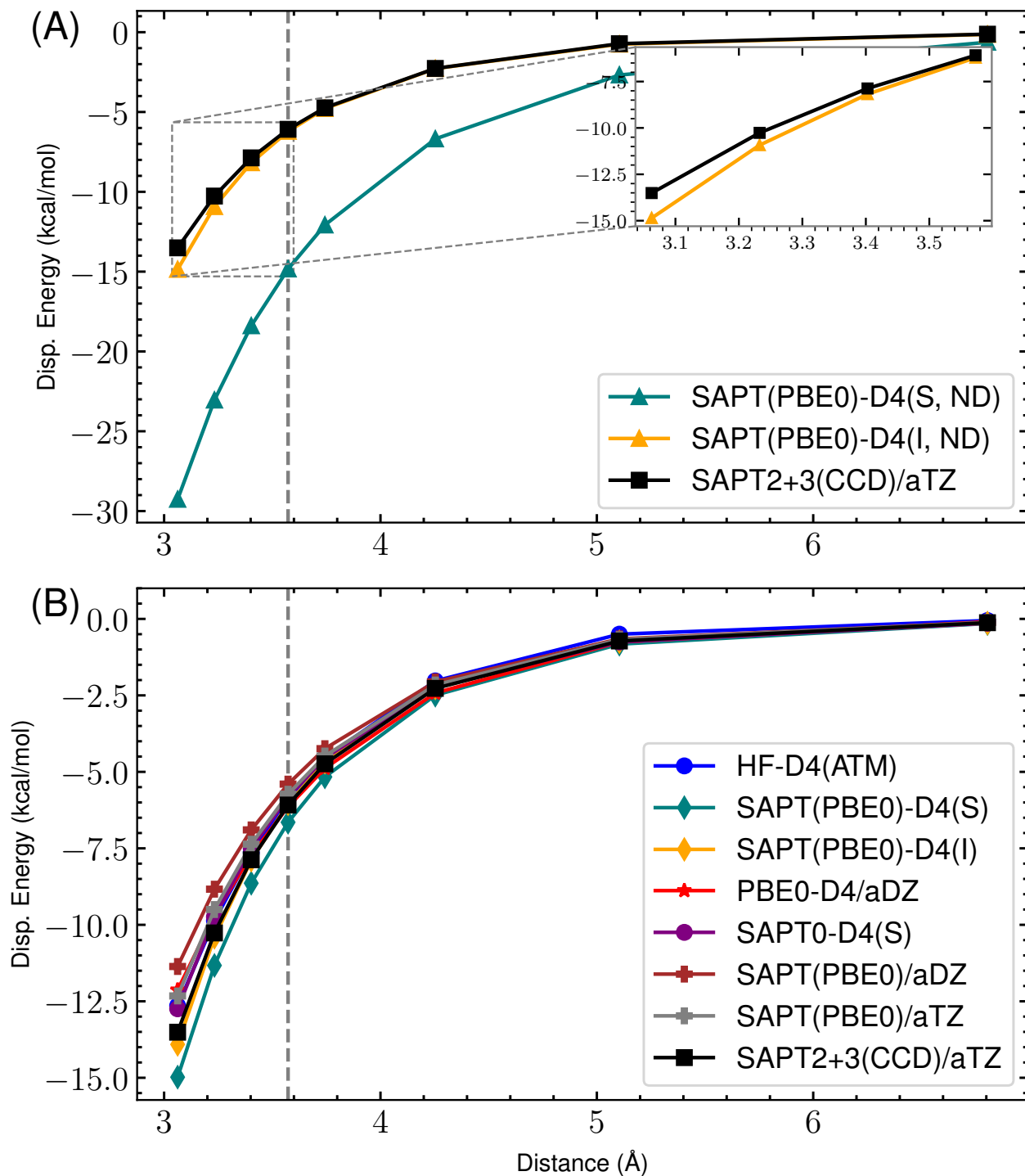


Figure 37: **s66x8 24_Benzene-Benzene**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

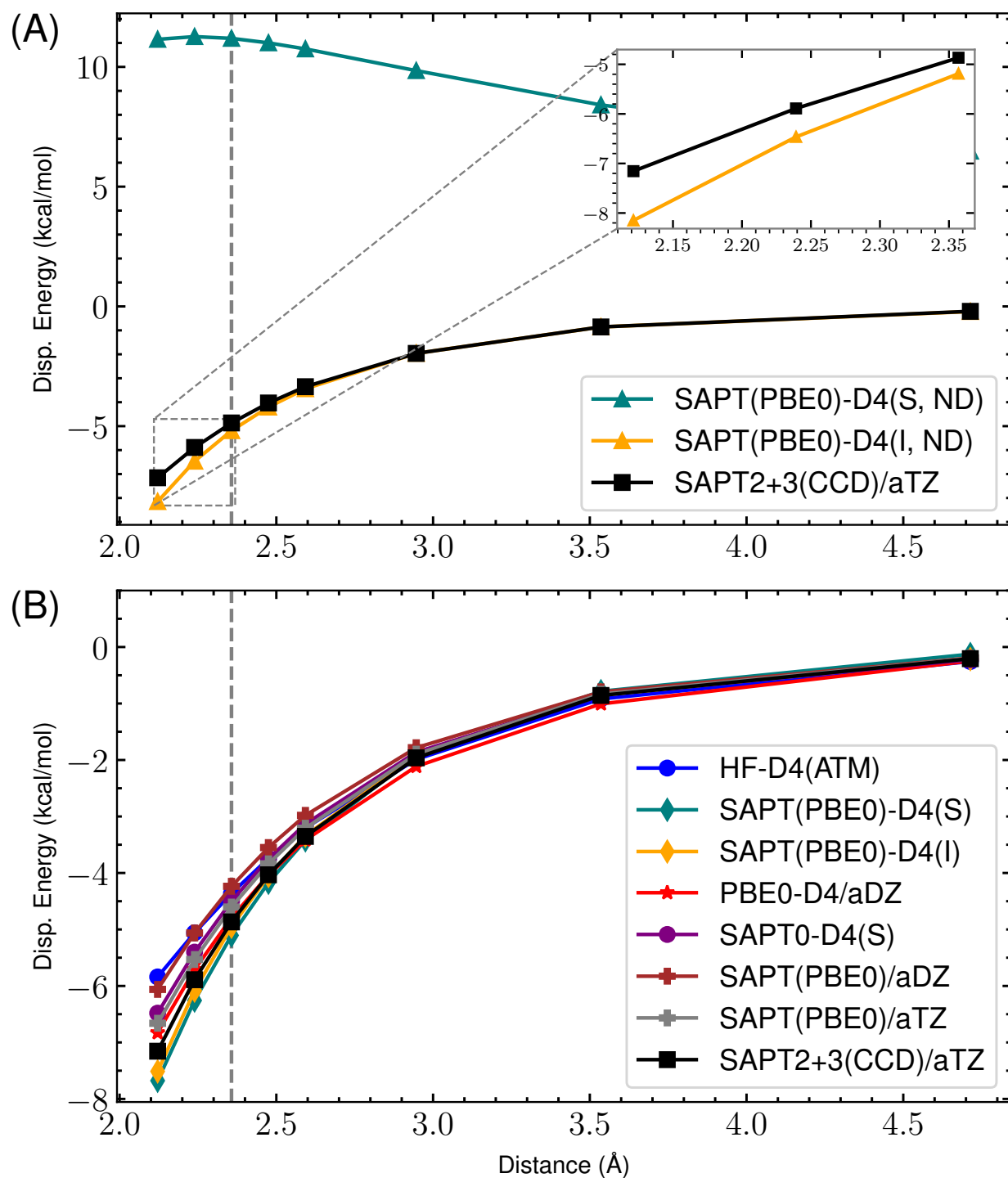


Figure 38: **s66x8 66_MeNH2-Pyridine**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

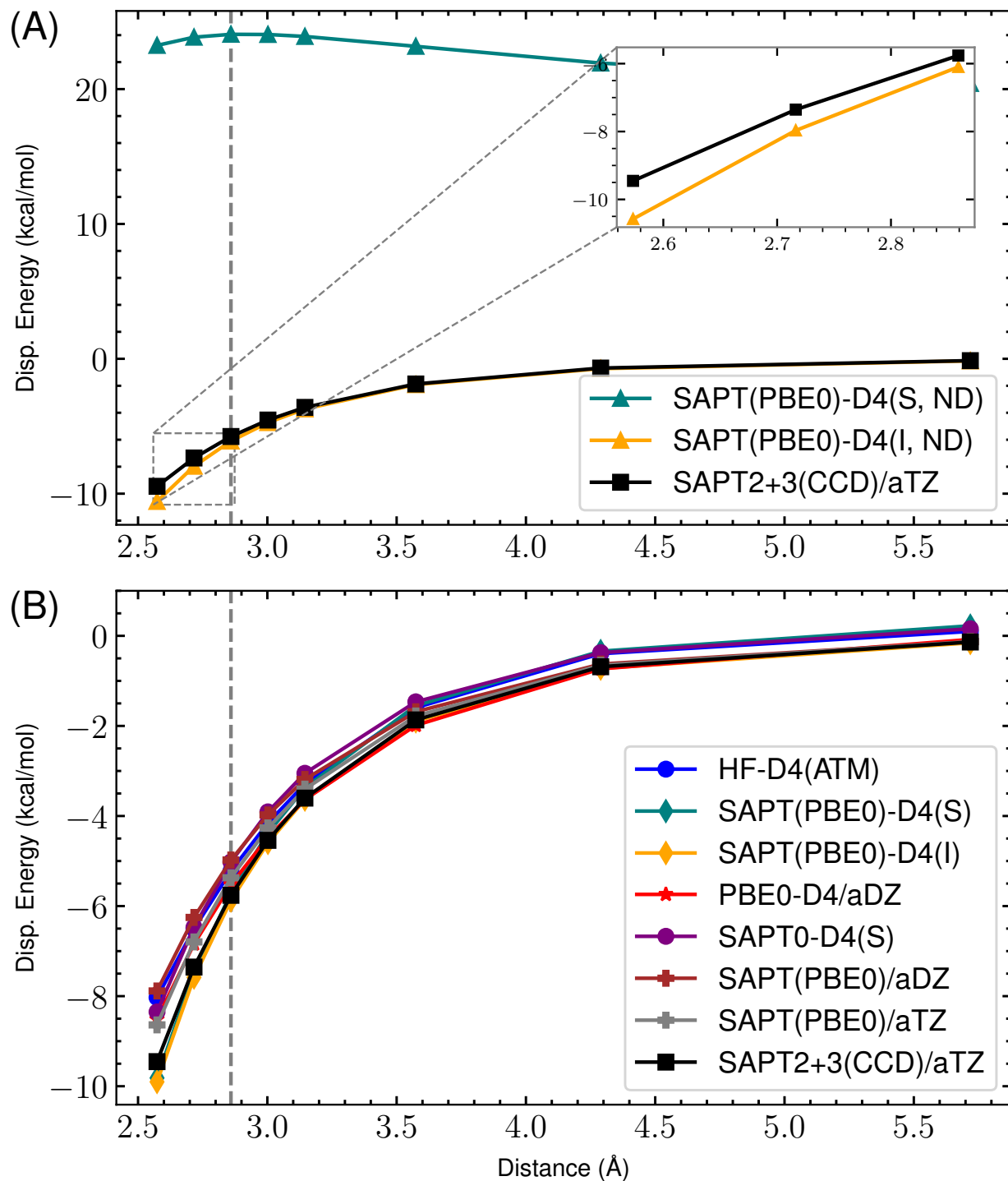


Figure 39: **s66x8 63_Benzene-AcOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

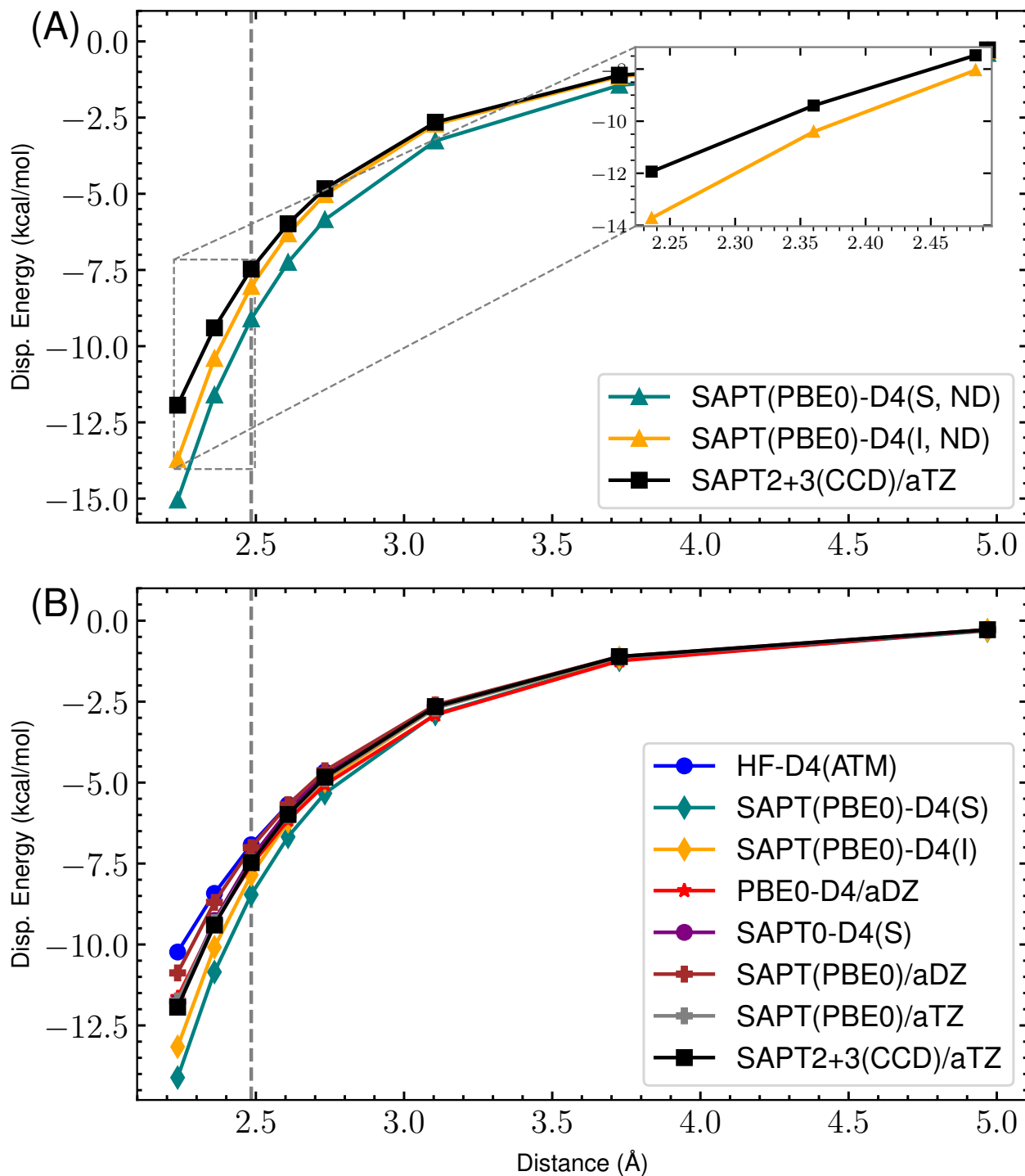


Figure 40: **s66x8 34. Pentane-Pentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

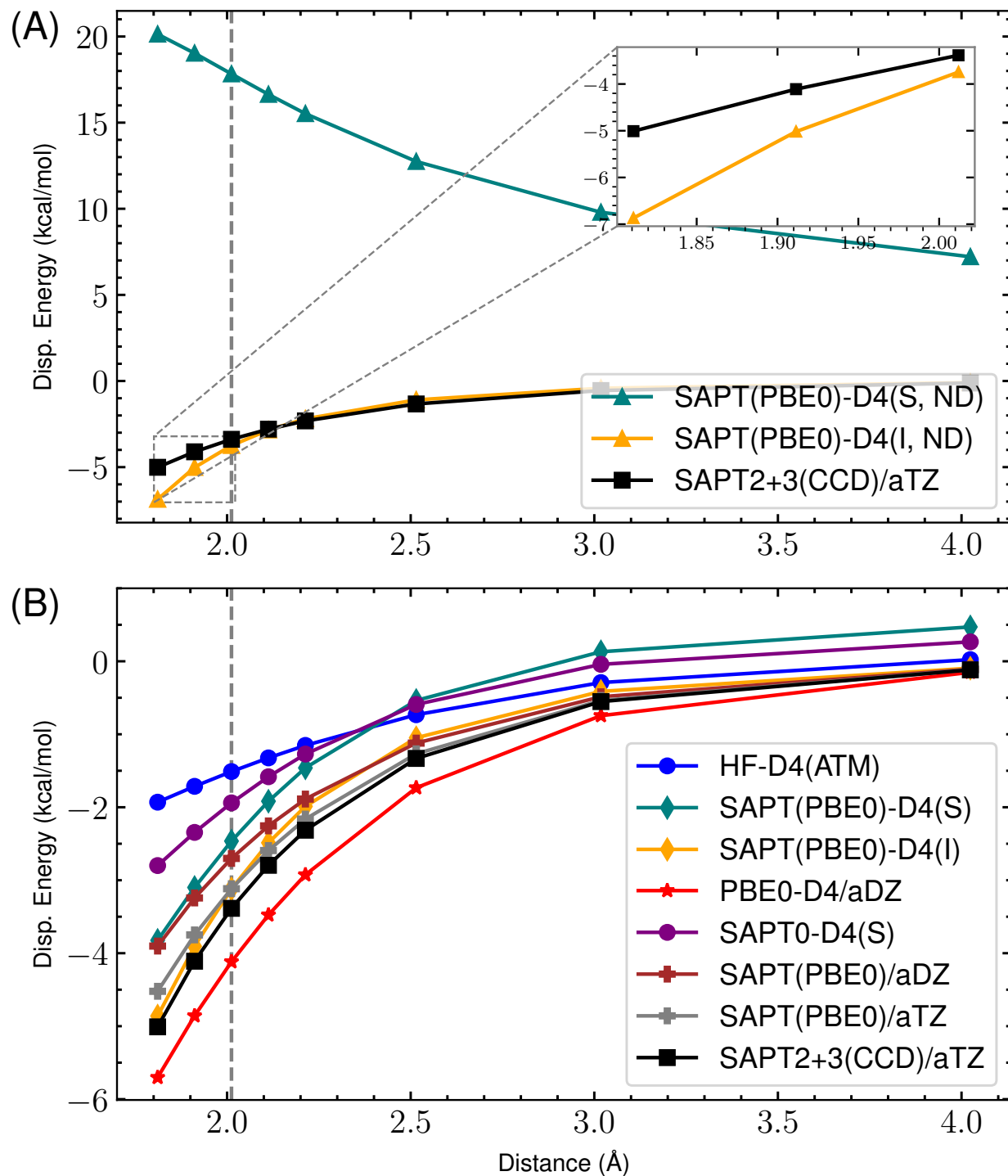


Figure 41: **s66x8 03_Water-MeNH₂**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

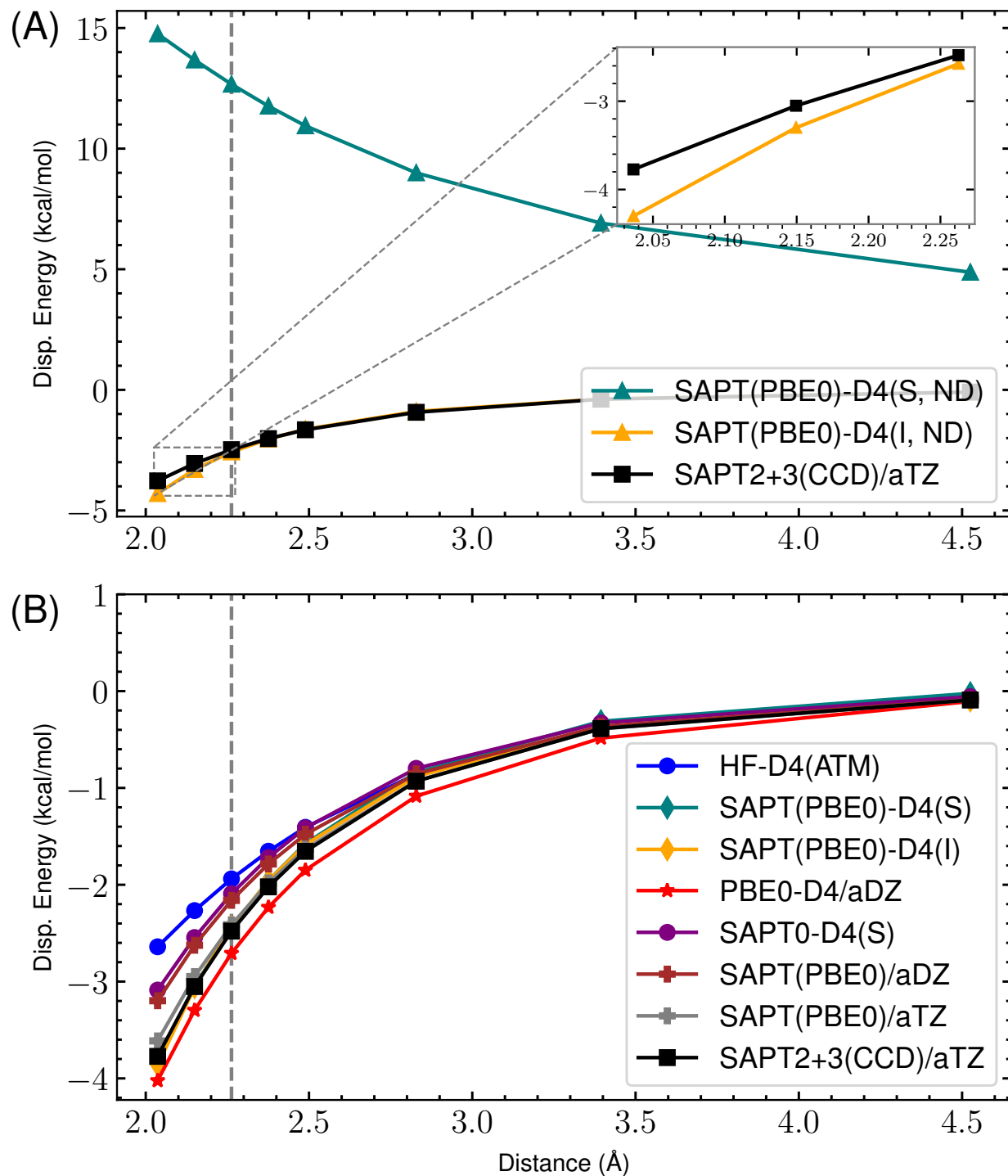


Figure 42: **s66x8 09_MeNH₂-MeOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

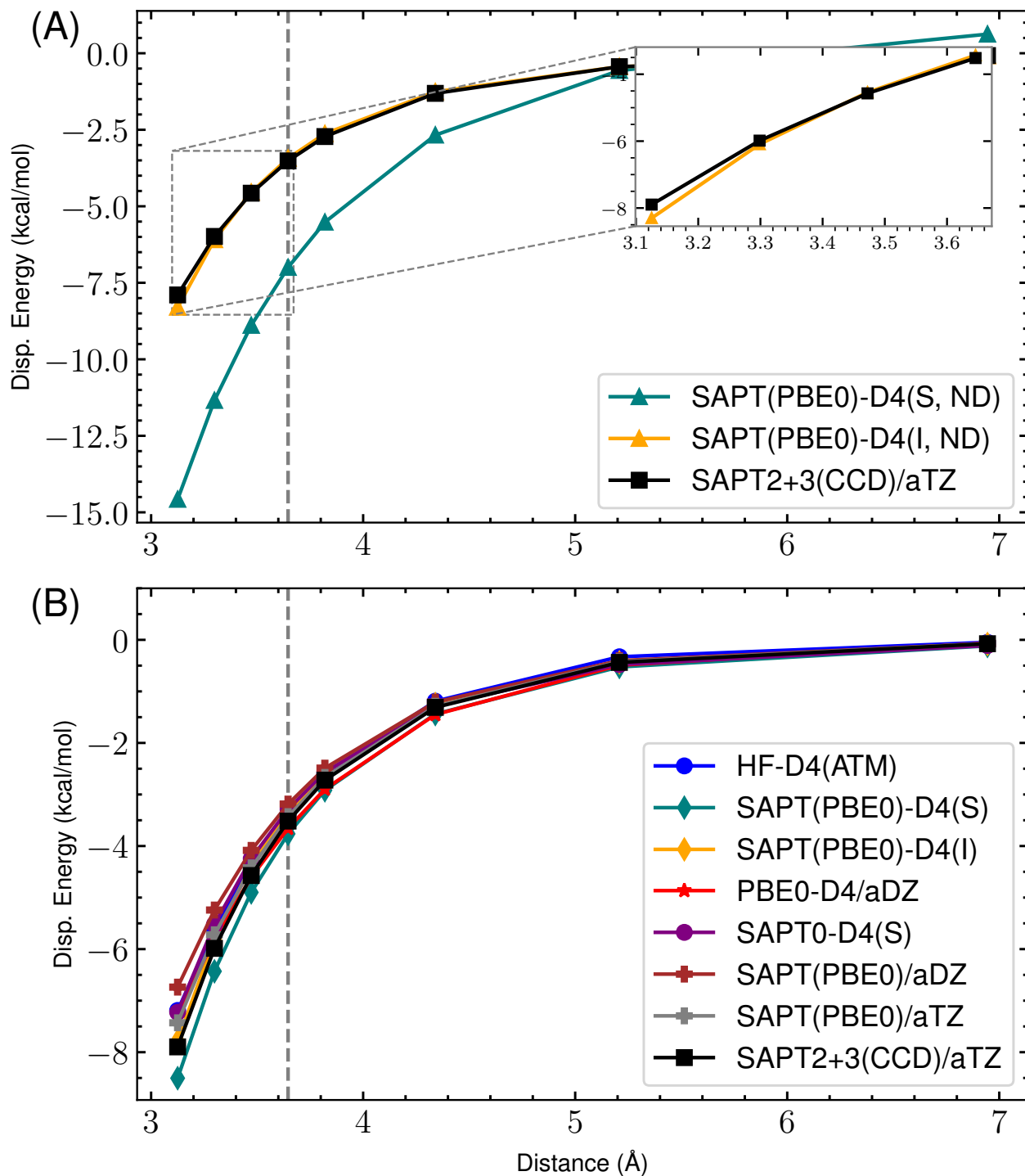


Figure 43: **s66x8 30_Benzene-Ethene**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

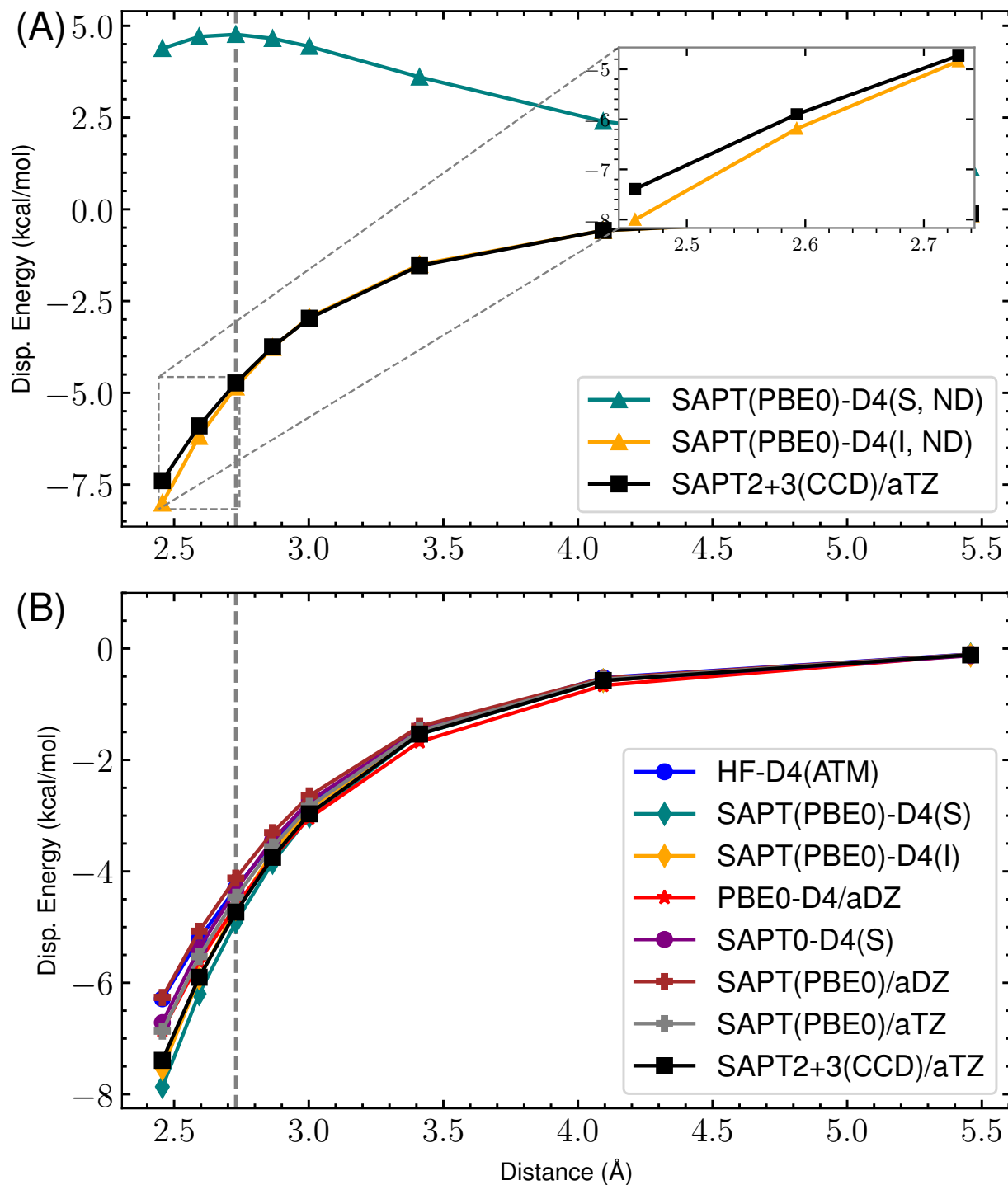


Figure 44: **s66x8 56_Benzene-MeNH₂**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

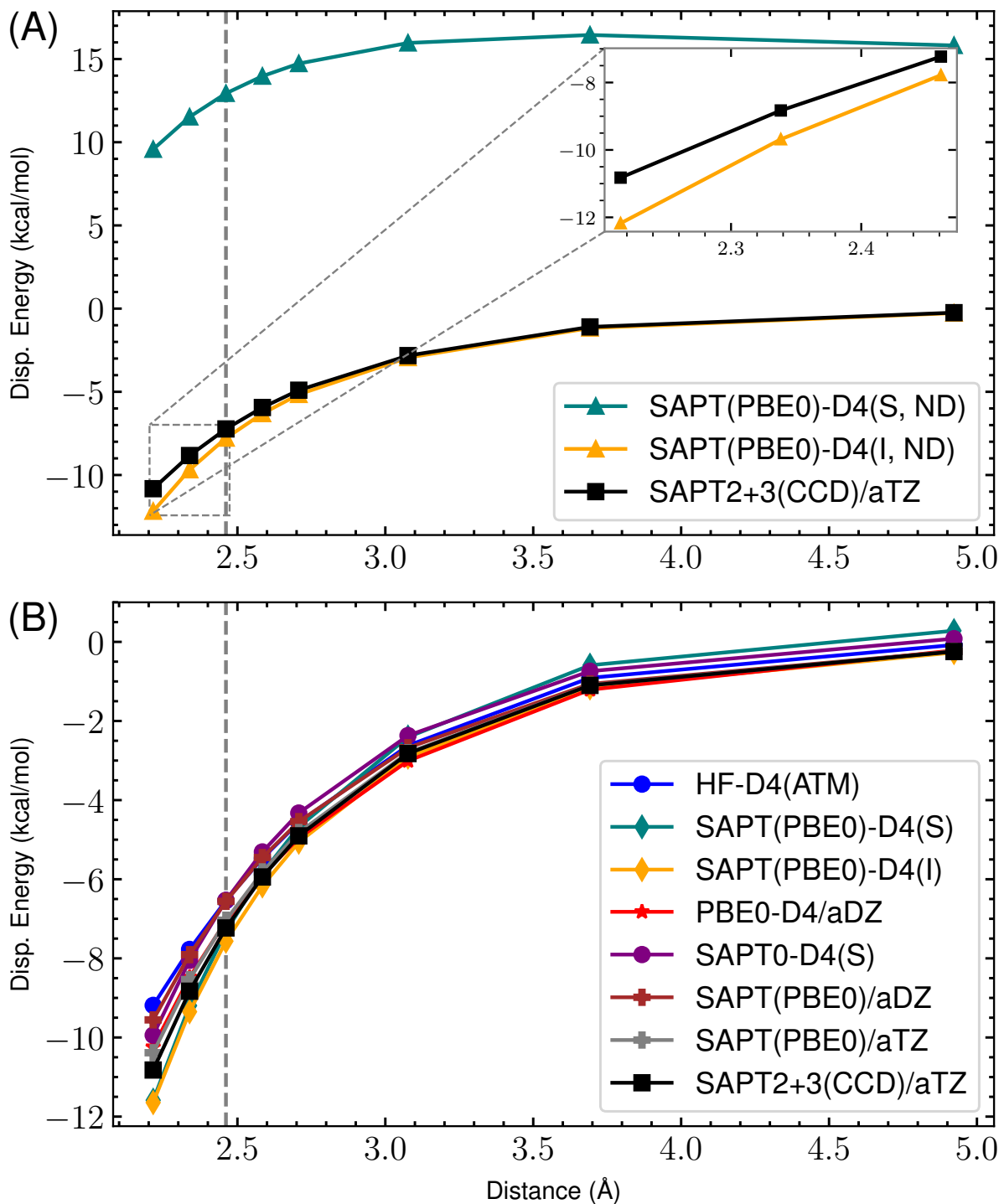


Figure 45: **s66x8 46_Peptide-Pentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

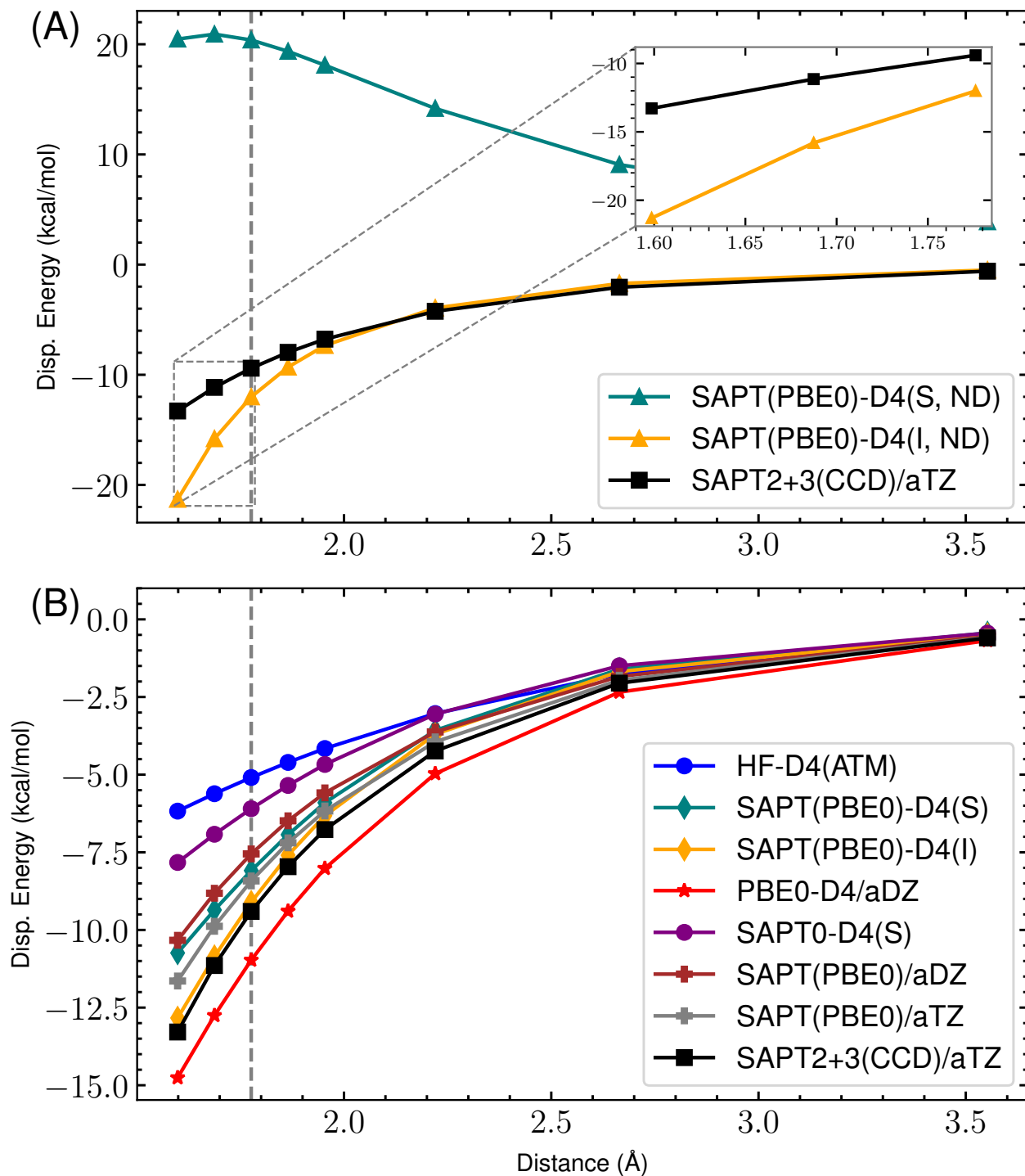


Figure 46: **s66x8 17_Uracil-Uracil**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

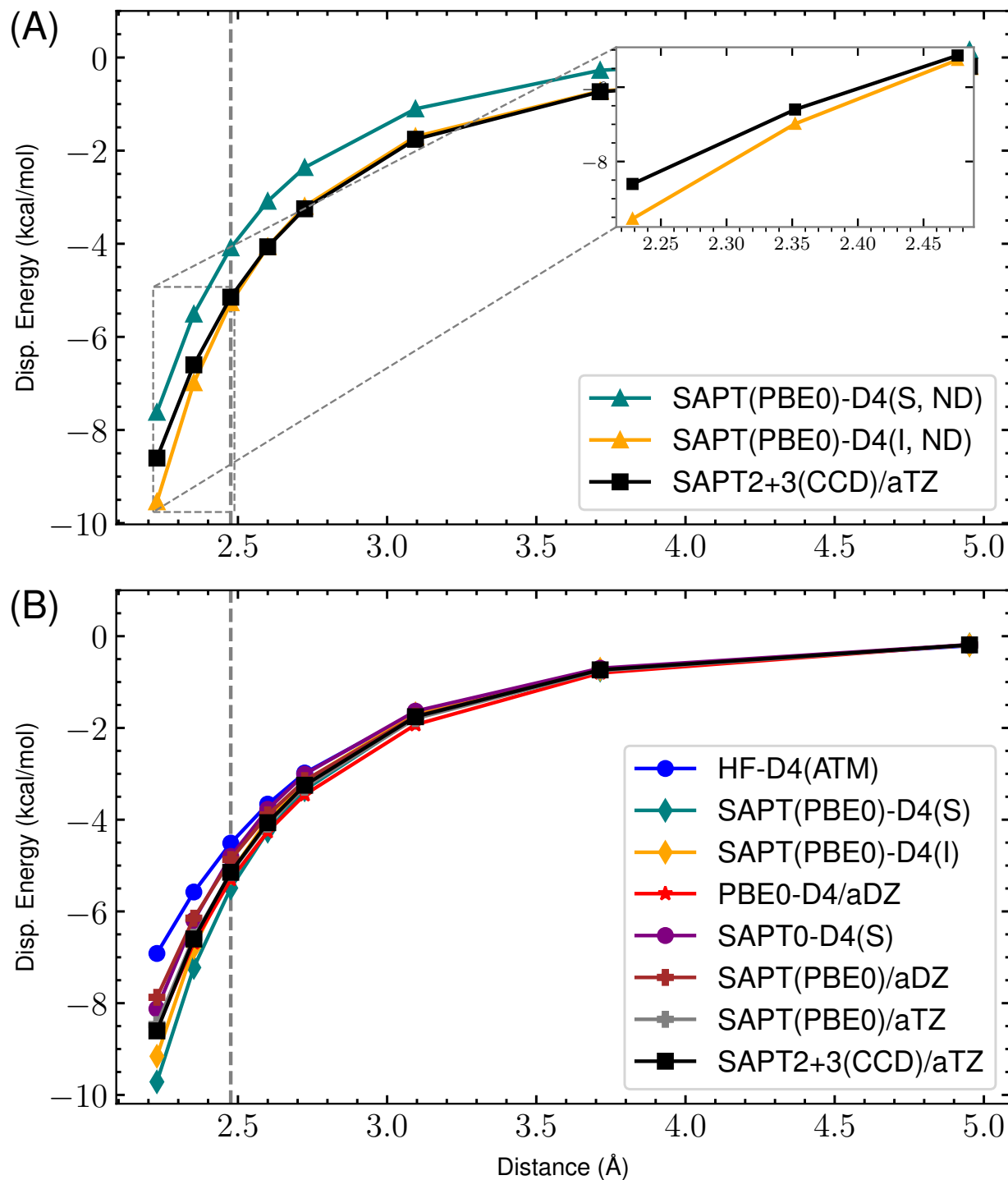


Figure 47: **s66x8 35_Neopentane-Pentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

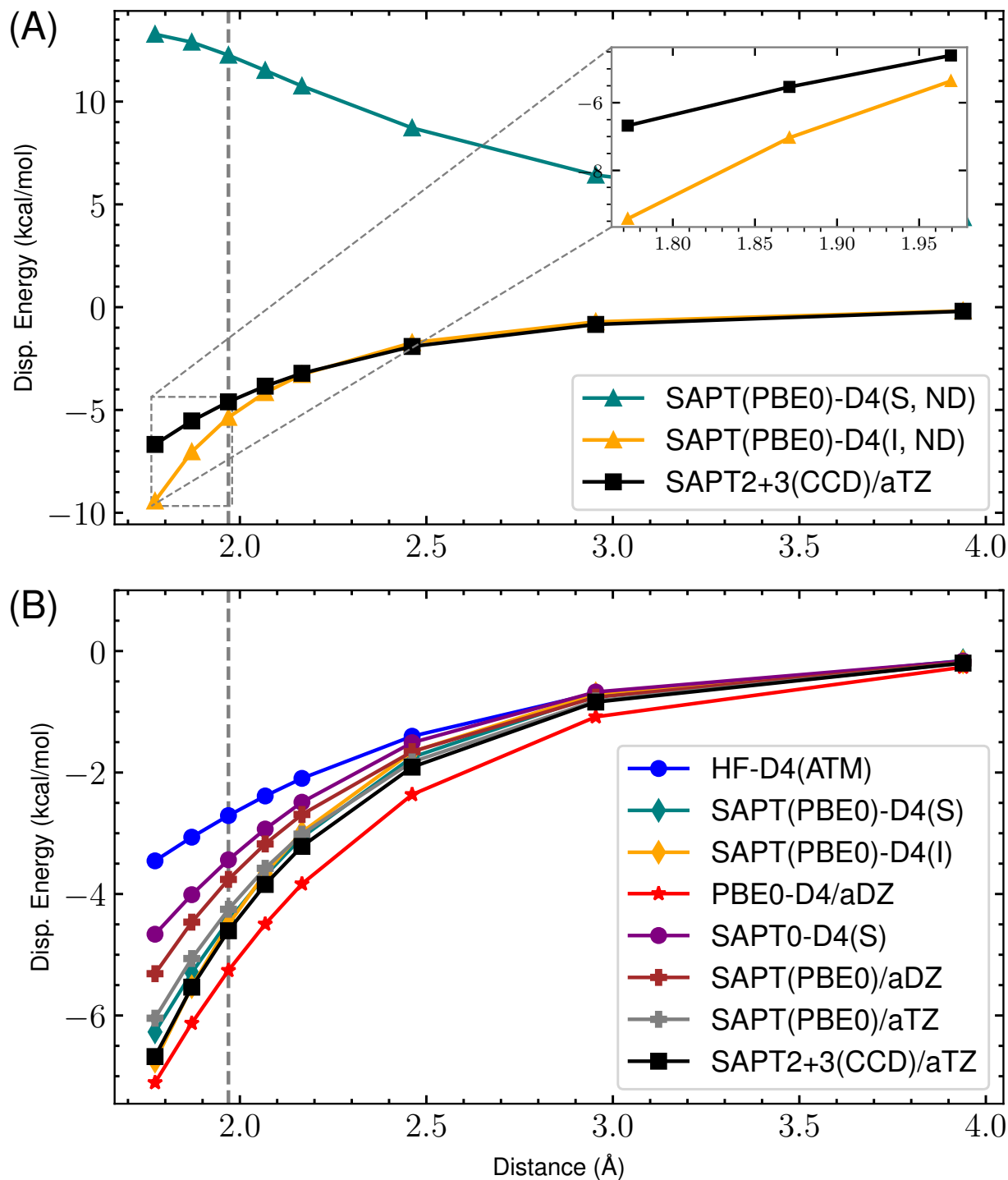


Figure 48: **s66x8 06_MeOH-MeNH2**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

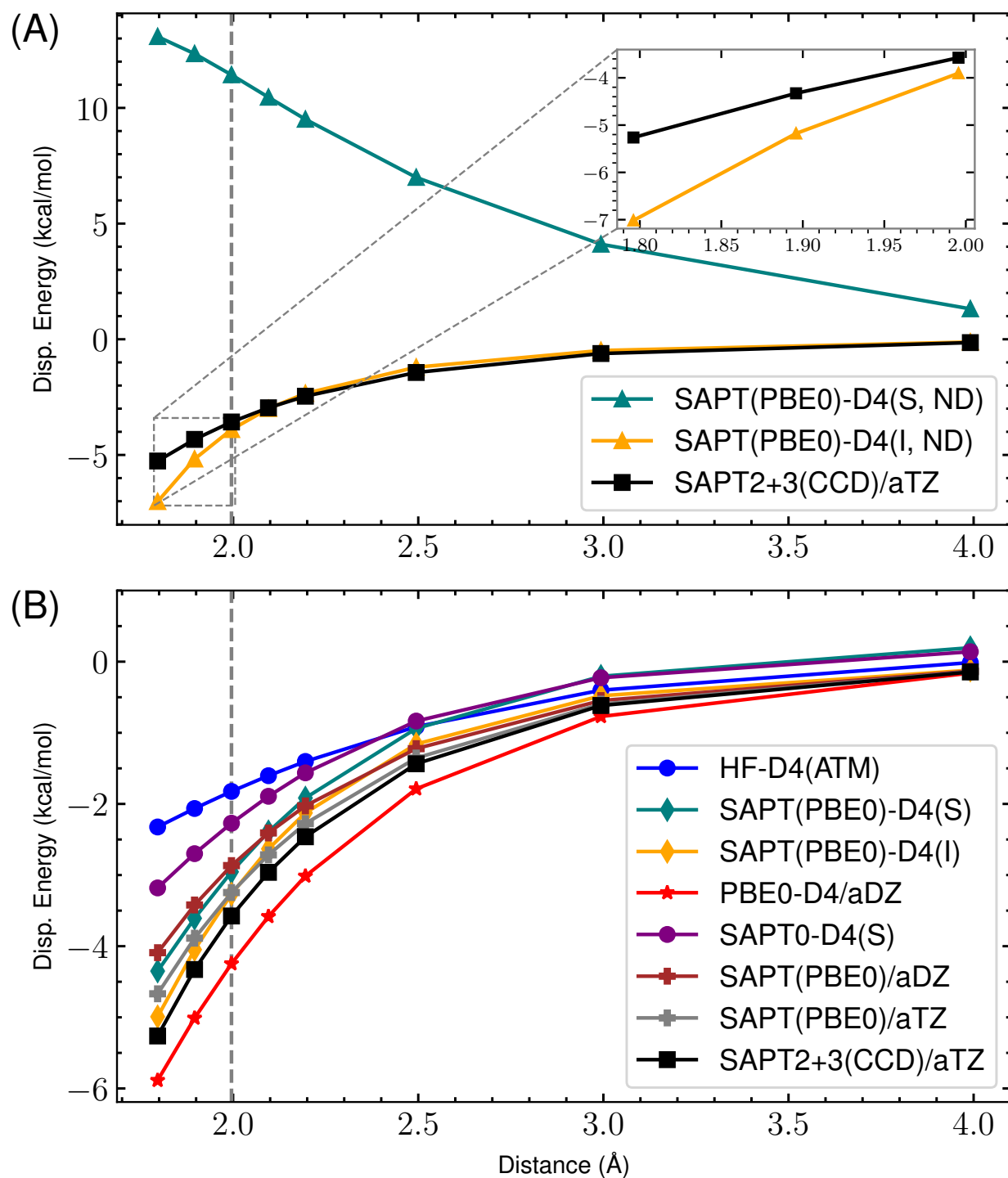


Figure 49: **s66x8 18_Water-Pyridine**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

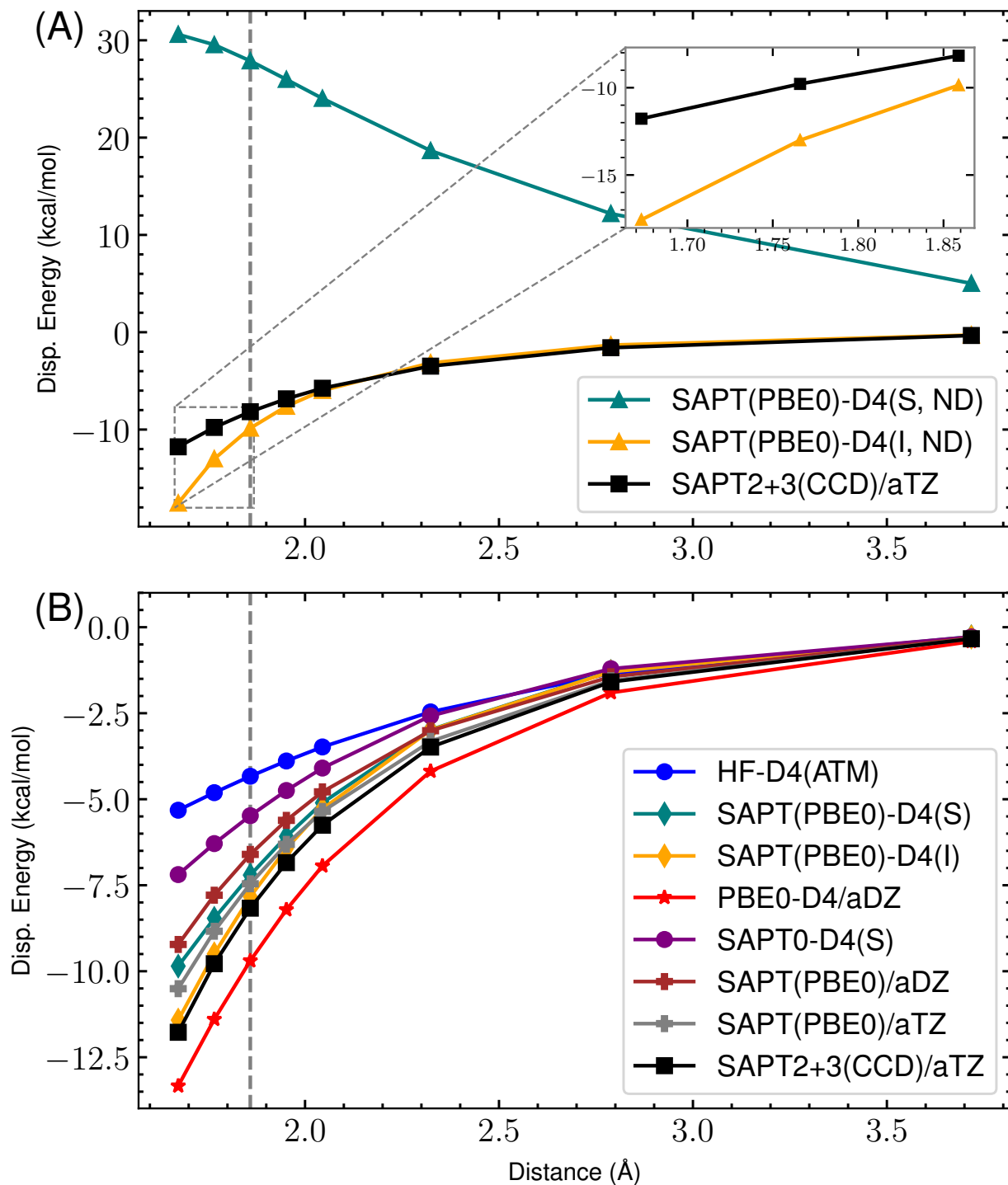


Figure 50: **s66x8 21_AcNH₂-AcNH₂**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

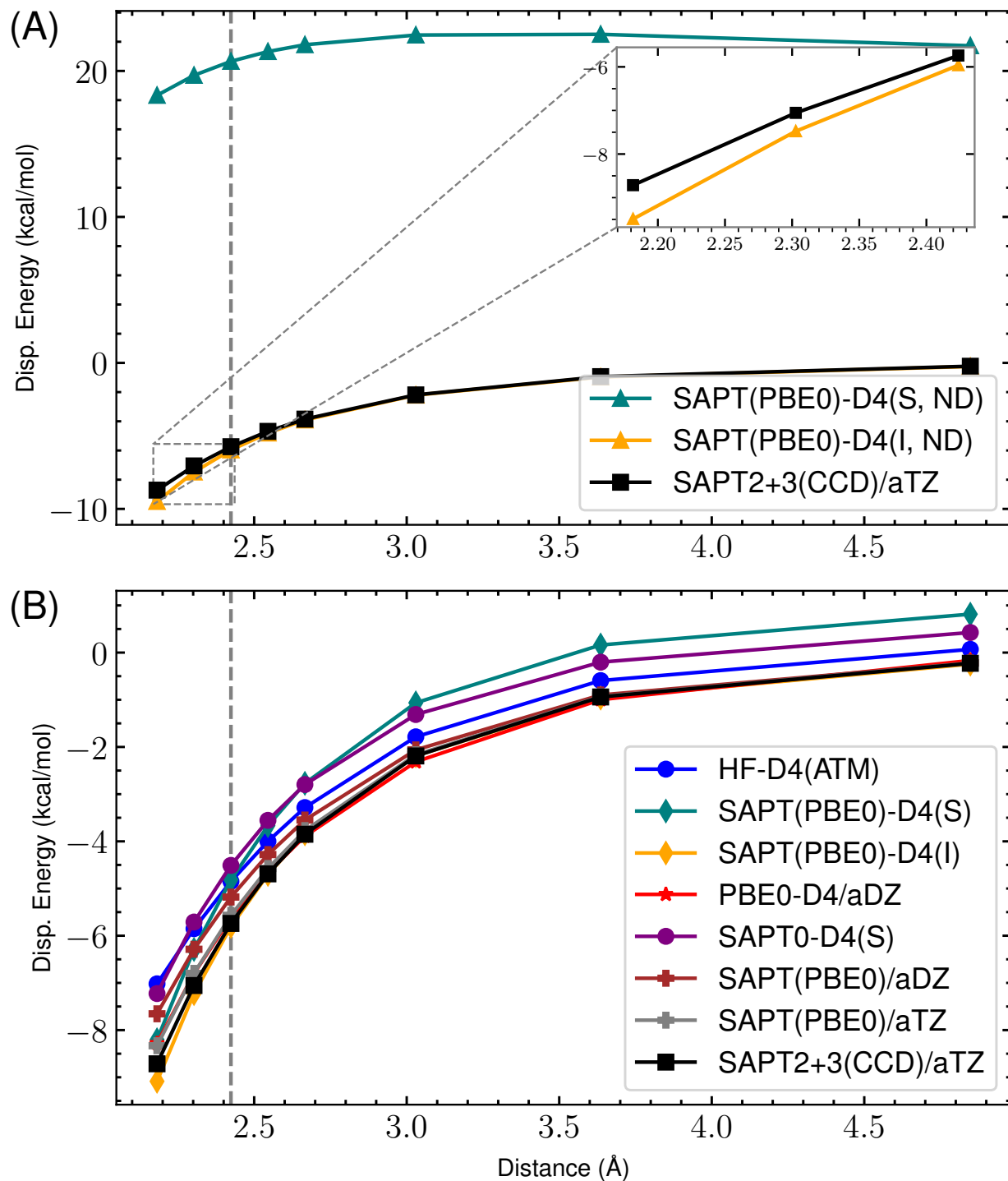


Figure 51: **s66x8 62_Pentane-AcNH2**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

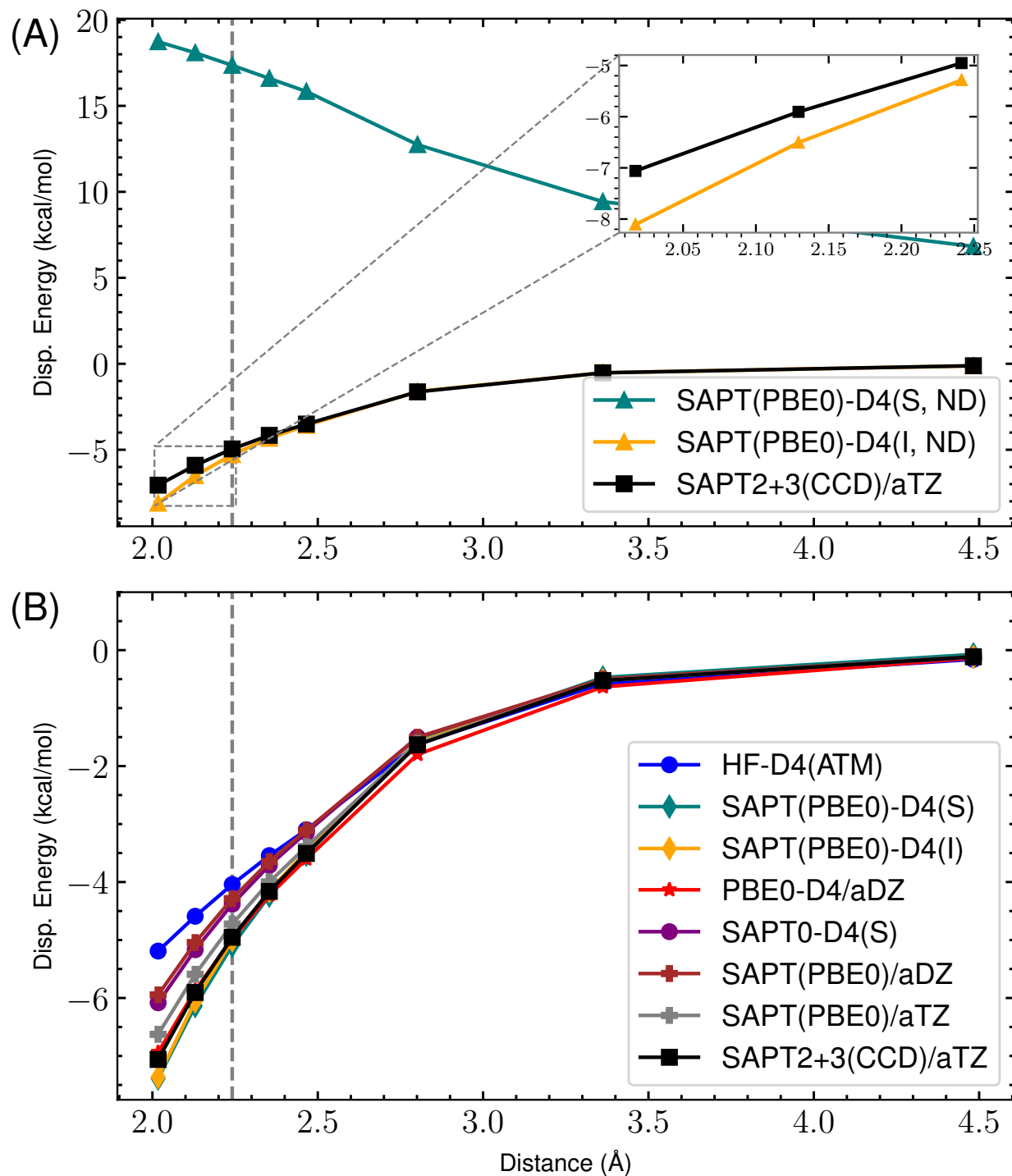


Figure 52: **s66x8 11_MeNH2-Peptide**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

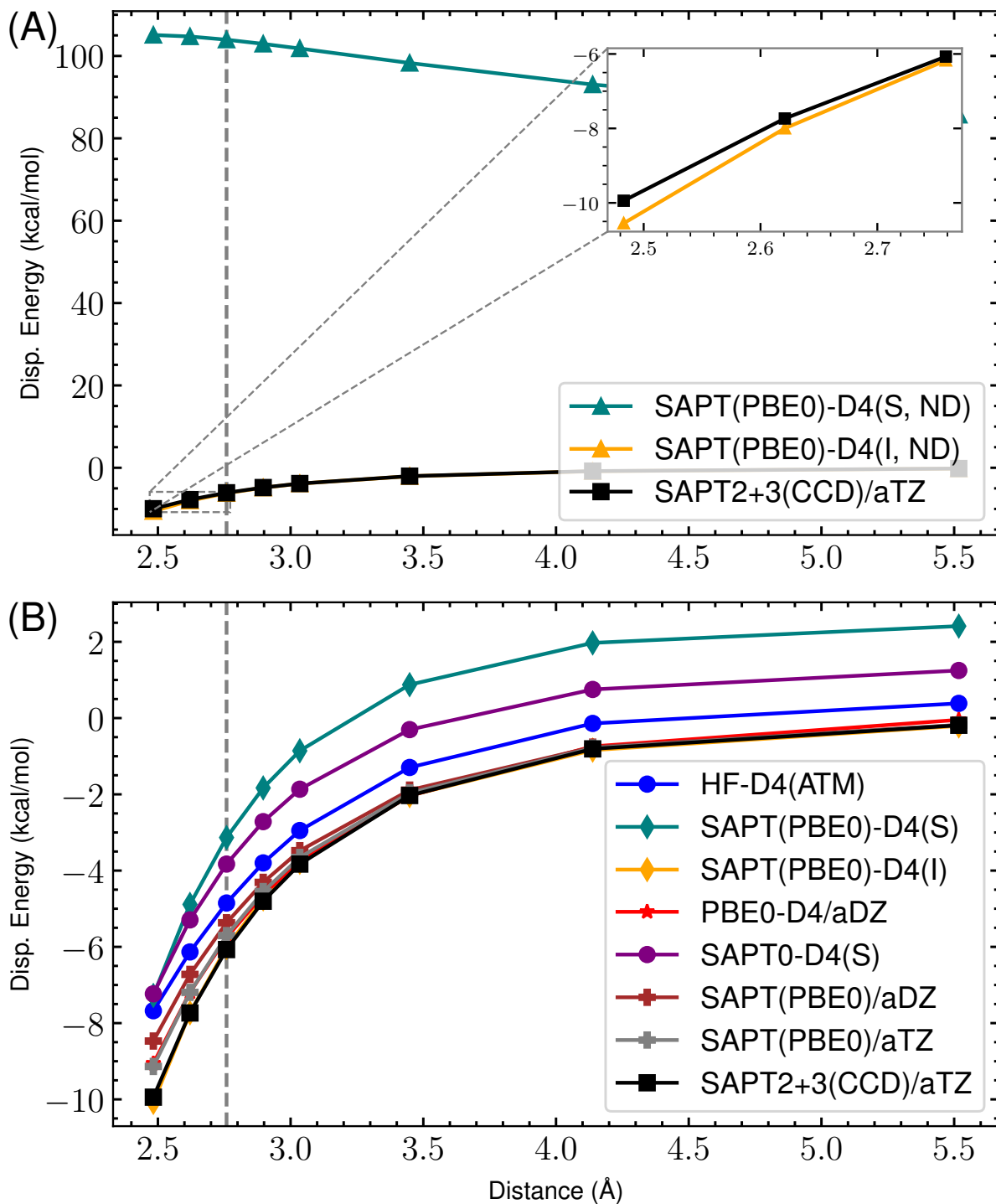


Figure 53: **s66x8 43_Uracil-Neopentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

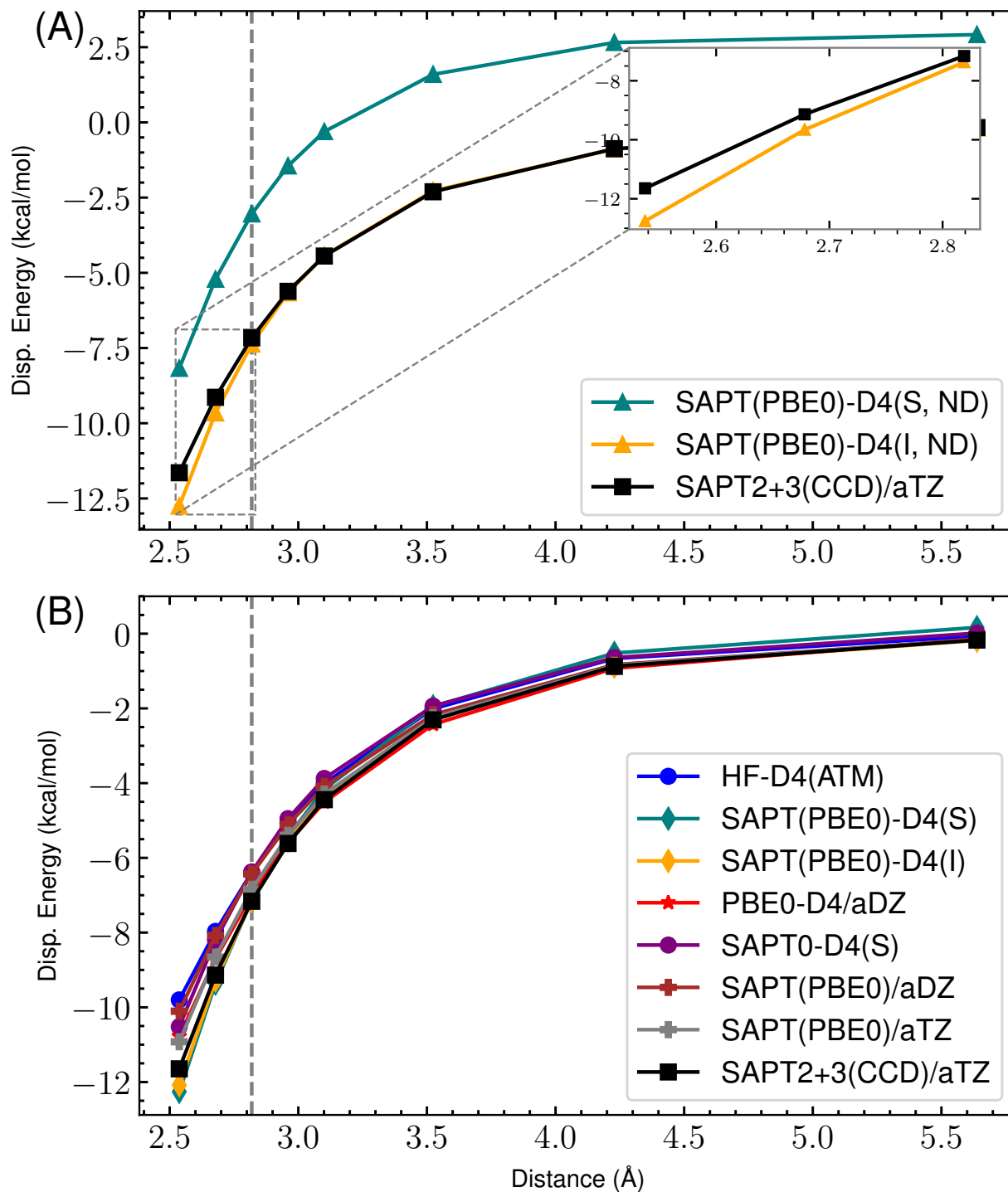


Figure 54: **s66x8 39_Benzene-Cyclopentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

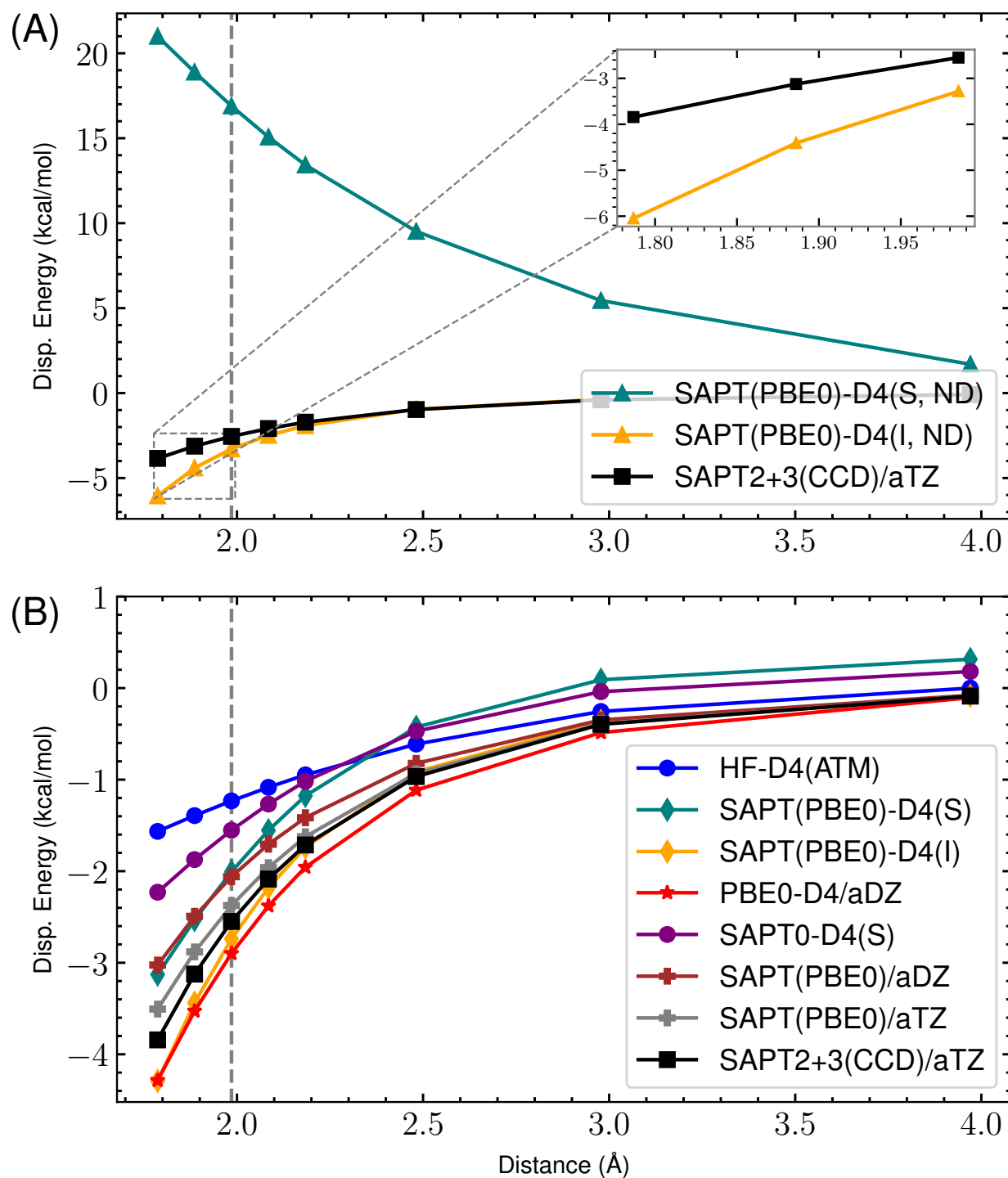


Figure 55: **s66x8 08_MeOH-Water**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

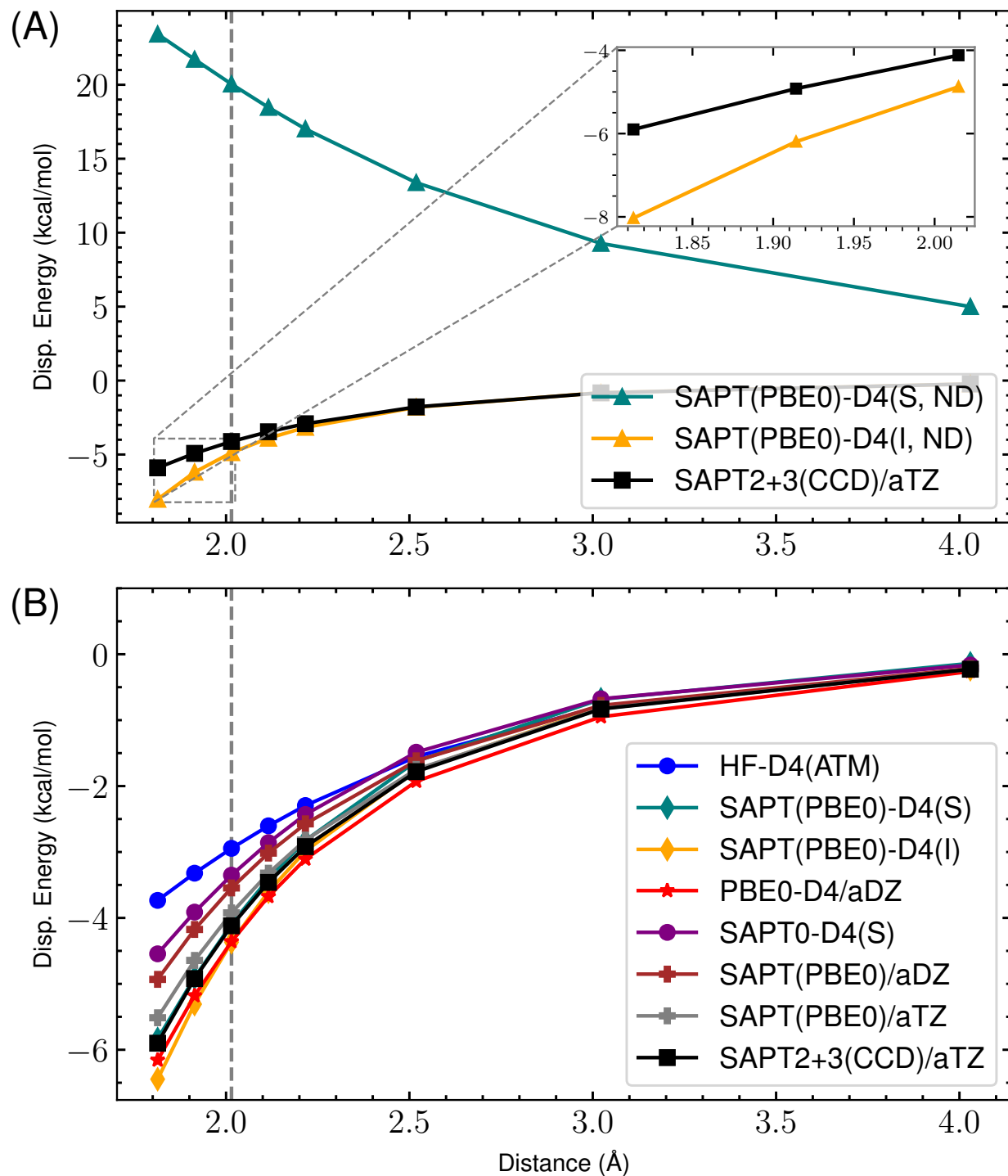


Figure 56: **s66x8 13_Peptide-MeOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

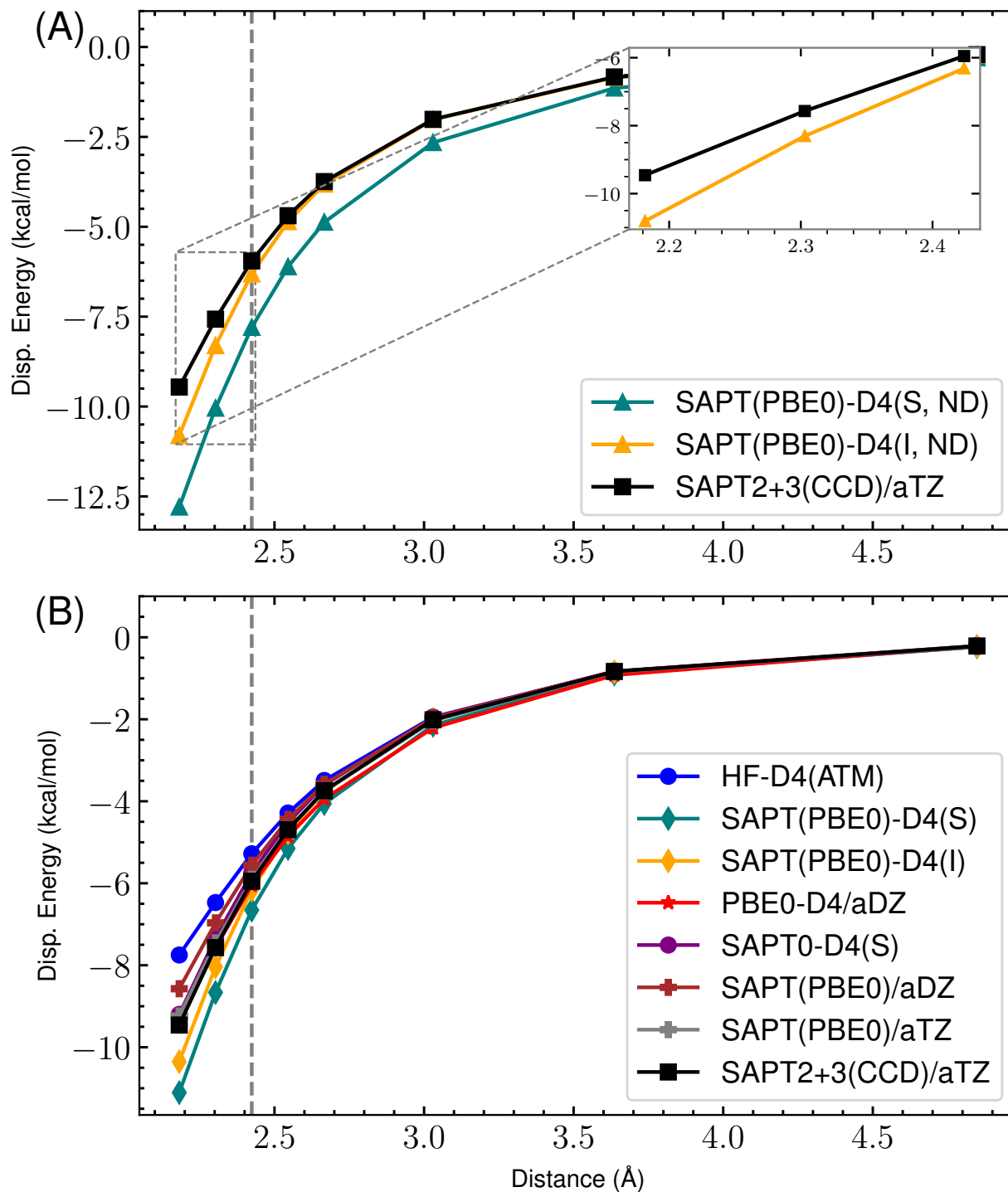


Figure 57: **s66x8 38_Cyclopentane-Cyclopentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

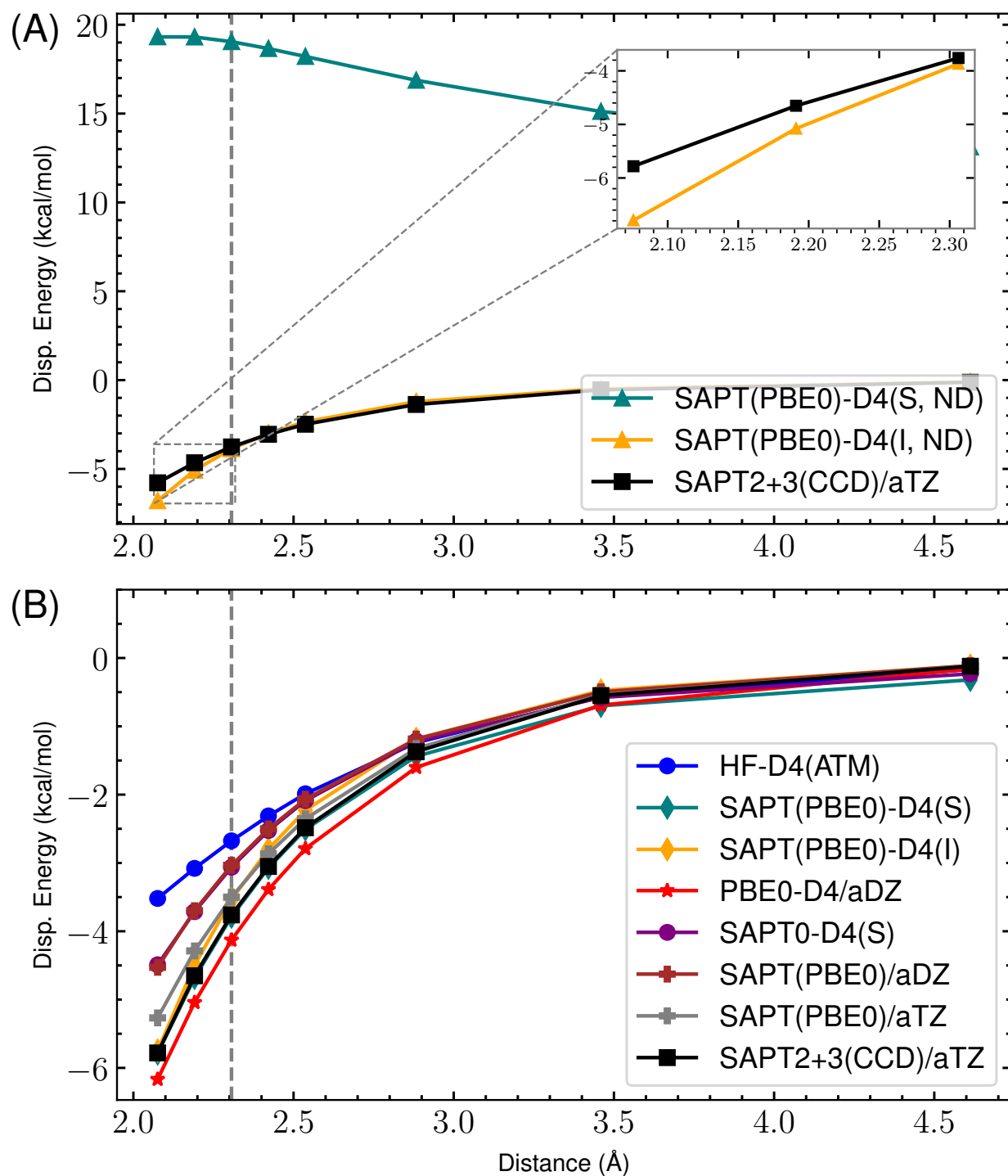


Figure 58: **s66x8 60_Ethyne-AcOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

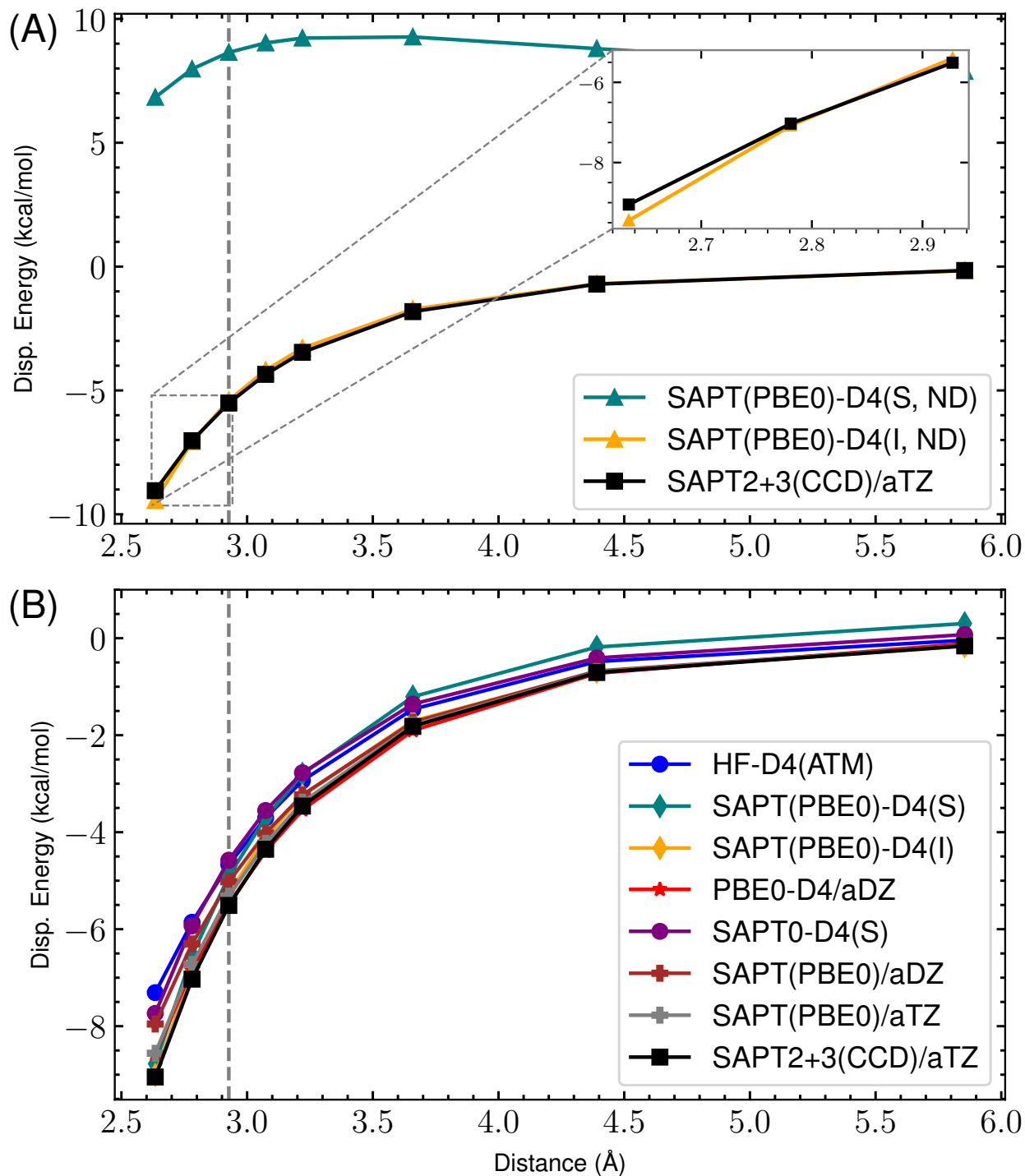


Figure 59: **s66x8 40_Benzene-Neopentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

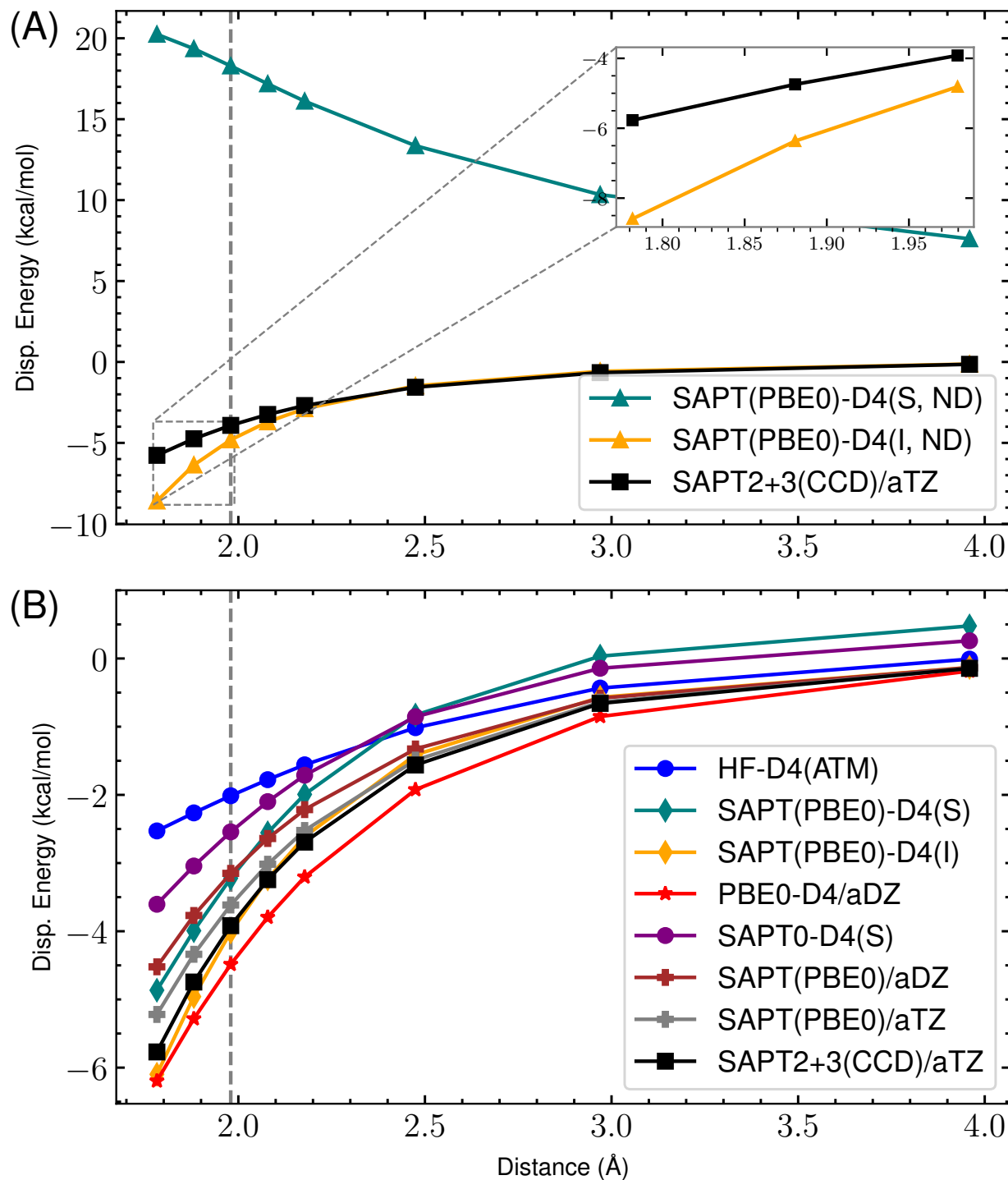


Figure 60: **s66x8 12_MeNH2-Water**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

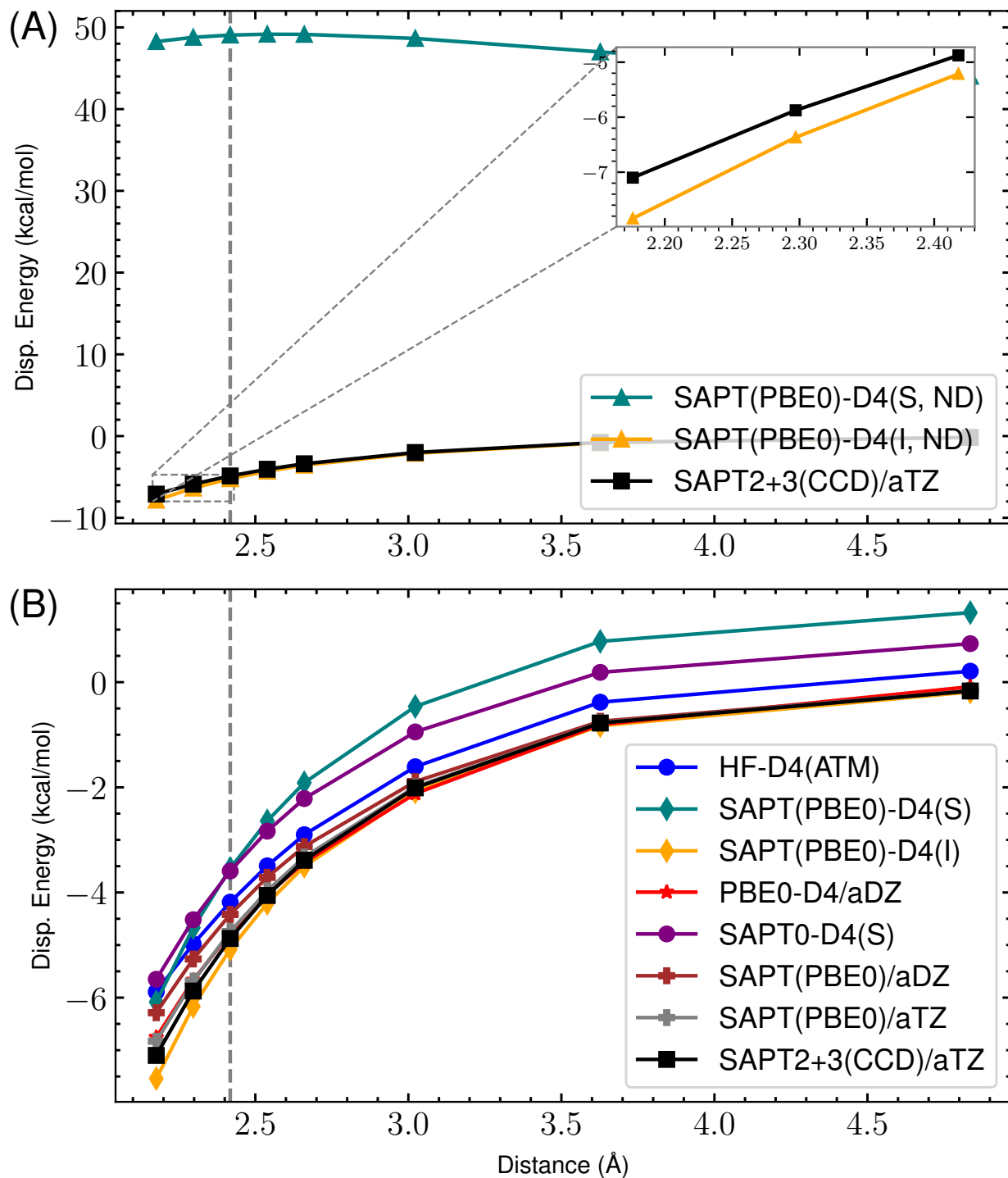


Figure 61: **s66x8 61_Pentane-AcOH**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

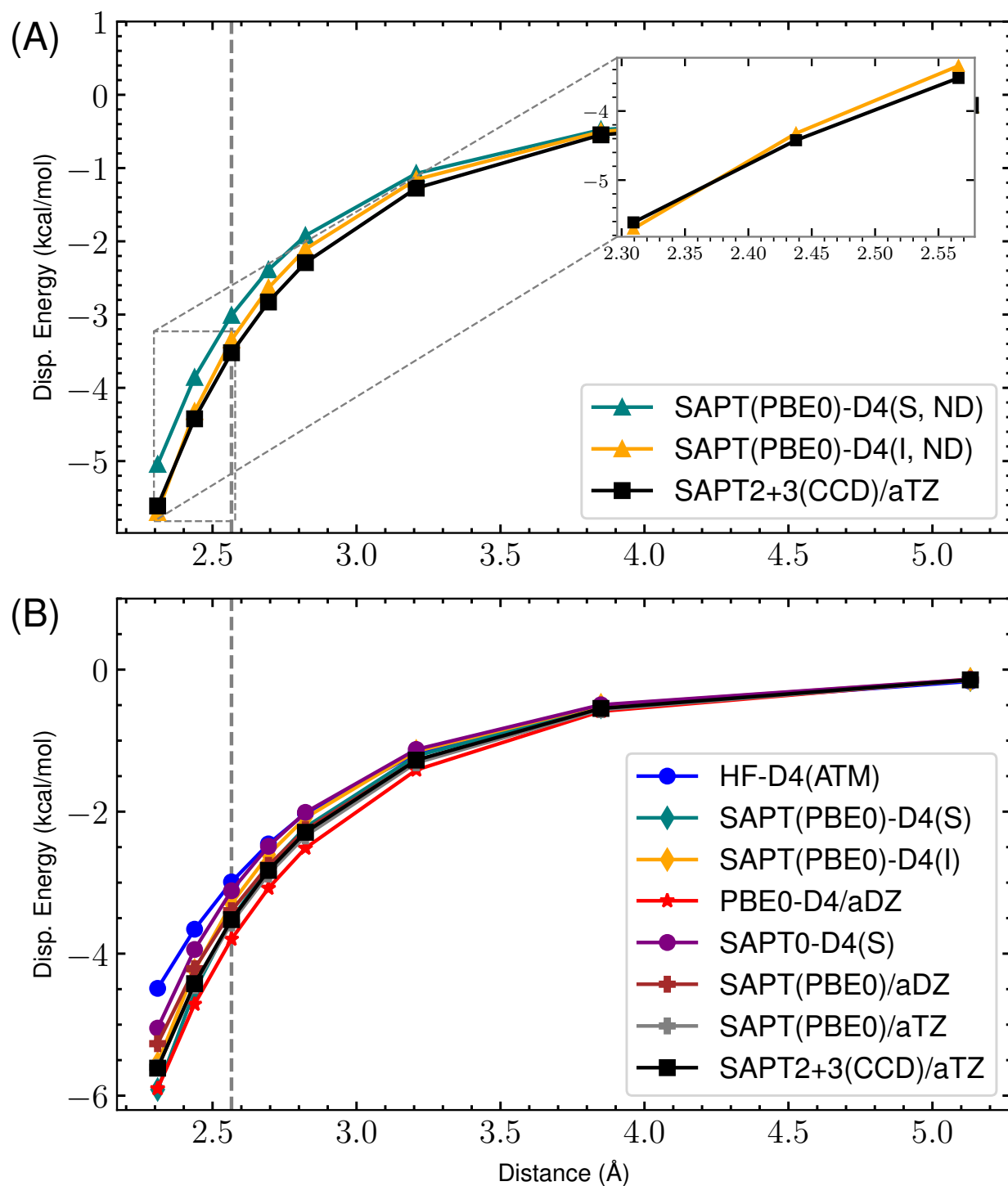


Figure 62: **s66x8 36_Neopentane-Neopentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

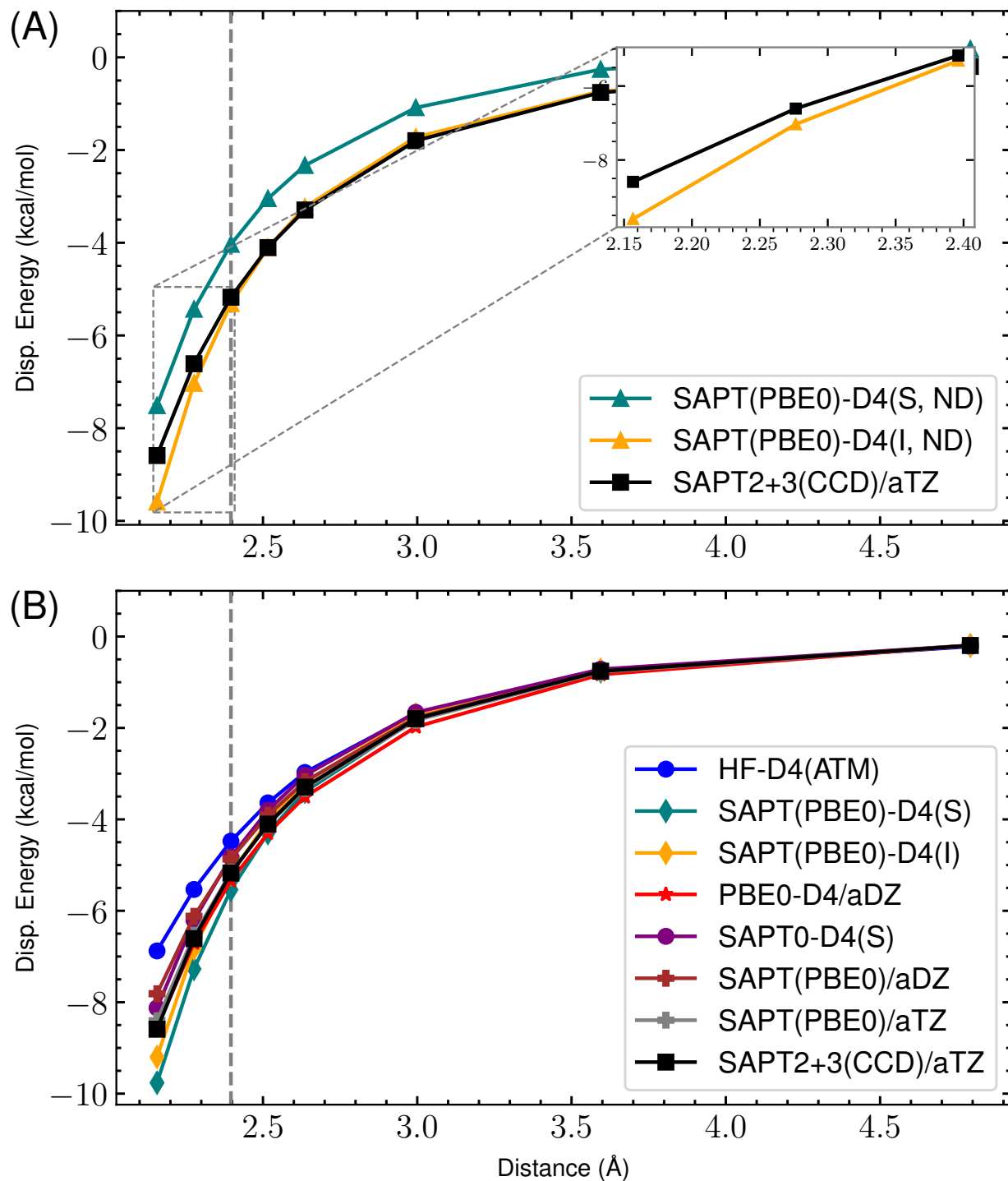


Figure 63: **s66x8 37_Cyclopentane-Neopentane**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

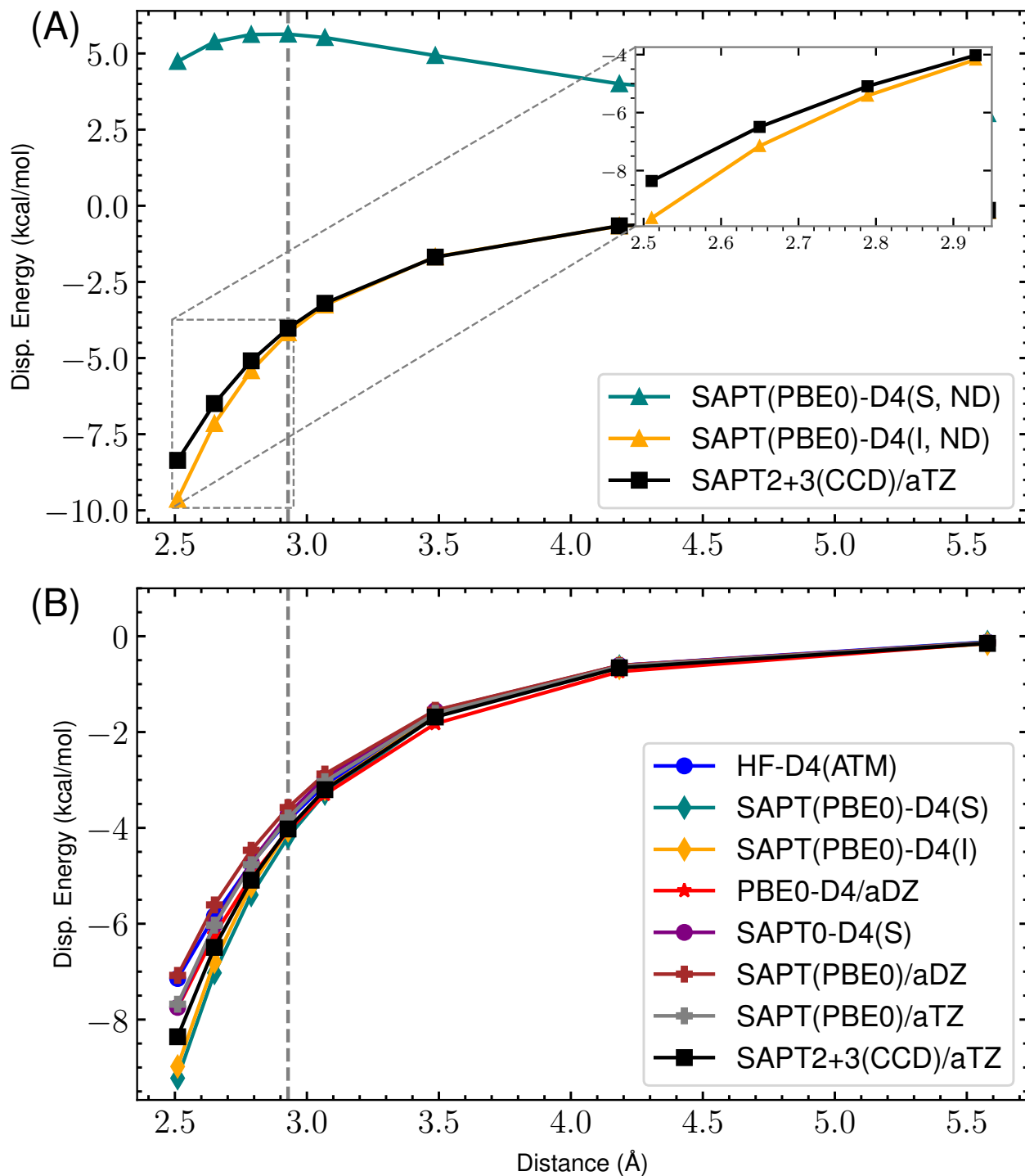


Figure 64: **s66x8 49_Benzene-Pyridine**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

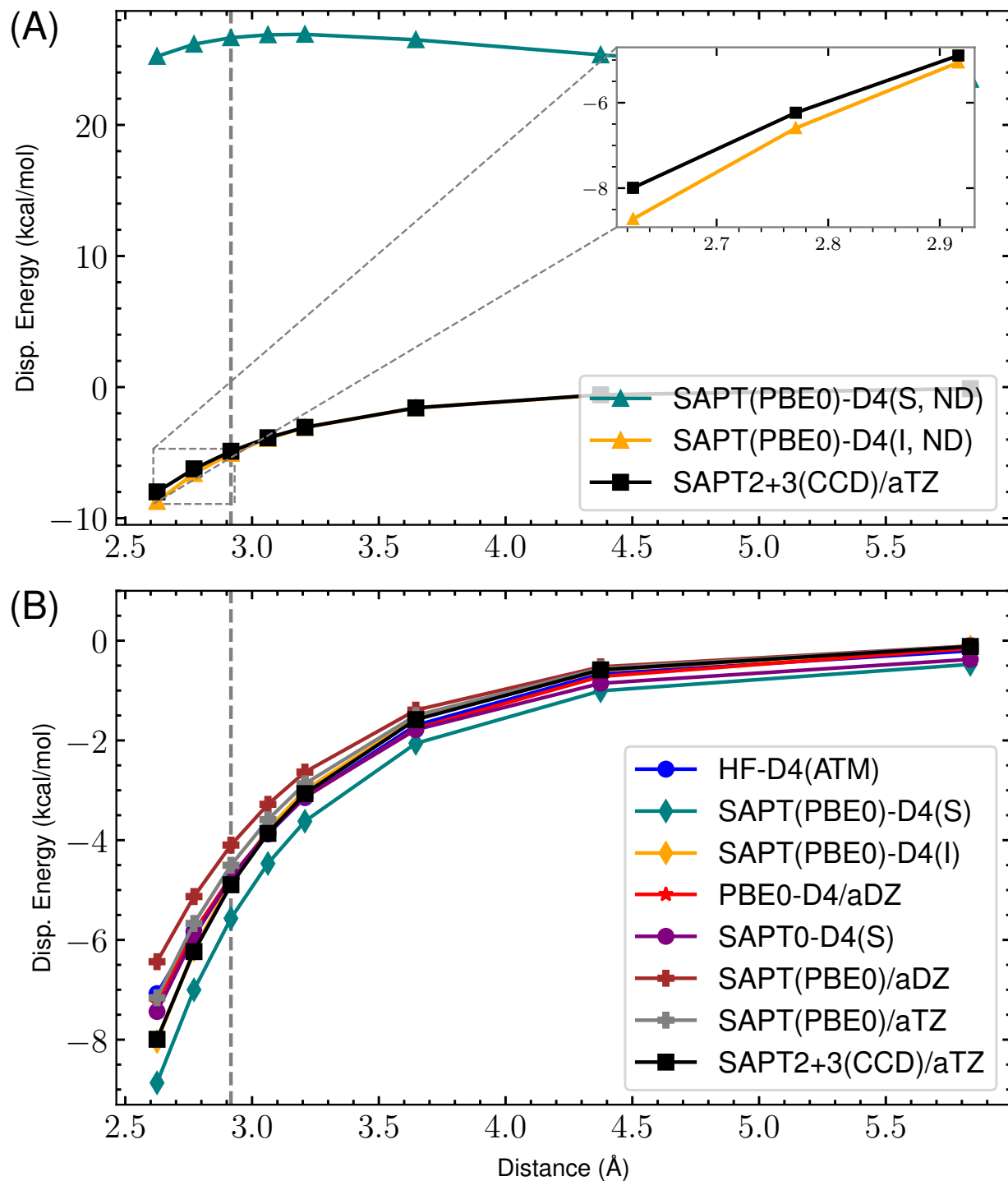


Figure 65: **s66x8 32_Uracil-Ethyne**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.

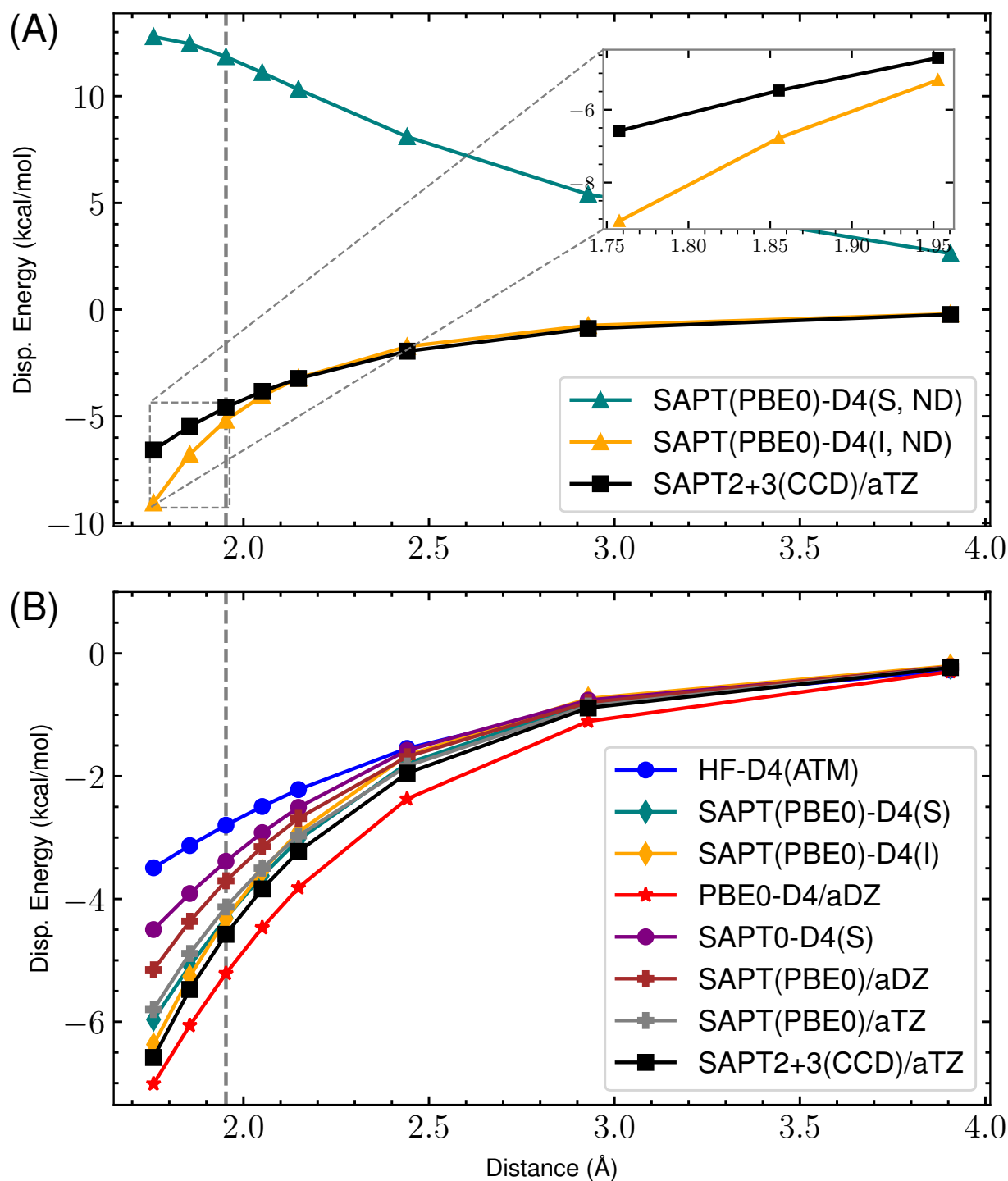


Figure 66: **s66x8 19_MeOH-Pyridine**: Dispersion potential Energy curve for SAPT(PBE0) with aDZ/aTZ and -D4 variants versus the reference SAPT2+3(CCD)/aug-cc-pVTZ dispersion energy. (A) highlights the completely non-damped (ND) D4 curves compared to the reference for both supermolecular (S) and intermolecular (I). The remaining $s_8 = 0.89529649$. (B) shows the damped -D4(S), -D4(I), HF-D4, SAPT(PBE0)/aDZ, SAPT(PBE0)/aTZ, and reference dispersion energies. Here, (S) and (I) refer to the supermolecular and intermolecular dispersion energy approaches, respectively. The “ND” states that no damping function is used. For the damped SAPT(PBE0)-D4(S) and SAPT(PBE0)-D4(I) terms, a_1 , a_2 , and s_8 are fit to the residual CCSD(T)/CBS energies on the 4470 dimer dataset.